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Interface circuits for a six-phase variable reluctance energy harvester: a comparative study of rectification and power conditioning topologies

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Per tutte le persone
per me speciali

Contents

Abstract

In recent years, more and more wireless sensor networks (WSN) and IoT devices are being used, and this has made their power supply a critical issue. Battery-based solutions have significant replacement and maintenance costs, especially when there are many sensor nodes or when they are hard to reach. Energy harvesting is the conversion of ambient energy, such as mechanical vibrations, heat, light, or RF signals, into usable electrical energy. It is proposed as a solution to this problem, since it allows the realisation of autonomous nodes that need almost no maintenance.

This thesis starts with an overview of the main transduction technologies used for harvesting from mechanical energy. These include piezoelectric, electromagnetic, electrostatic and triboelectric technologies, and their advantages and limitations are discussed. The thesis then focuses on the technology based on variable reluctance, called Variable Reluctance Energy Harvester (VREH). This technology works well for low-speed rotational applications, because both the magnet and the coil stay fixed, and the change in linked flux is caused only by the passage of a ferromagnetic toothed wheel. The state of the art of research on this technology is then reviewed, starting from the first prototypes built for railway monitoring, moving through single-pole topologies, and arriving at the multi-phase system with six units (MP-VREH) used as reference in this work. This system is designed to be integrated into large commercial vehicles, such as buses, at rotation speeds between 100 and 400 rpm (about 20-80 km/h).

Starting from Faraday's law, a complete electrical model of the VREH is then derived. This model gives the relations that link the extracted power to the geometric and electrical parameters of the transducer, such as the number of turns, the coil resistance, the flux difference between the aligned and unaligned positions, and the number of teeth of the wheel. From this model, the maximum power that can theoretically be transferred to the load is found, and the conditions for impedance matching are discussed as a function of the rotation speed.

The central part of the thesis focuses on the design of the power conditioning circuits for the six-phase system considered. Several rectification topologies are proposed and compared. They are first evaluated through circuit simulation, and then through an experimental test bench, which includes the design of a dedicated PCB and the calibration of the current sensors needed to accurately measure the power extracted from the system. Finally, the most suitable power management integrated circuit (PMIC) is chosen to regulate and store the harvested energy. This completes the conversion chain, from rotational mechanical energy to a stable power supply for the final load.

Sommario

Negli ultimi anni la crescente diffusione di reti di sensori wireless (WSN) e di dispositivi IoT ha reso critico il problema della loro alimentazione. Le soluzioni a batteria comportano infatti costi di sostituzione e manutenzione non trascurabili, specialmente quando i nodi sensore sono numerosi o di difficile accesso. L'*energy harvesting*, ossia la conversione di energia ambientale (vibrazioni meccaniche, calore, luce, segnali RF) in energia elettrica utilizzabile, si propone come soluzione a questo problema, permettendo la realizzazione di nodi autonomi e a manutenzione pressoché nulla.

Questo lavoro di tesi si apre con una panoramica delle principali tecnologie di trasduzione, piezoelettrica, elettromagnetica, elettrostatica e triboelettrica, per l'harvesting da energia meccanica, evidenziandone vantaggi e limiti, per poi concentrarsi sulla tecnologia basata sulla riluttanza variabile (*Variable Reluctance Energy Harvester*, VREH). Questa tecnologia è particolarmente adatta alle applicazioni rotazionali a bassa velocità, poiché magnete e bobina restano entrambi fissi e la variazione di flusso concatenato è generata unicamente dal passaggio di una ruota dentata ferromagnetica. Viene quindi ripercorso lo stato dell'arte della ricerca su questa tecnologia, dai primi prototipi per il monitoraggio ferroviario, alle topologie a singolo polo, fino al sistema multifase (MP-VREH) a sei unità preso come riferimento in questo lavoro, pensato per l'integrazione in veicoli commerciali di grandi dimensioni, quali autobus, a velocità di rotazione comprese tra 100 e 400 rpm (circa 20-80 km/h).

A partire dalla legge di Faraday viene poi sviluppato un modello elettrico completo del VREH, con la derivazione delle relazioni che legano la potenza estratta ai parametri geometrici ed elettrici del trasduttore, quali numero di spire, resistenza di bobina, differenza di flusso tra posizione allineata e disallineata, e numero di denti della ruota, fino alla determinazione della potenza massima teoricamente trasferibile al carico e alla discussione delle condizioni di adattamento di impedenza in funzione della velocità di rotazione.

La parte centrale della tesi è dedicata alla progettazione dei circuiti di condizionamento della potenza per il sistema a sei fasi considerato. Vengono proposte e confrontate diverse topologie di raddrizzamento, valutate dapprima tramite simulazione circuitale e successivamente tramite la realizzazione di un banco di prova sperimentale, comprensivo della progettazione di una scheda PCB dedicata e della taratura dei sensori di corrente per la misura accurata della potenza estratta dal sistema. Infine, viene affrontata la scelta del circuito integrato di power management (PMIC) più adatto a regolare e immagazzinare l'energia raccolta, completando così la catena di conversione dall'energia meccanica rotazionale fino all'alimentazione stabile del carico finale.

1 Introduction

1.1 Motivation

The number of wireless sensor networks (WSNs) and Internet of Things (IoT) devices has grown rapidly in recent years, raising a fundamental question: how to power all these devices without replacing batteries or requiring constant maintenance. Recent advances in low-power electronics have reduced the energy consumption of sensor nodes to the range of tens to hundreds of microwatts [12]. At such low power levels, it becomes possible to power these devices by extracting energy from the surrounding environment, removing the need for batteries and allowing the network to operate indefinitely.

Batteries are the most common power source for WSN nodes, but they have well-known limitations. In many applications, such as structural health monitoring, industrial machinery surveillance, or medical implants, replacing or recharging exhausted batteries is difficult or simply not possible [7]. Using larger batteries to extend the device lifetime increases the size of the system, which is often not acceptable. In addition, the environmental cost of batteries and the consequent maintenance of large-scale networks are serious obstacles to the widespread use of WSNs [10].

Energy harvesting offers a practical solution to these problems. Instead of storing a fixed amount of energy, harvesting systems collect energy from sources available in the environment such as sunlight, heat gradients, mechanical vibrations, or radio frequency signals and convert it into electrical power to supply an electronic system.

Among all available energy sources, mechanical vibrations have received the most attention in the research community, accounting for more than half of publications in this field [10]. Piezoelectric, triboelectric, and electromagnetic transducers are the most widely used mechanisms, each with different trade-offs in output voltage, power density, and complexity [12, 7]. Figure 1.1 shows the distribution of recent publications about transducer types, highlighting the dominance of mechanical energy harvesting approaches.

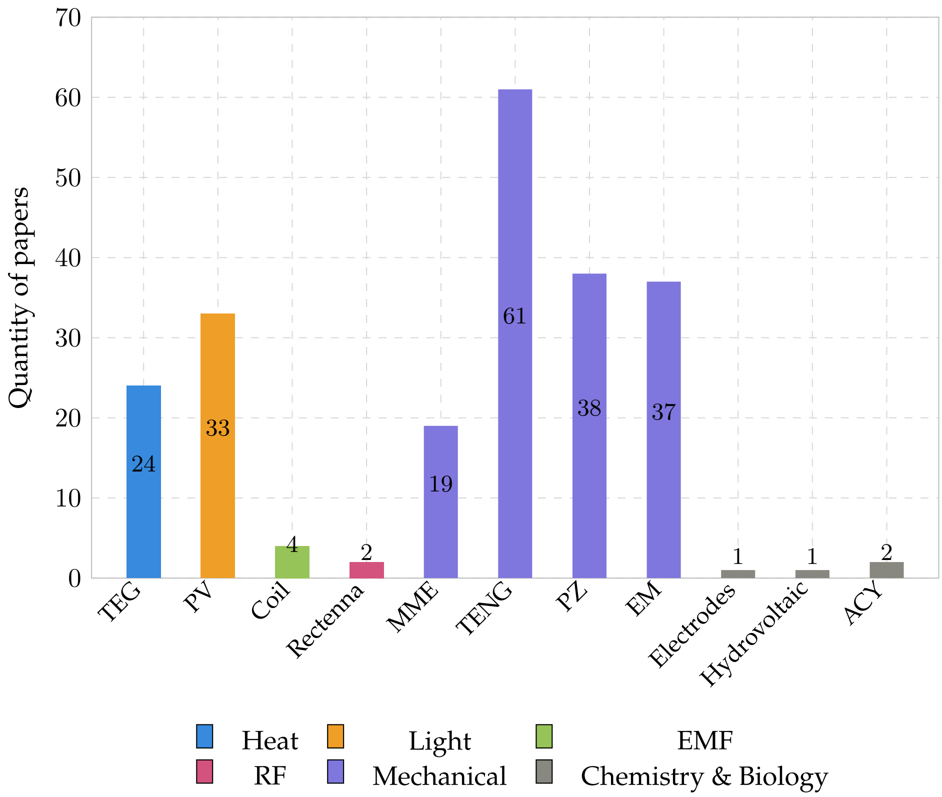


Figure 1.1: Distribution of papers based on transducer type [10].

Energy harvesting systems are generally divided into two categories [7]: systems that use the harvested energy immediately (*Harvest-Use*), and systems that store it first in a battery or supercapacitor before supplying it to the node (*Harvest-Store-Use*). The second approach is far more common, since most energy sources are not continuously available and some form of storage is needed to guarantee uninterrupted operation.

1.2 Energy Harvesting Systems

An energy harvesting system is made up of three components that work in sequence, as illustrated in Figure 1.2.

- *Transducer*: converts the ambient energy into an electrical signal;
- *Power conditioning circuit*: performs AC-DC conversion, impedance matching and voltage regulation. A typical implementation combines a rectifier circuit with a power management integrated circuit (PMIC) [19];
- *energy storage element*: is used to keep the device operating for a certain amount of time even if the source is weak or no longer present. Such an element can be a capacitor/super capacitor or a rechargeable battery [12].

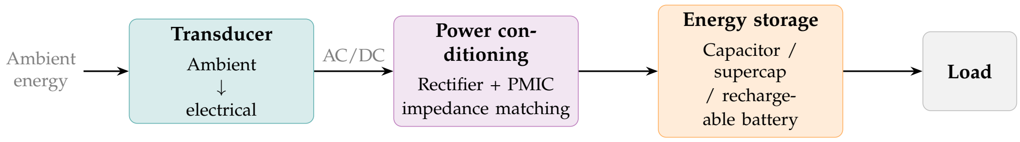


Figure 1.2: Block diagram of a typical energy harvesting system.

The performance of the entire system depends in particular on the transducer. Its design must be matched to the characteristics of the ambient source as well as to the power needs of the target device. A transducer that is well suited to one environment may perform poorly in another, which is why the choice of transduction mechanism is the central design decision in any harvesting application [7].

Designing a good harvesting system is not just about getting the most power out of it. The system must work reliably in real conditions, where the energy source can be irregular or unpredictable. At the same time, it must be small, cheap, and easy to integrate into the target device [7]. Finding the right balance between these requirements is the main challenge in this field, and it has led to the development of many different solutions, as discussed in the following sections.

1.3 Performance metrics

A key challenge in evaluating energy harvesting systems is choosing the right performance metrics. Xu et al. [19] proposed two metrics for this purpose. The first is the *power conversion efficiency* (PCE), which measures how much of the power entering the interface circuit actually reaches the output:

$$\text{PCE} = \eta_{\text{Rec}} \cdot \eta_{\text{DC-DC}} = \frac{P_{\text{out}}}{P_{\text{in}}}. \tag{1.1}$$

The second is the *power extraction efficiency* (PEE), which compares the output power to the maximum power the transducer could deliver under ideal conditions:

$$\text{PEE} = \frac{P_{\text{out}}}{P_{\text{max}}} = \frac{P_{\text{in}}}{P_{\text{max}}} \cdot \text{PCE}. \tag{1.2}$$

The difference between the two is important. A circuit can convert power efficiently (high PCE) while still wasting much of what the transducer could produce, if the circuit itself is not able to work at the optimal operating point. For this reason, PEE is a more useful metric when the goal is to maximise the total power delivered to the load [19].

1.4 Main Transduction Technologies

Mechanical vibrations can be converted into electrical energy through four main transduction mechanisms: piezoelectric, electromagnetic, electrostatic, and triboelectric. Each of them works in a different way and comes with its own advantages and limitations [11, 2].

Piezoelectric

Piezoelectric transducers produce a voltage when a material is mechanically deformed. The most common design is a cantilever beam with a piezoelectric layer on top: when the beam vibrates, the material deforms and generates an electric charge. This approach gives a high output voltage and does not need a magnet or moving parts, which makes it easy to integrate into small devices [1]. The main drawback is that piezoelectric materials are fragile and may crack under large deformations. Also, the output drops quickly when the vibration frequency moves away from the resonant frequency of the beam [12].

Electromagnetic

Electromagnetic transducers generate a voltage by moving a coil near a permanent magnet. This is the same principle used in large electrical generators, applied at a small scale. These harvesters tend to produce a low voltage but can deliver a relatively high current, which makes them a good choice when power output is the main goal. One limitation is that the coil and magnet must move relative to each other, which can be hard to achieve in applications involving large or slowly rotating objects [11].

Electrostatic

Electrostatic transducers work by changing the capacitance between two conductive plates as they move. When the plates get closer or further apart due to vibration, the stored electrical energy changes and can be extracted as useful power. This approach is easy to manufacture using standard microfabrication processes, making it attractive for very small devices. However, it requires an external voltage or a pre-charged material to get started, which adds complexity to the system [11, 2].

Triboelectric

Triboelectric transducers use contact between two different materials to generate surface charges. When the materials touch and then separate, charges build up and can drive a current through an external circuit. This type of harvester, known as a triboelectric nanogenerator (TENG), was first demonstrated by Fan et al.[6] and has attracted a lot of interest because it can be made from cheap, flexible materials and works well at low frequencies, such as those produced by human motion. The main weaknesses are sensitivity to humidity and wear of the contact surfaces over time [10].

Among these four mechanisms, no single technology is universally superior. The best choice depends on the specific application, the characteristics of the available vibration source, and the constraints on size and cost. This comparison motivates the investigation of alternative approaches, including variable reluctance transduction, which is discussed in the following section.

1.5 The Variable Reluctance Principle

Variable reluctance energy harvesting (VREH) is a type of electromagnetic energy harvesting based on Faraday's law of induction. The transducer consists of three main parts: a permanent magnet, a pickup coil wound around a ferromagnetic pole piece, and a toothed ferromagnetic wheel that rotates nearby. When a tooth on the wheel aligns with the pole piece, the magnetic flux through the coil reaches its maximum value. When the gap between two teeth faces the pole piece, the flux drops to its minimum. This periodic variation in flux induces an alternating voltage across the coil, with a frequency that depends on the rotational speed and the number of teeth on the wheel [17]:

$$\omega_c = \omega\, N_t, \tag{1.3}$$

where ω is the rotational speed in rad/s and N_t is the number of teeth on the wheel. From Faraday's law, the output power of the transducer delivered to a matched load can be expressed as [18]:

$$P_L = \frac{N_{\text{Coil}}^2}{8R_{\text{Coil}}}\Delta_\phi^2 \left[\frac{\omega N_{\text{Tooth}}}{2\pi}\right]^2, \tag{1.4}$$

where N_{Coil} is the number of coil turns, R_{Coil} is the coil resistance, Δ_ϕ is the magnetic flux difference between the aligned and unaligned positions, and N_{Tooth} is the number of teeth. This expression shows that the output power grows with the square of the rotational speed, and that it can be increased by maximising the flux difference and the number of coil turns, or by reducing the coil resistance.

What makes VREH different from most other electromagnetic harvesters is that neither the magnet nor the coil needs to move. The flux variation is caused entirely by the relative motion between the toothed wheel and the pickup units. This is an advantage, since it removes the need for a mechanical connection between the rotating part and the harvester. It also means that the device is easy to scale, so changing the shaft diameter mainly affects the size of the toothed wheel, while the pickup unit can stay the same [18]. In contrast, traditional electromagnetic generators require scaling of the magnets or coils, which becomes costly for large shafts or slow rotational speeds.

At low rotational speeds, the output voltage and frequency are both small, which makes the design of the interface circuit more critical [19].

Kroener et al. [9] were among the first to study a VREH specifically designed for energy harvesting in railway monitoring, showing that the principle could reliably power wireless sensor nodes from the motion of passing train wheels. Since then, the approach has been explored in a growing number of industrial and structural monitoring applications [17, 18].

1.6 State of the Art of VREH

As already mentioned, the first applications of the variable reluctance principle to energy harvesting appeared in the context of railway monitoring. Kroener et al. [9] demonstrated that a VREH placed beside a rail track could generate enough power from the passage of train wheels to supply a wireless sensor node, reporting a peak output power of around 5.9 mW at a simulated train speed of 81.5 km/h. This work showed for the first time that the variable reluctance principle could be practically used as an energy source rather than just a measurement transducer.

A systematic comparison of VREH topologies was later carried out by Xu et al. [17], who surveyed the structures proposed in the literature and evaluated them in terms of output power and power density for low-speed rotating applications. The study identified the *m-shaped* pickup unit, which uses two opposing permanent magnets and a centre pole piece, as the topology with the highest volumetric power density among all structures considered. This result motivated a detailed design study of the m-shaped VREH, in which the influence of three key geometric parameters was investigated: the coil height h_{Coil}, the tooth height h_{Tooth}, and the magnet height h_{M} [18]. The study showed that these parameters involve trade-offs under size constraints, and provided a numerical optimisation method based on finite element analysis (FEA). The optimised structure reached output powers between 0.12 mW and 22 mW at rotational speeds of 5 rpm to 60 rpm within a volume constraint of 43.5 cm ^{3}; these output power levels are sufficient to power a range of duty-cycled wireless sensor systems [18].

A practical challenge in VREH systems is that the transducer produces a low-voltage alternating output, which must be rectified and conditioned before it can supply any system. At low rotational speeds, the output voltage can be well below 1 V, which makes the choice of rectifier circuit critical. Xu et al. [19] compared three passive rectifier topologies: a full-wave bridge rectifier (FWR), a voltage doubler (VD), and a negative voltage converter (NVC), all on the same VREH system at speeds between 5 rpm and 20 rpm. The results showed that the VD achieved the best power extraction efficiency at low speeds, reaching a PEE of 40-60%, while the NVC outperformed it above 18 rpm. The FWR showed the lowest performance across all conditions, with PEE values of only 20-40%. These results confirm that the interface circuit is not just a passive element but an active one that affects how much power the transducer can deliver to the load.

These studies show that VREH is a mature and well-understood technology for low-speed rotational applications, with demonstrated power levels sufficient for a variety of wireless sensing applications. The most recent step forward in this direction is the multi-phase VREH proposed by Wu et al. [16], which integrates six m-shaped pickup units into a bearing hub unit for large commercial vehicles, reaching output powers between 0.46 W and 4.77 W at rotational speeds of 100 to 400 rpm. This significant increase in output power opens new possibilities for powering more demanding condition monitoring systems. However, it also introduces new challenges in the design of the power conditioning circuit, which must handle six independent AC outputs with different phases and varying voltage levels depending on the rotational speed. These challenges are discussed in the next section.

1.7 Open Challenges

Despite the progress described in the previous sections, several challenges remain open in the design of VREH systems. This section focuses on those that are most relevant to the work presented in this thesis.

Power conditioning for multi-phase outputs

Single-phase VREH systems produce one alternating voltage signal that must be rectified and converted into a stable DC supply. This problem has been studied in detail for low-speed single-phase harvesters, where the choice of rectifier topology has a significant impact on the power extraction efficiency [19].

Many studies have investigated the design of power conditioning circuits for single-phase energy harvesters. The most common approach combines a rectification stage with a DC-DC converter, such as a Buck or Boost circuit, to regulate the output voltage to a level suitable for the target load [20, 21, 8, 3]. However, these solutions are designed for a single AC input and cannot be directly applied to a multi-phase harvester such as the MP-VREH, where six independent AC signals must be processed simultaneously.

Furthermore, many of these solutions do not include a startup circuit or a control system, which are both essential for the reliable operation of the power conditioning circuit in practice. Without a startup circuit, the system is not able to start operating when the harvested voltage is initially too low to power the control electronics. Without a control system, the circuit cannot adapt to changes in the rotational speed or load conditions, which are essential in real-world applications. This gap in the literature motivates the investigation presented in this thesis.

When the number of pickup units increases, as in the multi-phase VREH proposed by Wu et al. [16], which uses six m-shaped pickup units arranged around a common toothed wheel, the power conditioning problem becomes considerably more complex. Each pickup unit produces an independent AC output with a different phase, and the six signals are evenly spaced in time by a $60°$ phase shift. This means that standard single-phase rectifier solutions cannot be directly applied, and that the interface circuit must be designed to handle multiple AC inputs efficiently.

Low output voltage at low speed

A fundamental limitation of VREH transducers is that the output voltage increases with rotational speed. At low speeds, the induced voltage may be well below 1 V, which makes rectification difficult because conventional diode bridges lose a significant fraction of the available voltage across their forward drops [19]. In a multi-phase system, this problem affects all six phases simultaneously. Finding a rectifier topology that works efficiently across a wide speed range, from the minimum speed at which the harvester can sustain the load, up to the maximum operating speed, is therefore a key design requirement.

Impedance matching under variable conditions

The maximum power that a VREH can deliver to a load is achieved when the load impedance is matched to the internal impedance of the pickup coil. This matching condition depends on the rotational speed, because the electrical frequency of the output signal changes with speed. For a single-phase harvester, Xu et al. [18] showed that a fixed resistive load leads to an approximately linear increase in output power with speed, whereas a complex conjugate matched load, which compensates for the inductive component of the coil, could further increase the output power, especially at higher speeds. In a six-phase system, achieving dynamic impedance matching for all phases simultaneously adds a further layer of complexity to the interface circuit design.

1.8 Harvester Specifications and Research Objective

The work presented in this thesis is based directly on an updated version of the MP-VREH proposed by Wu et al. [16]. The electrical characteristics of a single pickup unit, obtained from the transducer model, are summarised in Table 1.1. These values are used throughout this work as a preliminary reference for the operating conditions used in the design and evaluation of the proposed power conditioning circuits.

rpm	f (Hz)	V_{Lpk} (V)	R_{coil} (Ω)	L_{coil} (mH)	P_o (W)
100	36.7	0.91	6	21.68	0.41
200	73.3	1.35	6	21.46	0.91
300	110.0	1.59	6	21.27	1.26
400	146.7	1.74	6	21.12	1.51

Table 1.1: Electrical parameters of a single MP-VREH pickup unit at different rotational speeds.

All values in Table 1.1 were obtained under impedance-matched load conditions. V_{Lpk} refers to the peak voltage across the load, and P_o is the average power dissipated in the load.

The overall system consists of six pickup units arranged in space so that they are electrically phase-shifted by 60° from one another.

These challenges motivate the work presented in this thesis, which focuses on the design and evaluation of power conditioning circuits for the six-phase MP-VREH described above. The goal is to identify the interface circuit topology that maximises the power delivered to the load across the relevant operating speed range of 100 to 400 rpm (20-80 km/h), corresponding to typical driving conditions of large commercial vehicles, for example buses.

2 Theoretical Background

This chapter introduces the physical and circuit-level concepts required to motivate the design choices made in the following chapters. In this chapter, starting from Faraday's law, an electrical model of the variable-reluctance energy harvester (VREH) is derived. The model is then used to discuss the maximum power transfer theorem and the role of impedance matching.

2.1 Electrical Model of the Variable-Reluctance Harvester

2.1.a Faraday's law

Faraday's law, which relates the electromotive force $\varepsilon(t)$ (EMF) induced in a closed loop wire to the time rate of change of the magnetic flux $\Phi(t)$ linked to the loop, is the starting point for any electromagnetic transducer.

$$\varepsilon(t) = -\frac{\mathrm{d}\Phi(t)}{\mathrm{d}t} \tag{2.1}$$

For a coil made of N_c turns sharing the same flux, the total linked flux is $\Lambda(t) = N_c\Phi(t)$ and the induced EMF becomes:

$$\varepsilon(t) = -\frac{\mathrm{d}\Lambda(t)}{\mathrm{d}t} = -N_c\frac{\mathrm{d}\Phi(t)}{\mathrm{d}t}. \tag{2.2}$$

In case of a coil consisting of N_c turns mechanically spaced one respect to each other leading to different linked flux, the induce voltage can be described by

$$\varepsilon(t) = -\frac{d}{dt} \cdot \sum_{i=1}^{N_c} \Phi_i(t) \tag{2.3}$$

where Φ_i is the flux linkage associated with a single turn of the coil [17].

2.1.b Variable Reluctance Principle

Unlike conventional electromagnetic transducers, in which the EMF is induced by relative motion between a permanent magnet (PM) and a pickup coil, a variable-reluctance energy harvester (VREH) keeps both the magnet and the coil stationary. The magnetic flux linked to the coil is instead modulated by the passage of a rotating, toothed ferromagnetic wheel in front of a stationary pickup unit: as a tooth or a slot alternately faces the pickup unit, the reluctance of the magnetic circuit changes periodically, and it is this reluctance variation (rather than any motion of the magnet) that drives the flux change $\Phi_i(t)$ in (2.3) [17]. Because the rotating part carries no magnet, no coil and no electrical connection, the resulting transducer is mechanically simple, robust, and easily scalable to shafts of different diameters [18].

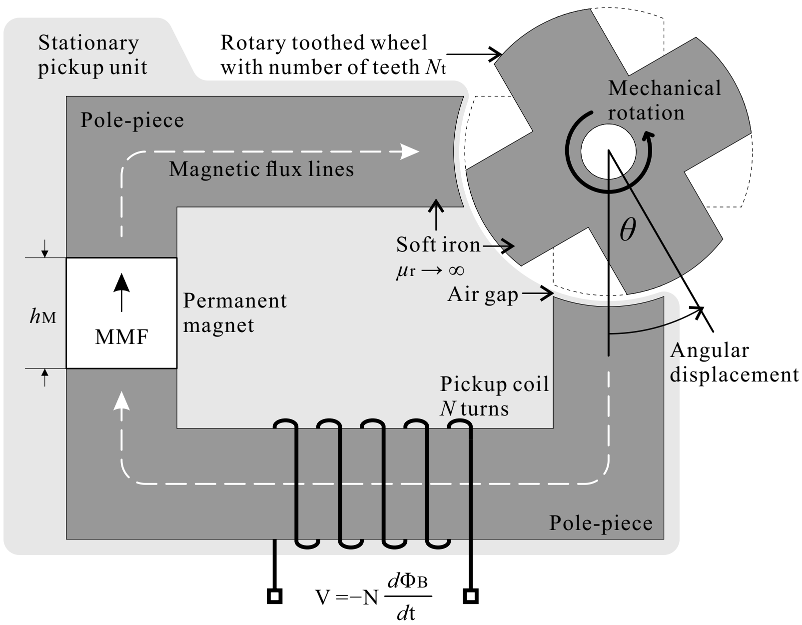


Figure 2.1: Structure of a variable-reluctance sensor: a stationary pickup unit, composed of a permanent magnet, a soft-ferromagnetic pole-piece and a pickup coil, faces a rotating toothed wheel across an air gap.

The pole-piece and the toothed wheel are both made of soft ferromagnetic material, while the PM is a hard ferromagnetic material providing a magneto-motive force (MMF) $\mathcal{F}_M$ that can be taken as constant. As the wheel rotates, the pickup unit cycles between two extreme relative positions: the *aligned* position, where a pole of the pickup unit faces a tooth of the wheel and the air-gap reluctance is minimum, and the *unaligned* position, where the pole faces a slot and the reluctance is maximum [17, 18]. The corresponding flux swings between a maximum Φ_{max} and a minimum Φ_{min}, so that the relevant design quantity is the flux difference $\Delta\Phi = \Phi_{max} - \Phi_{min}$ between the two extremes. Fig. 2.2 illustrates this swing directly: the air-gap flux density rises sharply to a plateau at the aligned position and nearly collapses at the unaligned position, confirming that the flux change is concentrated within a narrow angular window around each tooth-slot transition.

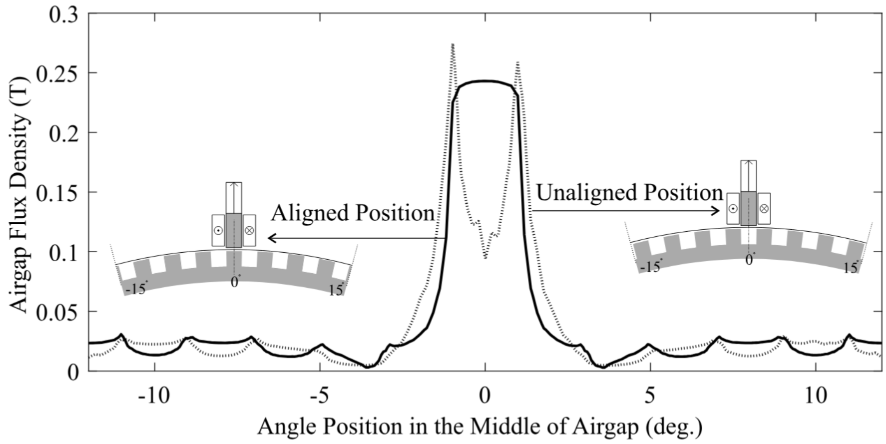


Figure 2.2: Airgap flux density distribution for aligned and unaligned positions. Reproduced from [17].

From flux swing to output power

For a wheel with N_t teeth rotating at angular velocity ω, the electrical frequency of the induced voltage is

$$\omega_c = \omega N_t. \tag{2.4}$$

At low rotational speeds the coil impedance is dominated by its resistance R_{Coil}, so that maximum power is transferred to a purely resistive load matched to it, $R_L = R_{Coil}$; under this condition the load voltage $V_L(t)$ is related to the open-circuit coil voltage $V_c(t)$ of (2.3) by

$$V_L(t) = \frac{1}{2} V_c(t). \tag{2.5}$$

Assuming the time-varying flux to be an ideal sinusoidal wave, uniformly distributed in the N_c turns of the coil, the peak-to-peak voltage developed across the matched load follows from (2.3) and (2.5) as [17]

$$V_{L,pp} = -\frac{1}{2}\, N_c\, \frac{\Phi_{max} - \Phi_{min}}{\frac{1}{2}T_c} = -N_c\, f_c\, \Delta\Phi, \tag{2.6}$$

where $\frac{1}{2}T_c$ is the time taken by the wheel to rotate from the aligned to the unaligned position, i.e. from a tooth to a slot centred on the pole-piece.

Because $V_{L,pp}$ in (2.6) is the peak-to-peak amplitude of an (assumed) sinusoidal load voltage, its root-mean-square value is $V_{L,rms} = V_{L,pp}/(2\sqrt{2})$, so that the RMS power dissipated in the matched load is [17]

$$P_L = \frac{V_{L,rms}^2}{R_L} = \frac{V_{L,pp}^2}{8\, R_{Coil}} = \frac{(\omega\, N_t\, N_c)^2}{8\, R_{Coil}}\, (\Delta\Phi)^2, \tag{2.7}$$

where the last equality follows from substituting (2.4) into (2.6). Equation (2.7) shows that the harvested power scales quadratically with the rotational speed ω, the flux swing $\Delta\Phi$, the number of coil turns N_c and the number of teeth N_t, while it is inversely proportional to the coil resistance R_{Coil}.

For a given application (i.e. fixed ω and N_t), the design of the pickup unit therefore splits into two largely independent optimisation problems [18]:

$$P_L \propto \frac{N_c^2}{R_{Coil}}, \tag{2.8a}$$

$$P_L \propto (\Delta\Phi)^2, \tag{2.8b}$$

Sub-function (2.8a) is a winding-design problem: for a coil of resistivity ρ_{wire} occupying a cross-section of height h_{Coil} and width W_{Coil} inside the pole-piece, it can be shown that (2.8a) reduces to [18]

$$P_L \propto \frac{h_{Coil}\, W_{Coil}\, k_w}{\rho_{wire}}, \tag{2.9}$$

with $k_w < 1$ the winding fill factor, defined as the fraction of the winding-window cross-section $h_{Coil} W_{Coil}$ actually occupied by copper [18]. For a fixed available cross-section, output power is thus maximized simply by maximizing the copper fill of the winding window. Sub-function (2.8b), in contrast, is a magnetic-circuit design problem. Once the magnet is chosen, its coercivity H_c and height h_M fix the MMF $\mathcal{F}_M = H_c\, h_M$, so $\Delta\Phi$ can no longer be increased by acting on the magnet alone. What remains under the designer's control is the shape of the pole-piece and toothed wheel around the air gap, thus the reluctance should be made as small as possible at the aligned position and as large as possible at the unaligned position, so that the two extremes are as far apart as possible and $\Delta\Phi$ is maximized. This magnetic-circuit optimization is detailed next.

Magnetic-circuit model of the reluctance swing

To turn $\Delta\Phi$ into a design formula, the magnetic circuit can be treated like an electrical one, with reluctance playing the role of resistance. The reluctance of a simple flux-path element of length l, cross-section A and relative permeability μ_r is

$$\mathcal{R} \;=\; \frac{l}{\mu_0\,\mu_r\,A}. \tag{2.10}$$

The pole-piece and the toothed wheel are made of soft ferromagnetic material, whose relative permeability is several orders of magnitude larger than that of air ($\mu_r \approx 1$) or of the PM ($\mu_r \approx 1.05$). Since a larger μ_r means a much smaller reluctance, the pole-piece and the wheel contribute essentially nothing to the total loop reluctance [18]. What is left, at either rotor position, is just two terms in series: the air-gap reluctance and the magnet's own reluctance,

$$\mathcal{R}_M \;=\; \frac{h_M}{\mu_0\,\mu_M\,A_M}. \tag{2.11}$$

Denoting by $\mathcal{R}_{g,a}$ and $\mathcal{R}_{g,u}$ the air-gap reluctance at the aligned and unaligned positions, Hopkinson's law gives the flux at each extreme simply as the MMF divided by the loop reluctance, $\Phi_{max} = \mathcal{F}_M/(\mathcal{R}_{g,a}+\mathcal{R}_M)$ and $\Phi_{min} = \mathcal{F}_M/(\mathcal{R}_{g,u}+\mathcal{R}_M)$. Subtracting the two gives the flux swing already introduced in (2.8b), now written out explicitly:

$$\Delta\Phi \;=\; \mathcal{F}_M \left(\frac{1}{\mathcal{R}_{g,a}+\mathcal{R}_M} - \frac{1}{\mathcal{R}_{g,u}+\mathcal{R}_M} \right). \tag{2.12}$$

Once the magnet material and the nominal air-gap length are fixed by mechanical constraints, (2.12) shows that only two geometric parameters are left to control $\Delta\Phi$:

- the **tooth height** h_{Tooth}, which affects only $\mathcal{R}_{g,u}$: a taller tooth raises the reluctance seen at the unaligned position, but this benefit saturates once the tooth is taller than about 3 mm, while the aligned-position reluctance $\mathcal{R}_{g,a}$ stays essentially unchanged [18];
- the **magnet height** h_M, which pulls in two opposite directions at once: it raises the MMF $\mathcal{F}_M = H_c h_M$ (good for $\Delta\Phi$), but it also raises the magnet's own reluctance $\mathcal{R}_M$ (bad for $\Delta\Phi$). Because of this trade-off, a non-trivial optimal value of h_M exists.

In a size-constrained design, what matters is not just P_L but the power delivered per unit volume, i.e. the volumetric optimisation of the pickup unit. Modelling the harvester as a hollow cylinder of overall diameter d_o, shaft diameter d_{Shaft} and depth l_o, the volumetric power density is

$$p_v = \frac{P_L}{V_{VREH}}, \qquad V_{VREH} = \frac{\pi}{4}\, l_o\, (d_o - d_{Shaft})^2. \tag{2.13}$$

Once the air-gap length and the (shared) flux-path width are fixed, the available radial space must be split between the coil and the tooth, $C = h_{Coil} + h_{Tooth}$. Growing h_{Coil} always helps P_L, through (2.9), whereas growing h_{Tooth} only helps up to the 3 mm knee point described above, through (2.12). Because of this asymmetry, for any given C there is an optimal split between h_{Coil} and h_{Tooth} that maximises p_v [18].

The same reasoning applies to the magnet height by setting $\frac{\mathrm{d}\Delta\Phi}{\mathrm{d}h_M} = 0$ in (2.12) yields a closed-form expression for the optimal magnet height,

$$h_M^* = \mu_M \, \frac{k_2}{k_1} \, \frac{W_{g,a}}{W_{g,u}} \, \sqrt{l_{g,a}\, l_{g,u}}, \tag{2.14}$$

where $l_{g,a}$, $l_{g,u}$, $W_{g,a}$, $W_{g,u}$ are the effective air-gap lengths and widths at the aligned and unaligned positions, and $k_1, k_2 > 1$ are correction factors that account, respectively, for flux leakage and for the MMF drop across the finite-permeability soft iron [18].

2.1.c Electrical Thévenin Model of the Pickup Unit

The derivation of Faraday's law in Section 2.1.a gives the ideal electromotive force induced in the coil, but it does not yet describe how that voltage appears at the physical terminals of the pickup unit. A real coil is made of a finite length of copper wire, so it has a nonzero winding resistance, and it carries a current that produces its own magnetic flux, so it also has a nonzero self inductance. Both effects are in series with the ideal source, since the same wire that generates the electromotive force also carries the resistance and the flux linkage that define R_{Coil} and L_{Coil}.

Circuit equation

Let $e(t)$ denote the ideal electromotive force predicted by (2.3), and let $i(t)$ be the current flowing out of the coil into the external circuit. Applying Kirchhoff's voltage law around the loop formed by the source, the winding resistance, the winding inductance and the external circuit gives

$$v(t) \;=\; R_{Coil}\, i(t) \;+\; L_{Coil}\, \frac{\mathrm{d}i(t)}{\mathrm{d}t} \;+\; e(t), \tag{2.15}$$

where $v(t)$ is the voltage measured at the coil terminals, available to the power conditioning circuit. In other words, (2.15) models the pickup unit as an ideal voltage source $e(t)$ in series with a resistance R_{Coil} and an inductance L_{Coil}, feeding the external circuit at the coil terminals; this series generator model is used throughout the rest of this thesis.

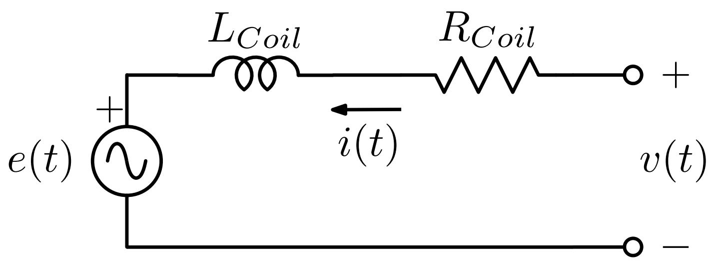


Figure 2.3: Series generator model of the pickup unit: an ideal source in series with the winding resistance R_{Coil} and inductance L_{Coil}.

Winding resistance

The winding resistance follows directly from the resistivity of the wire, its total length and its cross sectional area,

$$R_{Coil} \;=\; \rho_{wire}\, \frac{N_c\, l_t}{A_w}, \tag{2.16}$$

where N_c is the number of turns, l_t is the mean length of a single turn, and A_w is the cross sectional area of the wire. If the coil occupies a winding window of height h_{Coil} and width W_{Coil} with a copper fill factor k_w, the wire cross section can be written as $A_w = k_w\, h_{Coil}\, W_{Coil}/N_c$, so that (2.16) becomes

$$R_{Coil} \;=\; \rho_{wire}\, \frac{N_c^2\, l_t}{k_w\, h_{Coil}\, W_{Coil}}. \tag{2.17}$$

Treating l_t as approximately constant for a fixed pole piece geometry, (2.17) reproduces the proportionality $N_c^2/R_{Coil} \propto k_w\, h_{Coil}\, W_{Coil}/\rho_{wire}$ already used in (2.9) to optimise the winding cross section [18].

Self inductance and its dependence on rotor position

The self inductance of a coil linked to a magnetic circuit of total reluctance $\mathcal{R}$ is, by the same reluctance used for Hopkinson's law in (2.10),

$$L_{Coil} = \frac{N_c^2}{\mathcal{R}}. \tag{2.18}$$

The pickup unit is a variable-reluctance device, so the reluctance seen by the coil's own flux changes with the rotor angle too, in the same way as the reluctance that shapes the induced EMF. This means the self inductance is not constant: it swings between two extremes,

$$L_{max} = \frac{N_c^2}{\mathcal{R}_{g,a} + \mathcal{R}_M}, \qquad L_{min} = \frac{N_c^2}{\mathcal{R}_{g,u} + \mathcal{R}_M}, \tag{2.19}$$

reached at the aligned and unaligned positions, just like the flux extremes of (2.12). For circuit-level analysis, such as the impedance matching study of Section 2.1.d, it is more practical to replace this varying inductance with a single, constant value L_{Coil}. This value can be obtained either by averaging over one electrical period, or by measuring it directly with an impedance analyser at the frequency of interest.

Phasor representation

For a sinusoidal steady state at the electrical frequency f_c of (2.4), the time domain relation (2.15) can be written in phasor form as

$$V = I\,(R_{Coil} + j\,\omega_e\,L_{Coil}) + E, \tag{2.20}$$

where E, I and V are the phasors associated with $e(t)$, $i(t)$ and $v(t)$, and $\omega_e = 2\pi f_c$. Equation (2.20) shows that, seen from its electrical terminals, the pickup unit behaves as an ideal source E in series with the impedance

$$Z_s = R_{Coil} + j\,\omega_e\,L_{Coil}, \tag{2.21}$$

which is precisely the Thévenin source impedance used in (2.22) for the impedance matching analysis of Section 2.1.d. The series generator, resistance, and inductance model derived in this section therefore provides the electrical foundation on which the maximum power transfer condition of the next section, and the power conditioning topologies compared in Chapter 3, are built.

2.1.d Maximum Power Transfer and Impedance Matching

The previous section modelled the pickup unit as a Thévenin source, an open circuit EMF $e(t)$ in series with the coil's own impedance. At low electrical frequency this impedance is essentially the winding resistance R_{Coil}, and matching a resistive load to it, $R_L = R_{Coil}$, is enough to extract the maximum available power, as already used in (2.7). As the rotational speed increases, however, the coil inductance L_{Coil} can no longer be neglected, and the condition for maximum power transfer must be generalized. This section derives that general condition and explains why it becomes a central design constraint for a harvester that must operate between 100 and 400 rpm (the target range of this thesis).

2.1.e Thévenin model of the source and the load

Let the coil be represented, at the electrical frequency of interest, by its Thévenin equivalent, an RMS open circuit phasor voltage V_c in series with the source impedance

$$Z_s \;=\; R_{Coil} \;+\; j\,\omega_e\,L_{Coil}, \tag{2.22}$$

where $\omega_e = 2\pi f_c$ is the electrical angular frequency and f_c is the electrical frequency defined in (2.4). The load presented to the coil by the power conditioning circuit is likewise represented, at the same frequency, by a generic complex impedance

$$Z_L \;=\; R_L \;+\; j\,X_L. \tag{2.23}$$

Here R_L is the resistive part of the load and X_L its reactive part, which can be inductive ($X_L > 0$) or capacitive ($X_L < 0$) depending on how the load is realised.

2.1.f Power delivered to the load

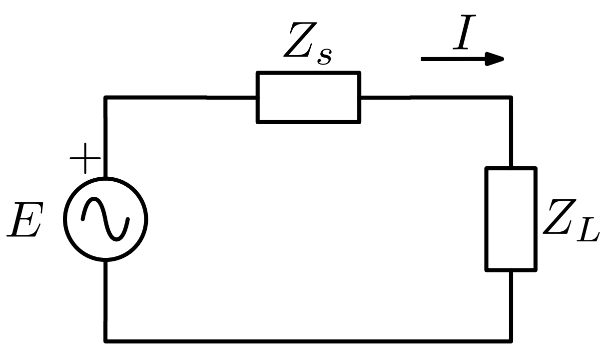


Figure 2.4: Thévenin equivalent circuit used for the impedance matching analysis: source E with source impedance Z_s, driving a generic load impedance Z_L.

With Z_s and Z_L in series, the RMS current flowing in the loop is

$$I = \frac{E}{Z_s + Z_L}, \tag{2.24}$$

and the average power dissipated in the resistive part of the load is

$$P_L = |I|^2 R_L = \frac{|E|^2 R_L}{(R_{Coil} + R_L)^2 + (\omega_e L_{Coil} + X_L)^2}. \tag{2.25}$$

Equation (2.25) is the generalization of (2.7) to a complex load. Setting $L_{Coil} = 0$ and $X_L = 0$ recovers the purely resistive case already used earlier in this chapter.

2.1.g The complex conjugate matching condition

Equation (2.25) can be maximised in two independent steps, since R_L and X_L appear in separate terms of the denominator.

First, for any fixed R_L, the denominator is minimised, and P_L is therefore maximised, by cancelling the reactive term. This requires

$$X_L = -\omega_e L_{Coil}. \tag{2.26}$$

With this choice, (2.25) reduces to

$$P_L = \frac{|E|^2 R_L}{(R_{Coil} + R_L)^2}. \tag{2.27}$$

Second, differentiating (2.27) with respect to R_L and setting the result to zero,

$$\frac{\partial P_L}{\partial R_L} = |E|^2 \frac{R_{Coil} - R_L}{(R_{Coil} + R_L)^3} = 0 \implies R_L = R_{Coil}. \tag{2.28}$$

Combining (2.26) and (2.28) gives the well known maximum power transfer theorem in its general, complex form,

$$Z_L = Z_s^* = R_{Coil} - j\,\omega_e L_{Coil}. \tag{2.29}$$

The load must present a resistance equal to the coil resistance and a reactance equal in magnitude but opposite in sign to the coil reactance. Physically, the load must absorb energy from the coil (resistive part) while returning, every half cycle, exactly the reactive energy that the coil inductance would otherwise store (capacitive part). Substituting (2.29) into (2.25) gives the maximum available power,

$$P_{L,\max} = \frac{|E|^2}{4\,R_{Coil}}, \tag{2.30}$$

which is independent of L_{Coil}. Once the reactance is properly cancelled, the maximum extractable power depends only on the coil resistance, exactly as in the low frequency case of (2.7).

2.1.h Realising the reactive compensation

Condition (2.26) shows that an inductive coil requires a load with an equal and opposite *capacitive* reactance. The simplest way to obtain this is a compensation capacitor C_m placed in series with the resistive load, whose reactance at ω_e is $X_{C_m} = -1/(\omega_e C_m)$. Imposing $X_{C_m} = -\omega_e L_{Coil}$ gives the required capacitance

$$C_m = \frac{1}{\omega_e^2\,L_{Coil}}. \tag{2.31}$$

Once the capacitor C_m is fixed according to (2.31), it cancels the coil's reactance and therefore restores the resistive matching condition of (2.28) and the maximum power of (2.30) only at the single electrical frequency ω_e for which it was chosen.

2.2 Multi-Phase System Model

2.2.a System Overview

The harvester considered in this thesis is the six-phase multi-phase variable-reluctance energy harvester (MP-VREH) proposed by Wu et al. [16] and already introduced in Section 1.8, whose electrical specifications are summarised in Table 1.1. It consists of six m-shaped pickup units, each electrically equivalent to the series generator derived in Section 2.1.c, arranged around a common toothed wheel and integrated into a single bearing hub. As the wheel rotates, each pickup unit produces an independent AC voltage at the electrical frequency $f_c = \omega N_t$ of (2.4). The six pickup units are evenly spaced around the wheel, so that their induced voltages are mutually phase-shifted by $60°$, one from the next.

From an electrical standpoint, the harvester can therefore be represented as six identical Thévenin sources, each with open-circuit voltage $e_k(t)$, series resistance R_{Coil} and series inductance L_{Coil}, whose only difference is a fixed phase offset. This six-source representation is the starting point for the power conditioning circuits developed in Chapter 3, which must process six independent AC inputs rather than a single one.

2.2.b Multi-Phase Electrical Model

Let $\omega_e = 2\pi f_c$ be the electrical angular frequency common to all six phases, and let $k = 0, 1, \ldots, 5$ index the pickup units in the order in which they are met by a tooth of the wheel. Under the ideal assumption of even angular spacing, the open-circuit EMF of phase k is a sinusoid of amplitude $E_{pk}(\omega_e)$, itself a function of ω_e as in (2.6), delayed with respect to phase 0 by $k \cdot 60°$,

$$e_k(t) \;=\; E_{pk}(\omega_e)\,\sin\!\left(\omega_e t - k\,\frac{\pi}{3}\right), \qquad k = 0, 1, \ldots, 5. \tag{2.32}$$

Each phase drives the power conditioning circuit through its own series impedance $Z_s = R_{Coil} + j\,\omega_e L_{Coil}$, exactly as derived for the single pickup unit in (2.22). Because R_{Coil}, L_{Coil} and $E_{pk}(\omega_e)$ are the same for all six units at a given speed, the six branches differ only in the phase term of (2.32), so the same circuit design can be reused for every phase, each one simply delayed in time by $60°$.

2.2.c Design Requirements for the Interface Circuit

The specifications summarised in Table 1.1 fix the operating range that the interface circuit must handle. The load voltages V_{Lpk} reported there are measured under the impedance-matched condition of (2.5), i.e. half of the open-circuit source amplitude E_{pk} of (2.32) used throughout this section; over the 100-400 rpm target speed range, this relation puts E_{pk} between about 1.8 V and 3.5 V. The rectifier must remain efficient at these low input voltages. Because the six phases are electrically identical but time-shifted, as shown in (2.32), the interface circuit must either replicate the same rectification stage six times, one per phase, or combine the six inputs in a way that preserves the impedance matching condition of Section 2.1.d for each of them individually. These two requirements, low-voltage operation and per-phase impedance matching across a variable speed range, motivate the comparison of rectifying topologies carried out in Chapter 3.

3 Proposed Rectifying Circuits

3.1 Overview

Section 1.8 showed that, over the target speed range, the peak voltage available at each phase of the MP-VREH stays low, which makes the choice of rectifier circuit critical. Implementing a design that wastes a large, fixed voltage drop at every conduction cycle can lose a substantial fraction of the already limited available power. Xu et al. [19] compared three passive rectifier topologies for this purpose: a full-wave bridge rectifier (FWR), a voltage doubler (VD), and a negative voltage converter (NVC), using the PCE and PEE metrics of (1.1) and (1.2), and found that no single topology is best across the whole speed range. This chapter focuses on the two topologies at the extremes of that comparison, the FWR, as the simplest and cheapest baseline, and the NVC, as the best performer at higher speed, and discusses how each one must be adapted to rectify the six phase-shifted inputs of (2.32) rather than a single-phase source.

Rather than rectifying each of the six phases independently, the six inputs can be grouped so as to reuse existing, well-documented multi-phase rectifier bridges. For the FWR, this thesis groups the six phases into two three-phase subsystems, each rectified by a three-phase bridge, a topology already well documented in the literature. For the NVC, which was identified above as the best performer at higher speed, the six phases are instead grouped into three two-phase subsystems, each pairing two phases that are $180°$ apart and connecting them in anti-series so that their voltages add instead of cancelling. These two configurations are described in detail in the dedicated sections of this chapter and in Chapter 4.

3.2 Full-Wave Rectifier (FWR)

Figure 3.1 shows the three-phase bridge used to rectify one of the two three-phase groups introduced in the overview of this chapter. Taking every other pickup unit of (2.32) (phases $k = 0, 2, 4$, or equivalently $k = 1, 3, 5$) gives exactly such a group: three sources $120°$ apart, the spacing a three-phase bridge rectifier is designed for. The three input terminals are labelled A, B and C in the figure. Diodes D_1, D_3 and D_5 share a common cathode at OUT+, with their anodes tied to A, B and C respectively; diodes D_2, D_4 and D_6 share a common anode at OUT-, with their cathodes tied to A, B and C respectively.

At any instant, only the top diode connected to whichever of the three phases is momentarily the highest is forward biased, conducting that phase's current to OUT+; likewise, only the bottom diode connected to whichever phase is momentarily the lowest is forward biased, returning current from OUT- to that phase. The remaining four diodes are reverse biased and block. The current path therefore always includes exactly two diodes in series, one from the top group and one from the bottom group, so the voltage across OUT+ and OUT- is always lower than the instantaneous difference between the highest and lowest phase voltages by two forward voltage drops. A filter capacitor at the output reduces the ripple left by this waveform.

3.2.a Star or Delta Connection of the Three-Phase Group

Figure 3.1 fixes how A, B and C are rectified, but not how the three independent pickup coils of the group are wired together to produce those three terminals in the first place. Two configurations are considered for a group of phases k, $k + 2$, $k + 4$.

In a *star* (Y) connection, one terminal of each of the three coils is joined at a common internal node, left floating and not brought out to the bridge; the other, free terminal of each coil becomes A, B or C. Denoting the floating node voltage $v_S(t)$, the terminal voltages are $v_A(t) = v_S(t) + e_k(t)$, $v_B(t) = v_S(t) + e_{k+2}(t)$ and $v_C(t) = v_S(t) + e_{k+4}(t)$; the line-to-line voltage seen by the bridge does not depend on $v_S(t)$, since it cancels in the difference,

$$v_A(t) - v_B(t) \;=\; e_k(t) - e_{k+2}(t) \;=\; \sqrt{3}\, E_{pk} \cos\!\left(\omega_e t - k\,\frac{\pi}{3} - \frac{\pi}{3}\right), \tag{3.1}$$

an amplitude $\sqrt{3}$ times larger than that of a single phase, and phase shifted from it by $30°$.

In a *delta* (Δ) connection, the three coils are instead connected end to end in a closed loop, so that each pair of terminals A-B, B-C, C-A coincides exactly with the two terminals of one coil. The line-to-line voltage is then simply the coil's own EMF,

$$v_A(t) - v_B(t) \;=\; e_k(t), \tag{3.2}$$

with no amplitude increase. The closed loop must carry the sum of the three EMFs, which is identically zero for a balanced set ($e_k(t) + e_{k+2}(t) + e_{k+4}(t) = 0$ for any t, since the three phases are $120°$ apart), so no circulating current flows around the loop from the sources alone.

The star connection therefore delivers a $\sqrt{3}$ times larger voltage to the bridge for the same pickup units, which helps mitigate the low-voltage operation issue discussed below, at the cost of an inaccessible internal node whose potential is fixed only by the balance of the three branch currents. The delta connection avoids this floating node, at the cost of losing the $\sqrt{3}$ gain. Both configurations are carried forward as candidates for the FWR stage in the simulations of Chapter 4.

Because the two-diode drop of the FWR is essentially constant regardless of the phase voltages, the lower those voltages are, the larger the fraction of the signal it removes. Given the phase peak voltage range of Section 1.8, the MP-VREH operates exactly in this low-voltage regime, where the fixed diode drop is expected to dominate the losses of the FWR stage; the star connection of (3.1) mitigates this by a factor of $\sqrt{3}$, while the delta connection of (3.2) does not. The FWR's main advantage remains simplicity: it needs no active devices, no gate drive and no threshold voltage to overcome, which makes it a useful baseline against which the NVC of Section 3.3 is compared.

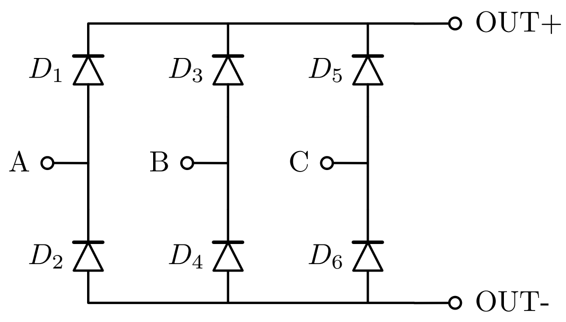


Figure 3.1: Three-phase full-wave bridge rectifier used for one of the two three-phase groups of pickup units, with input terminals A, B, C.

3.3 Negative Voltage Converter (NVC)

Figure 3.2 shows the two-phase NVC used to rectify one of the three two-phase groups introduced in the overview of this chapter. Rather than pairing adjacent phases, each group combines two pickup units that are electrically in antiphase: from (2.32), phases k and $k+3$ satisfy

$$e_{k+3}(t) \;=\; E_{pk}\,\sin\Big(\omega_e t - (k+3)\,\frac{\pi}{3}\Big) \;=\; E_{pk}\,\sin\Big(\omega_e t - k\,\frac{\pi}{3} - \pi\Big) \;=\; -e_k(t), \tag{3.3}$$

since they are 180° apart. Connecting the two coils in anti-series, i.e. joining their same-polarity terminals together and leaving the other two terminals free, gives a combined open-circuit voltage across the free terminals A and B of

$$v_A(t) - v_B(t) \;=\; e_k(t) - e_{k+3}(t) \;\overset{(3.3)}{=}\; 2\,e_k(t), \tag{3.4}$$

twice the amplitude of a single phase, at the cost of a doubled series impedance $2Z_s$. Connecting the same pair in straight series, by contrast, would simply cancel the two antiphase voltages to zero, since $e_k(t) + e_{k+3}(t) = 0$. The three NVC groups therefore pair phases $k = 0, 3$; $k = 1, 4$; and $k = 2, 5$, each combined in anti-series at the terminals A and B of Figure 3.2. The bridge replaces the diodes of the FWR with four MOSFETs S_1-S_4, each with a Schottky diode (D_1-D_4) connected in parallel. S_1 and S_2 connect terminal A to OUT+ and OUT- respectively, while S_3 and S_4 connect terminal B to OUT+ and OUT- respectively. The bridge is *gate cross-coupled*: the gates of S_1 and S_2 are tied together and driven directly by terminal B, while the gates of S_3 and S_4 are tied together and driven directly by terminal A. This removes the need for a separate gate-drive or polarity-detection circuit, since each pair of switches is simply driven by the other AC terminal.

When A is higher than B, the gate voltage B shared by S_1 and S_2 turns S_1 on and keeps S_2 off, while the gate voltage A shared by S_3 and S_4 turns S_4 on and keeps S_3 off. The load current therefore flows from A through S_1 to OUT+, and returns from OUT- through S_4 to B. When B is higher than A, the roles swap, and the current instead flows through S_3 and S_2. In either case, the forward voltage drop across the bridge is the sum of the drain-source voltages of the two conducting MOSFETs, rather than a fixed diode drop as in the FWR.

This switching mechanism only works once the gate-source voltage across the conducting pair is large enough to turn the MOSFETs fully on. Below a critical voltage difference $V_{\text{off}} \approx 1.2$ V between A and B, set by the largest threshold voltage among the four MOSFETs, the devices cannot switch properly, and the parallel Schottky diodes D_1-D_4 carry the current instead: in this regime the NVC behaves like an FWR, with the same two-diode drop loss. Above V_{off}, the MOSFETs conduct with an on-resistance much smaller than a diode's forward drop, and the output voltage tracks the input with very low losses.

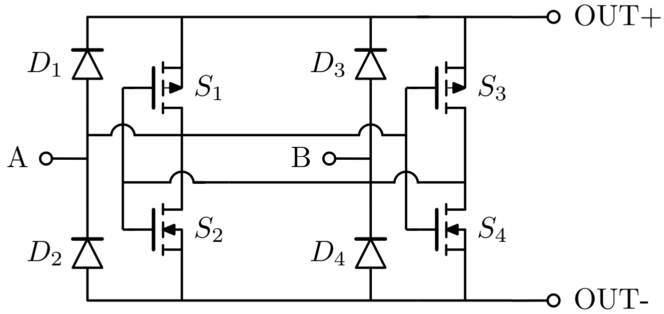


Figure 3.2: Negative voltage converter used for one of the three two-phase groups of pickup units, with input terminals A, B.

3.4 Application to the Six-Phase System

Sections 3.2 and 3.3 already present the FWR and the NVC in the multi-phase form adopted in this thesis. The FWR groups the six pickup units of Section 2.2 into two three-phase subsystems, taking phases $k = 0, 2, 4$ and $k = 1, 3, 5$ of (2.32) respectively. The NVC, instead, groups them into three two-phase subsystems, each pairing two phases $180°$ apart and connecting them in anti-series, as derived in (3.4) ($k = 0, 3$; $k = 1, 4$; $k = 2, 5$). Because all six phases share the same R_{Coil}, L_{Coil} and peak voltage and differ only by the $60°$ time shift, every group of a given topology sees identical operating conditions, only shifted in time, so the performance of one group is representative of all the others. This is the configuration carried forward into the simulations of Chapter 4.

4 Simulations

4.1 Overview

This chapter presents the Simulink simulations used to evaluate the FWR and NVC topologies of Chapter 3, following the star/delta and anti-series configurations introduced in Sections 3.2 and 3.3. Before going through each configuration individually, this section describes how the simulations are run and, in particular, how the phase sources and the load are parametrised so that the simulated circuit matches the real prototype as closely as possible.

The real quantities used for this purpose, the phase peak voltages and the current-sensing series resistance described below, come from preliminary measurements taken specifically to support these simulations, not from the full experimental characterisation of the prototype. The complete measurement setup and the actual validation measurements are presented later, in Chapter 6.

The phase model of (2.32) assumes an ideal sinusoidal EMF of amplitude E_{pk} for every phase, but the voltage actually measured on the prototype is not a perfect sinusoid. Driving the simulation directly with the peak voltage of the real waveform would therefore not reproduce the same power as the real system, since a non-sinusoidal waveform and an ideal sinusoid sharing the same peak value do not, in general, have the same RMS value. For this reason, the peak voltage of each of the six phases was measured directly on the prototype at seven rotational speeds spanning the target range ($100, 150, 200, 250, 300, 350, 400$ rpm), giving a 6×7 table of real peak voltages $V_{\text{pk,real}}$, one per phase and speed. The peak voltage of a given phase is not constant from one revolution of the harvester to the next, so $V_{\text{pk,real}}$ is taken as the average of the peak voltage measured over 100 revolutions at that speed. Since the acquired waveform is a sequence of discrete samples rather than a continuous signal, the RMS value of each measured waveform was computed from the discretised version of its standard definition,

$$V_{\mathrm{rms}} = \sqrt{\frac{1}{N}\sum_{n=1}^{N} v[n]^2}, \tag{4.1}$$

where $v[n]$ are the N samples acquired over multiple full periods of the waveform, corresponding to 100 revolutions. This made it possible to compute a crest factor CF ≈ 1.51 as the ratio between the peak and the RMS value of the measured waveforms, close to but slightly above the value $\sqrt{2} \approx 1.414$ that an ideal sinusoid would give, reflecting the fact that the real EMF is not perfectly sinusoidal. Since CF $> \sqrt{2}$, this implies that, at the same peak value, the real EMF delivers less power than an ideal sinusoidal waveform.

Figures 4.1 and 4.2 show, as a representative example, the open-circuit voltage measured on all six phases at 100 rpm. Figure 4.1 covers half a revolution of the harvester and shows the general shape of the measured waveform, visibly different from ideal sinusoids. Figure 4.2 focuses on a short time window to highlight the $60°$ phase shift between consecutive phases. The peak-to-peak voltage is not identical across the six phases: this is not a measurement artefact, but a consequence of manufacturing tolerances between the six nominally identical pickup units of the prototype, which is also why each phase is treated individually in Table 4.1 rather than assumed to be identical.

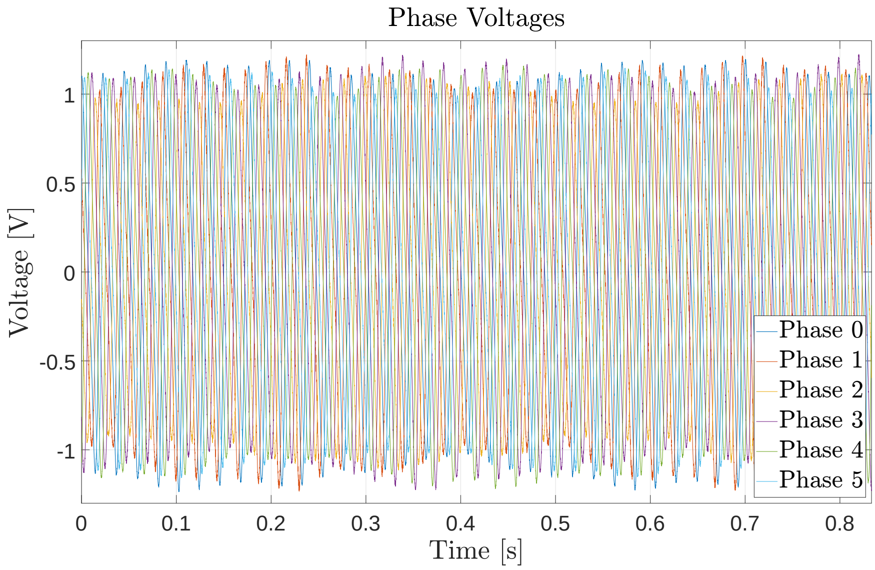


Figure 4.1: Open-circuit voltage measured on all six phases over half a revolution of the harvester at 100 rpm.

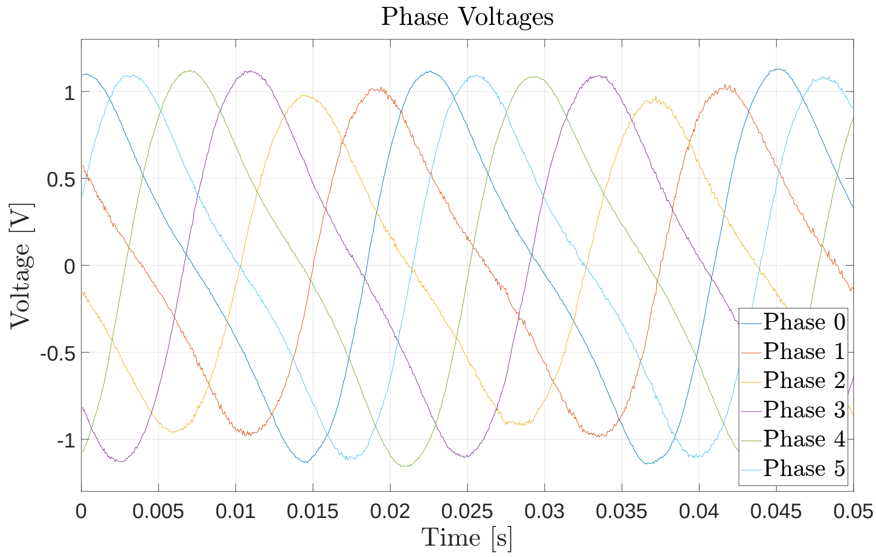


Figure 4.2: Detail of the same measurement, showing non ideal sinusoidal waveforms and the $60°$ phase shift between the six phase voltages.

Each phase is still represented in the Simulink model as an ideal sinusoidal source, consistently with (2.32), but rather than using $V_{\text{pk,real}}$ directly as its amplitude, every tabulated value is rescaled by $\sqrt{2}/\text{CF}$,

$$V_s = V_{\text{pk,real}} \cdot \frac{\sqrt{2}}{\text{CF}} = \sqrt{2}\, V_{\text{rms,real}}, \tag{4.2}$$

for every phase and every simulated speed, since CF is, by definition, the ratio of the real peak to the real RMS value. V_s is therefore the peak of the ideal sinusoid whose RMS value, and therefore whose power, equal that of the real measured phase voltage.

Table 4.1 lists the resulting value of V_s used for each of the six coils at each of the seven simulated speeds.

Figure 4.3 illustrates this correction for phase 0 (coil 0) across the whole target speed range, comparing three quantities: the peak voltage predicted by the finite-element (COMSOL) model behind the reference parameters of Table 1.1, the raw measured peak voltage $V_{\text{pk,real}}$ used directly as if the phase were an ideal sinusoid, and the crest-factor-corrected equivalent sinusoidal voltage V_s of (4.2). At the low-speed end of the range, both the raw and the crest-factor-corrected voltage remain well below the FEM prediction, and the correction changes a little. Moving towards the high-speed end, both curves rise past the FEM prediction, but the raw measurement crosses it earlier, around 340 rpm, and overshoots it the most by 400 rpm; the crest-factor correction delays this crossing to near

	Rotational speed (rpm)						
Coil	100	150	200	250	300	350	400
0	1.055	1.480	1.913	2.326	2.744	3.179	3.601
1	1.041	1.458	1.862	2.255	2.652	3.036	3.344
2	0.964	1.352	1.722	2.092	2.464	2.847	3.239
3	1.052	1.463	1.872	2.288	2.712	3.128	3.542
4	1.020	1.418	1.815	2.230	2.627	3.030	3.349
5	0.997	1.393	1.803	2.197	2.592	3.006	3.408

Table 4.1: Equivalent sinusoidal peak voltage V_s (V) used for each of the six coils in the Simulink simulations, computed from (4.2).

the end of the range and leaves a much smaller gap at 400 rpm, bringing the corrected voltage into closer agreement with the FEM prediction. As already noted, the real voltage is lower than the FEM prediction for most of the speed range, so the power it corresponds to is lower too.

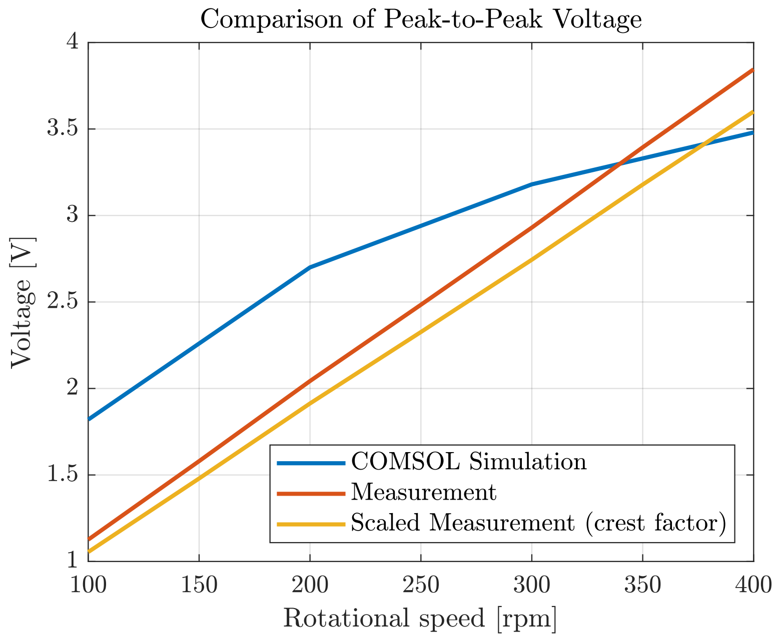


Figure 4.3: Peak voltage of phase 0 across the target speed range: the COMSOL finite-element prediction, the raw measured value used directly as an idealised sinusoid, and the same value corrected by the crest factor.

The simulated circuit also accounts for the current-sensing system used in the experimental setup of Chapter 6. The shunt resistor, the INA114AU sensing board and the connecting cables together introduce a series resistance that is not part of the idealised topologies; measured directly, this resistance is approximately $1.5\ \Omega$. Since this value is not negligible compared to the coil resistance $R_{\text{Coil}} = 6\ \Omega$ of Table 1.1, it is included in the Simulink model as an additional series resistance in the same branch as each coil and in the branch feeding the load, so that the simulated circuit is loaded in the same way as the real prototype during the measurements.

4.2 Full-Wave Rectifier (FWR)

4.2.a Star Connection

Figure 4.4 shows the Simulink circuit used to simulate one of the two three-phase groups in the star connection of Section 3.2. Each phase is modelled as an ideal sinusoidal source in series with the coil impedance and the current-sensing resistance introduced in Section 4.1, followed by the compensation capacitor C_m of (2.31). The three phases then reach the input terminals A, B, C of the three-phase bridge. The total input power is obtained with the two-wattmeter method, using voltage and current sensors placed on two of the three lines. At the output, the bridge is connected to a filter capacitor C_r and the load resistor R_{LY} in parallel, with a voltmeter measuring the resulting load voltage. The same circuit is used for both the first bridge (FWR1, phases $k = 0, 2, 4$) and the second bridge (FWR2, phases $k = 1, 3, 5$), each fed with the corresponding row of Table 4.1.

Compensation Capacitor (C_m) Sweep

The compensation capacitance C_m of (2.31) cancels the coil's reactance only at the single electrical frequency ω_e for which it is computed, so the value that maximises the output power at 100 rpm is not the value that maximises it at 400 rpm. To find these two values, C_m was swept independently at each end of the target speed range, with the filter capacitor and the load fixed at $C_r = 1480\ \mu$F and $R_L = 14.00\ \Omega$.

At 100 rpm, Figure 4.5 shows the output power rising steeply for small C_m and then flattening into a broad, almost flat maximum that extends from about $400\ \mu$F to at least $550\ \mu$F; within this plateau, the detail view of Figure 4.5 shows the power peaking around $C_m = 480$-$500\ \mu$F, so $C_m = 490\ \mu$F was chosen as the value giving the maximum output power. At 400 rpm, by contrast, Figure 4.6 shows a sharp, narrow peak at $C_m = 55\ \mu$F, with the output power falling off quickly on either side. In both cases, the second bridge (FWR2) delivers slightly more power than the first (FWR1) across the whole sweep, consistent with the

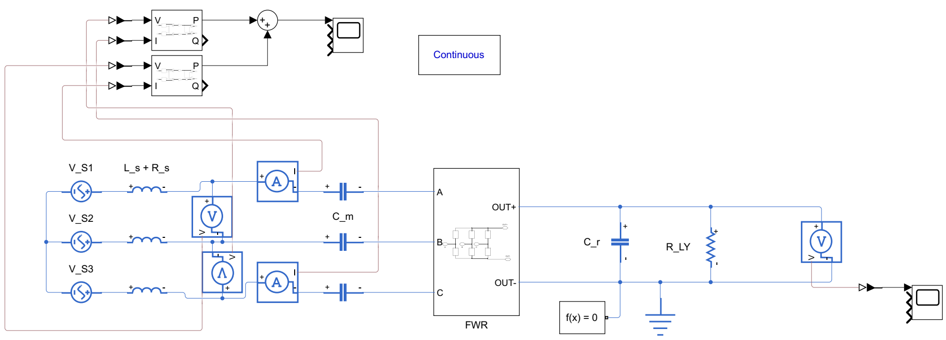
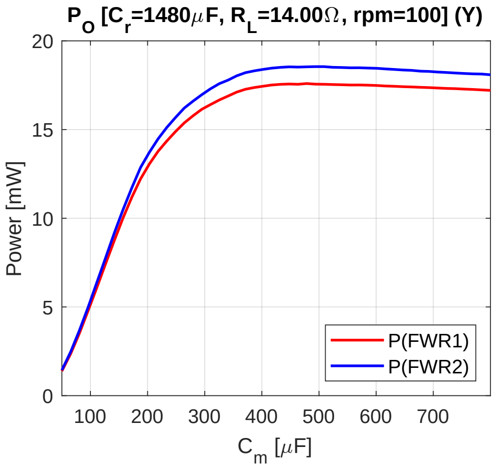


Figure 4.4: Simulink circuit used to simulate one three-phase group in the star connection, from the phase sources through the compensation capacitors C_m, the three-phase bridge, to the output filter C_r and load R_{LY}.

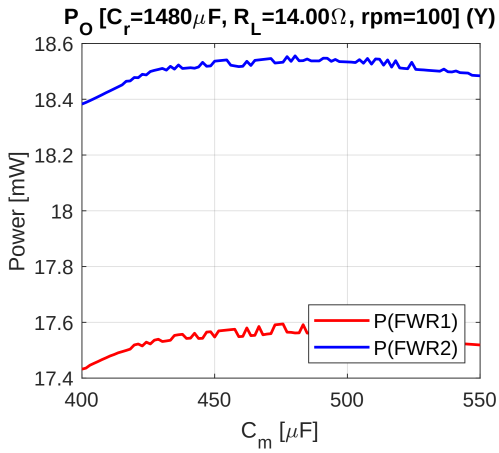
slightly higher average phase voltage of coils 1, 3, 5 in Table 4.1. The combined power of the two bridges, shown below, peaks at the same C_m chosen from the individual curves, confirming that these two values, $C_m = 490$ μF for 100 rpm and $C_m = 55$ μF for 400 rpm, also maximise the total power and are the two candidates carried forward to the load resistance and frequency sweeps below.

Load Resistance (R_L) Sweep

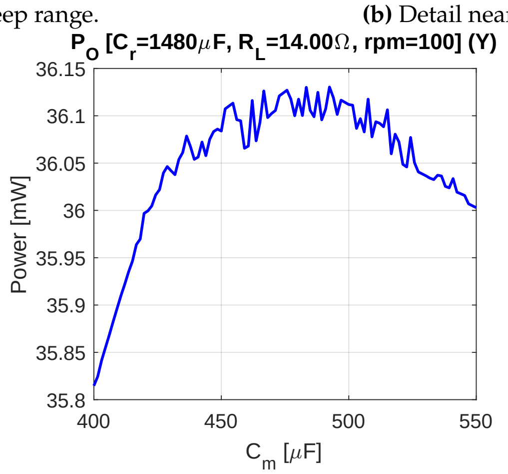
With $C_m = 490$ μF fixed at the value found optimal for 100 rpm in the previous sweep, and $C_r = 1480$ μF, the load resistance R_L was swept to find the value that maximises the output power at this speed. Figure 4.7 shows the resulting curve: the output power increases from $R_L = 10$ Ω, stays close to its highest value between about 13 and 15 Ω, and then slowly decreases towards $R_L = 18$ Ω. As in the compensation capacitor sweep, the second bridge (FWR2) delivers slightly more power than the first (FWR1) across the whole range, again consistent with the higher average phase voltage of coils 1, 3, 5 in Table 4.1. The value $R_L = 14.25$ Ω was chosen inside this broad maximum, and is the value used in



(a) Full sweep range.



(b) Detail near the maximum.



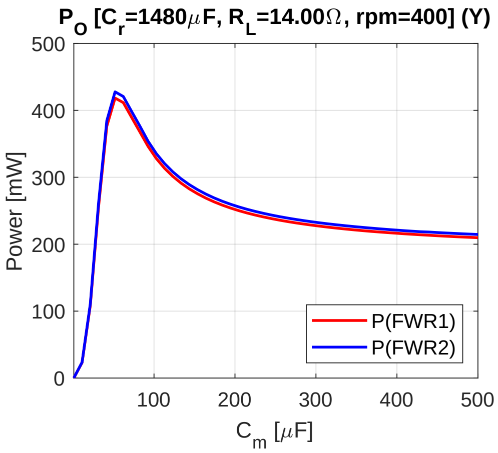
(c) Detail, combined power.

Figure 4.5: Output power of the star-connected first and second bridges as a function of the compensation capacitor C_m at 100 rpm ($C_r = 1480\ \mu\text{F}$, $R_L = 14.00\ \Omega$), individually (a, b) and combined (c).

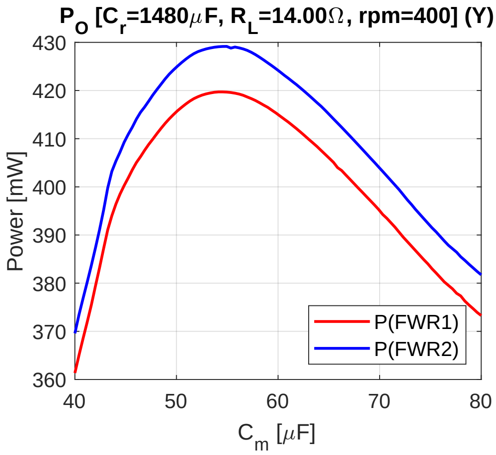
the frequency sweep and efficiency evaluation that follow.

Frequency Sweep

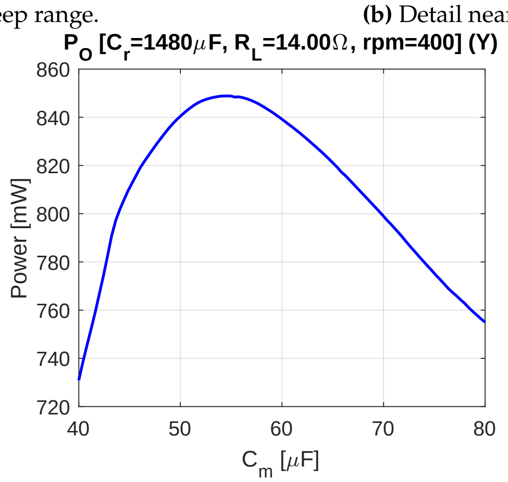
Figure 4.8 sweeps six fixed values of C_m, from 55 μF (optimal at 400 rpm) to 490 μF (optimal at 100 rpm), across the whole target speed range, with $C_r = 1480\ \mu\text{F}$ and $R_L = 14.25\ \Omega$ fixed, and plots the resulting combined output power of the first and second bridges. At the low-speed end of the range, the larger capacitors deliver noticeably more power than the smaller ones, while $C_m = 55\ \mu\text{F}$ performs the worst. As speed increases, the curves converge and cross around 300 rpm; beyond this point $C_m = 55\ \mu\text{F}$ quickly overtakes every other choice, reaching



(a) Full sweep range.



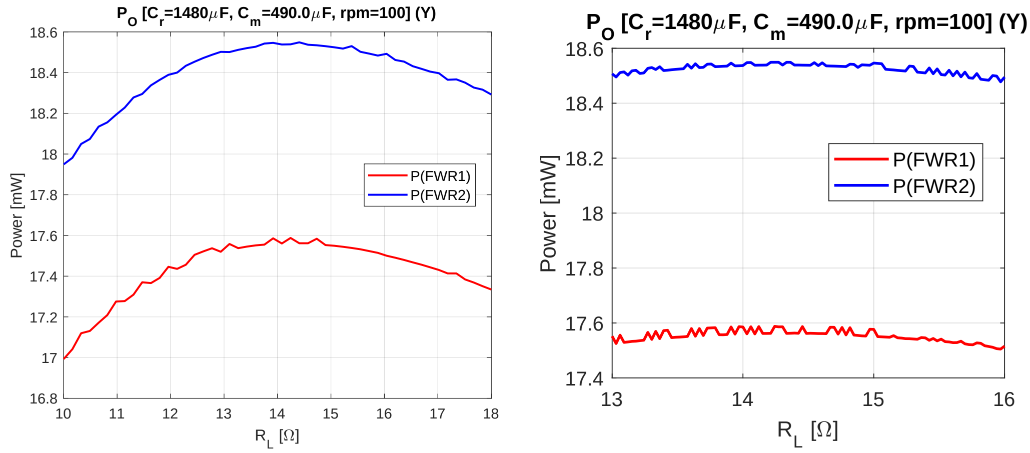
(b) Detail near the maximum.



(c) Detail, combined power.

Figure 4.6: Output power of the star-connected first and second bridges as a function of the compensation capacitor C_m at 400 rpm ($C_r = 1480$ μF, $R_L = 14.00$ Ω), individually (a, b) and combined (c).

about 850 mW at 400 rpm, noticeably more than every other value tested. No single value of C_m is therefore best across the whole range: a small C_m favours the high-speed end at the expense of the low-speed end, and vice versa for a large C_m. Since 100 rpm is the speed at which the harvester delivers the least power overall, and therefore sets the lower bound on the power available across the whole range, this trade-off favours choosing one of the larger capacitors, close to the value found optimal at 100 rpm in the compensation capacitor sweep, so as to keep this lower bound as high as possible even at the cost of some of the power that would otherwise be available at higher speed.



(a) Full sweep range. **(b)** Detail near the maximum.

Figure 4.7: Output power of the star-connected first and second bridges as a function of the load resistance R_L at 100 rpm ($C_r = 1480\ \mu$F, $C_m = 490\ \mu$F).

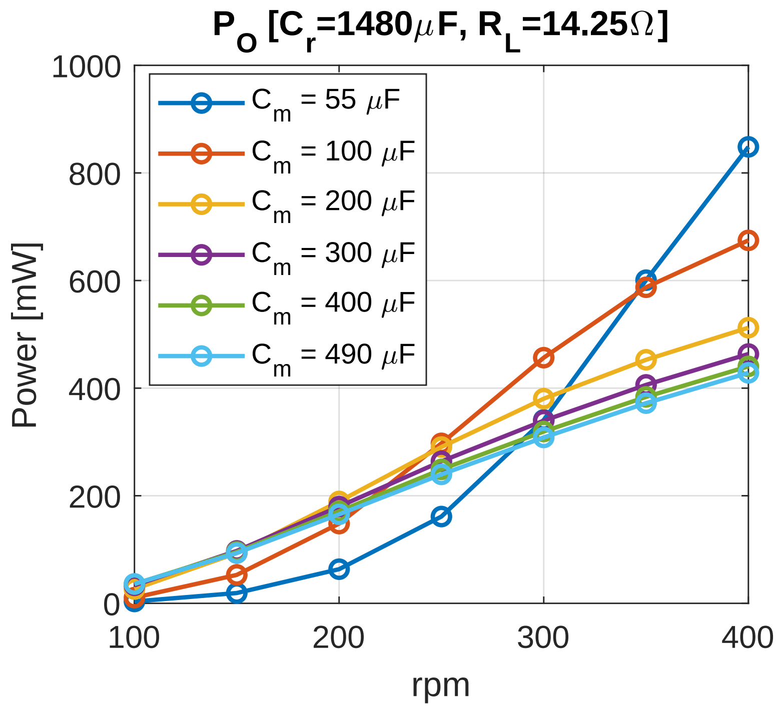


Figure 4.8: Combined output power of the first and second bridges across the target speed range, for six fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 14.25\ \Omega$).

Efficiency

Figures 4.9 and 4.10 show the rectifier efficiency $\eta_{\text{Rec}} = P_{\text{out}}/P_{\text{in}}$ and the power extraction efficiency (PEE, (1.2)) for the same six values of C_m across the speed range. By (1.1), the overall power conversion efficiency is PCE $= \eta_{\text{Rec}} \cdot \eta_{\text{DC-DC}}$. These FWR-only simulations do not include a DC-DC converter stage after the rectifier, so η_{Rec} is the quantity measured here. It coincides with the full PCE once such a stage, with its own efficiency $\eta_{\text{DC-DC}}$, is added. The η_{Rec} curves are less sensitive to the choice of C_m than the PEE curves, but the difference is still visible at low speed. At 400 rpm all six curves converge towards $\eta_{\text{Rec}} \approx 0.70$-$0.76$. At 100 rpm, however, η_{Rec} ranges from about 0.20 for the smallest capacitor ($C_m = 55\ \mu$F) to about 0.43 for the largest ones ($C_m \geq 300\ \mu$F). The PEE curves are far more sensitive to C_m. Each one peaks at the single speed for which that value of C_m is closest to (2.31), and falls off on either side, reaching at most PEE ≈ 0.49-0.71 depending on C_m. This mirrors the crossover already seen in the frequency sweep of Figure 4.8. To see why, note that (1.2) writes PEE as PEE $= (P_{\text{in}}/P_{\text{max}}) \cdot$ PCE, where $P_{\text{in}}/P_{\text{max}}$ compares the real input power to the maximum power the coil could theoretically deliver. Since η_{Rec}, and therefore PCE, barely change with C_m, almost all of the large swings in PEE come from $P_{\text{in}}/P_{\text{max}}$. In other words, PEE mainly reflects how close C_m is to cancelling the coil's reactance at that speed, not how efficiently the rectifier itself converts power.

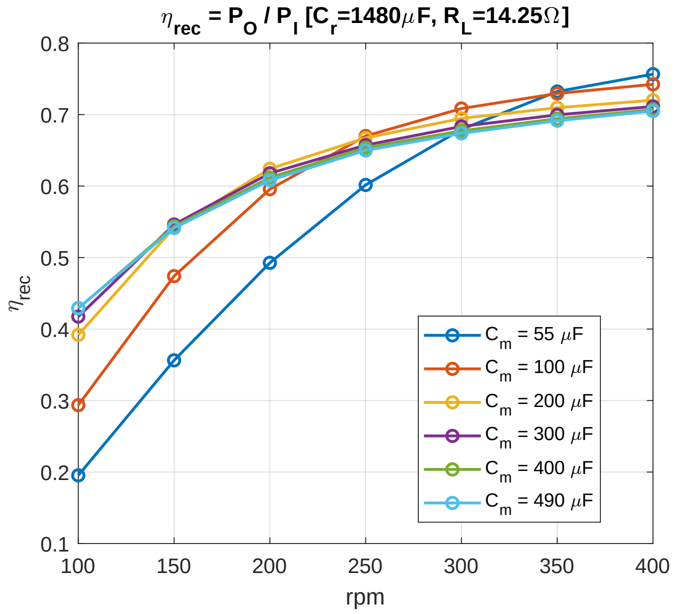


Figure 4.9: Rectifier efficiency η_{Rec} of the first and second bridges combined, across the target speed range, for six fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 14.25\ \Omega$).

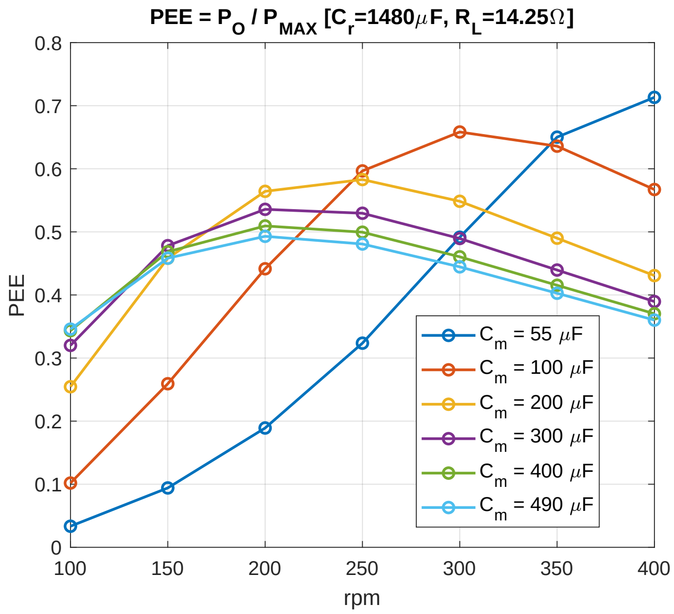


Figure 4.10: Power extraction efficiency (PEE) of the first and second bridges combined, across the target speed range, for six fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 14.25\ \Omega$).

4.2.b Delta Connection

Figure 4.11 shows the Simulink circuit used to simulate one of the two three-phase groups in the delta connection of Section 3.2. The circuit is almost identical to the star-connected one of Figure 4.4, in which each phase is modelled as an ideal sinusoidal source in series with the coil impedance and the current-sensing resistance introduced in Section 4.1, followed by the compensation capacitor C_m of (2.31), before reaching the input terminals A, B, C of the three-phase bridge. As with the star connection the total input power is again obtained with the two-wattmeter method, using voltage and current sensors placed on two of the three lines. At the output, the bridge is connected to a filter capacitor C_r and the load resistor R_{LD} in parallel, with a voltmeter measuring the resulting load voltage. The same circuit is used for both the first bridge (FWR1, phases $k = 0, 2, 4$) and the second bridge (FWR2, phases $k = 1, 3, 5$), each fed with the corresponding row of Table 4.1.

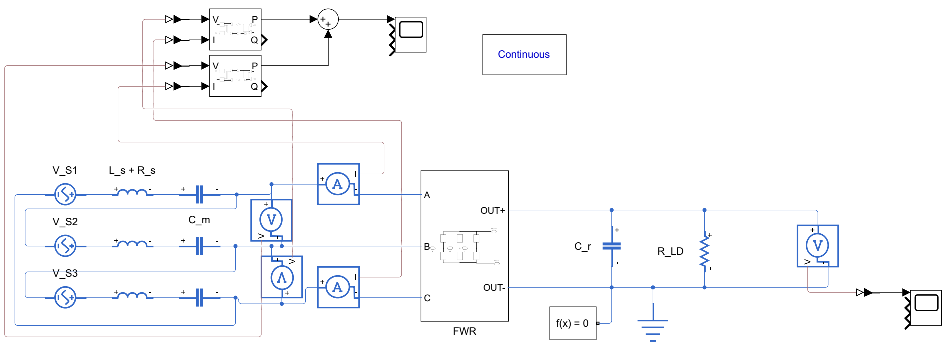


Figure 4.11: Simulink circuit used to simulate one three-phase group in the delta connection, from the phase sources through the compensation capacitors C_m, the three-phase bridge, to the output filter C_r and load R_{LD}.

Compensation Capacitor (C_m) Sweep

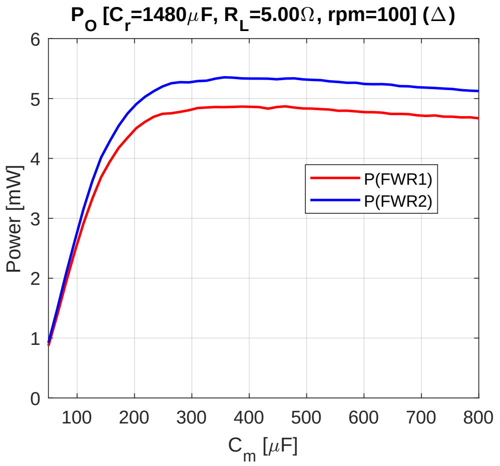
As for the star connection, C_m was swept independently at 100 and 400 rpm for the delta-connected group, with $C_r = 1480\ \mu$F and $R_L = 5.00\ \Omega$ fixed for this sweep. At 100 rpm, Figure 4.12 shows the same kind of behaviour already seen for the star connection: an almost flat maximum, here extending from about 250 μF to at least 500 μF; within this plateau, the detail view shows the power peaking around $C_m = 455\ \mu$F, so this value was chosen as the one giving the maximum output power. At 400 rpm, Figure 4.13 shows a sharp peak at $C_m = 53\ \mu$F, close to the value found for the star connection (55 μF), consistent with C_m of (2.31) depending only on the coil inductance and the electrical frequency, not on how the coils are interconnected. As in the star connection, the second bridge (FWR2) delivers slightly more power than the first (FWR1) across the whole sweep. The combined power of the two bridges, shown below, peaks at the same C_m chosen from the individual curves, confirming that these two values, $C_m = 455\ \mu$F for 100 rpm and $C_m = 53\ \mu$F for 400 rpm, also maximise the total power and are the two candidates carried forward to the load resistance and frequency sweeps below.

Load Resistance (R_L) Sweep

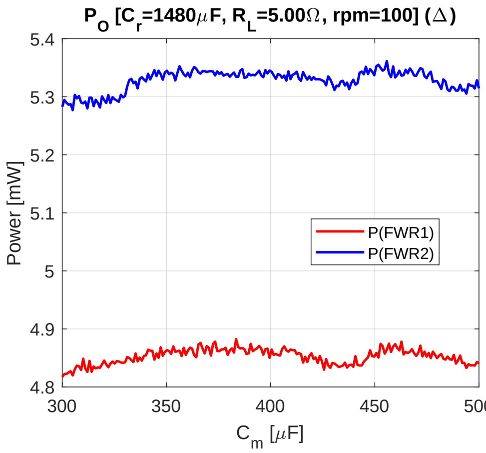
With $C_m = 455\ \mu$F fixed at the value found optimal for 100 rpm in the previous sweep, and $C_r = 1480\ \mu$F, the load resistance R_L was swept to find the value that maximises the output power at this speed. Figure 4.14 shows the resulting curve: the output power increases from $R_L = 2\ \Omega$, stays close to its highest value between about 5 and 6.5 Ω, and then slowly decreases towards $R_L = 10\ \Omega$. As in the compensation capacitor sweep, the second bridge (FWR2) delivers slightly more power than the first (FWR1) across the whole range, again consistent with the higher average phase voltage of coils $1, 3, 5$ in Table 4.1. The value $R_L = 5.75\ \Omega$ was chosen inside this broad maximum, and is the value used in the frequency sweep and efficiency evaluation that follow.

Frequency Sweep

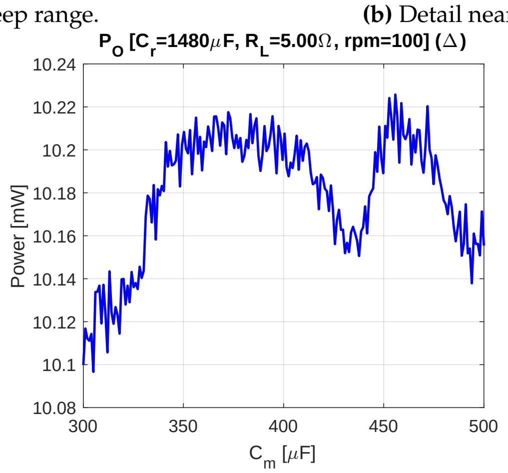
Figure 4.15 sweeps six fixed values of C_m, from 53 μF (optimal at 400 rpm) to 455 μF (optimal at 100 rpm), across the whole target speed range, with $C_r = 1480\ \mu$F and $R_L = 5.75\ \Omega$ fixed, and plots the resulting combined output power of the first and second bridges. Across the low end of the range, $C_m = 53\ \mu$F delivers the least power. At 100 rpm itself, all six curves are barely distinguishable. The gap only opens up above that speed. Unlike the star connection, however, the best performer over most of the middle of the range is not the largest capacitor:



(a) Full sweep range.



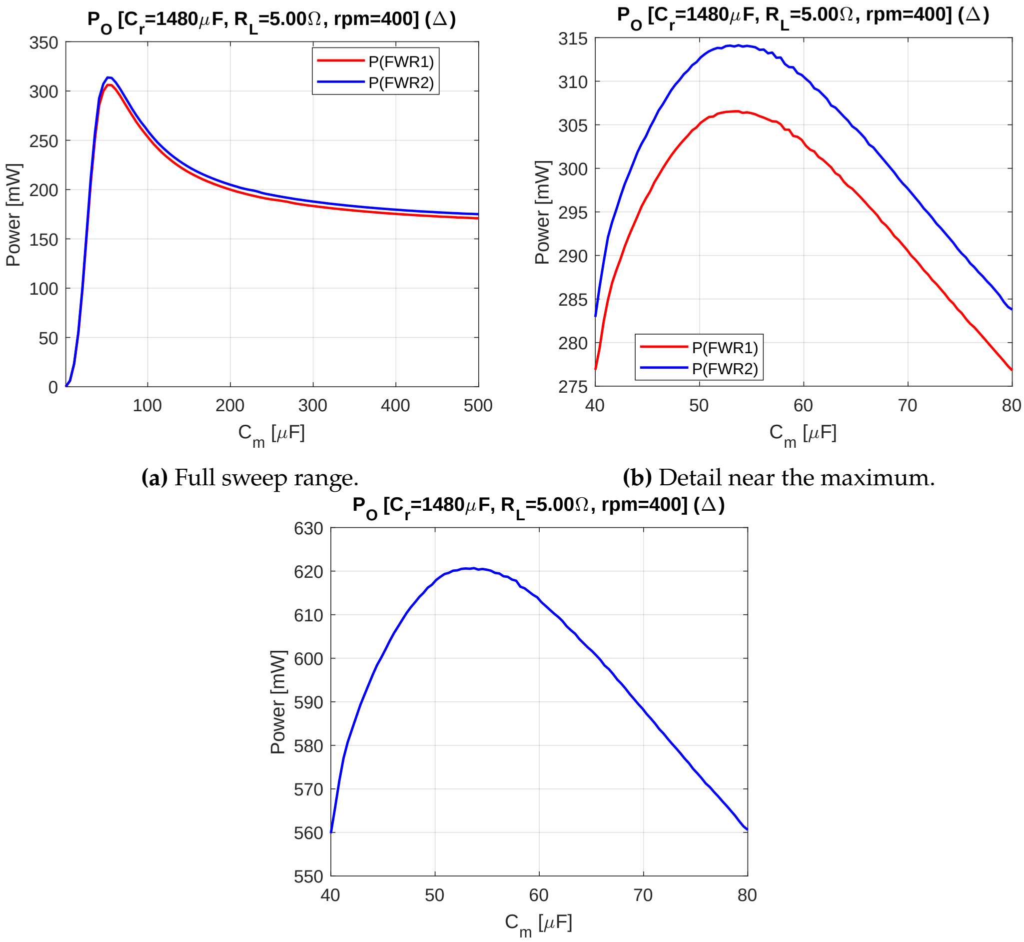
(b) Detail near the maximum.



(c) Detail, combined power.

Figure 4.12: Output power of the delta-connected first and second bridges as a function of the compensation capacitor C_m at 100 rpm ($C_r = 1480\ \mu$F, $R_L = 5.00\ \Omega$), individually (a, b) and combined (c).

$C_m = 100\ \mu$F delivers the most power from about 150 to 300 rpm, exceeding even $C_m = 455\ \mu$F in this range. Only beyond about 340 rpm does $C_m = 53\ \mu$F overtake every other choice, reaching about 620 mW at 400 rpm. As with the star connection, no single value of C_m is therefore best across the whole range, and since 100 rpm still sets the lower bound on the power available across the whole range, one of the larger capacitors, close to the value found optimal at 100 rpm in the compensation capacitor sweep, remains the preferred choice to keep this lower bound as high as possible.



(a) Full sweep range.

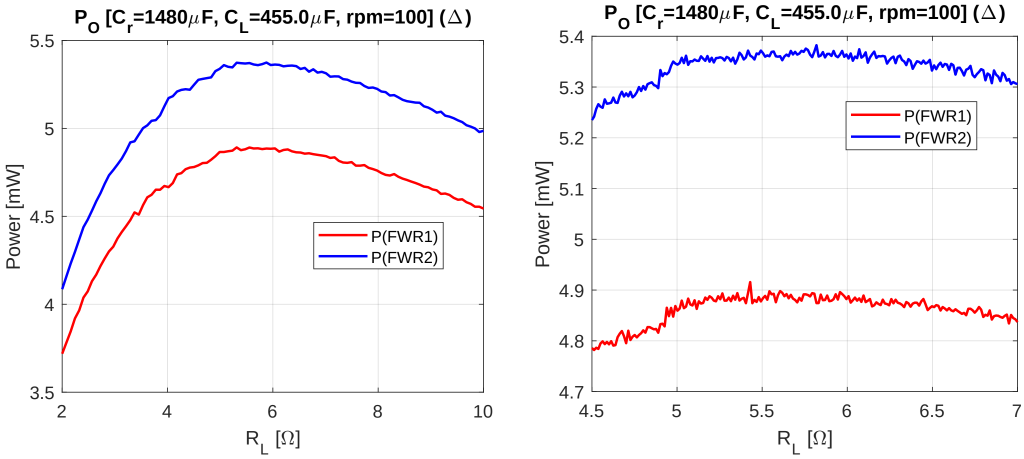
(b) Detail near the maximum.

(c) Detail, combined power.

Figure 4.13: Output power of the delta-connected first and second bridges as a function of the compensation capacitor C_m at 400 rpm ($C_r = 1480\ \mu$F, $R_L = 5.00\ \Omega$), individually (a, b) and combined (c).

Efficiency

Figures 4.16 and 4.17 show the rectifier efficiency η_{Rec} and the power extraction efficiency (PEE) for the same six values of C_m across the speed range. As with the star connection, the η_{Rec} curves are less sensitive to C_m than the PEE curves, converging to $\eta_{\text{Rec}} \approx 0.57$-$0.62$ by 400 rpm and ranging from about 0.11 to 0.21 at 100 rpm. These values are noticeably lower than the star connection's $\eta_{\text{Rec}} \approx 0.70$-$0.76$ at 400 rpm and 0.20-0.43 at 100 rpm, consistent with Section 3.2: without the star connection's $\sqrt{3}$ voltage gain, the FWR's fixed two-diode drop loss removes a larger fraction of the lower delta line voltage. The PEE curves again peak each at the speed where its C_m is closest to the matching value of (2.31), reaching at



(a) Full sweep range. **(b)** Detail near the maximum.

Figure 4.14: Output power of the delta-connected first and second bridges as a function of the load resistance R_L at 100 rpm ($C_r = 1480$ μF, $C_m = 455$ μF).

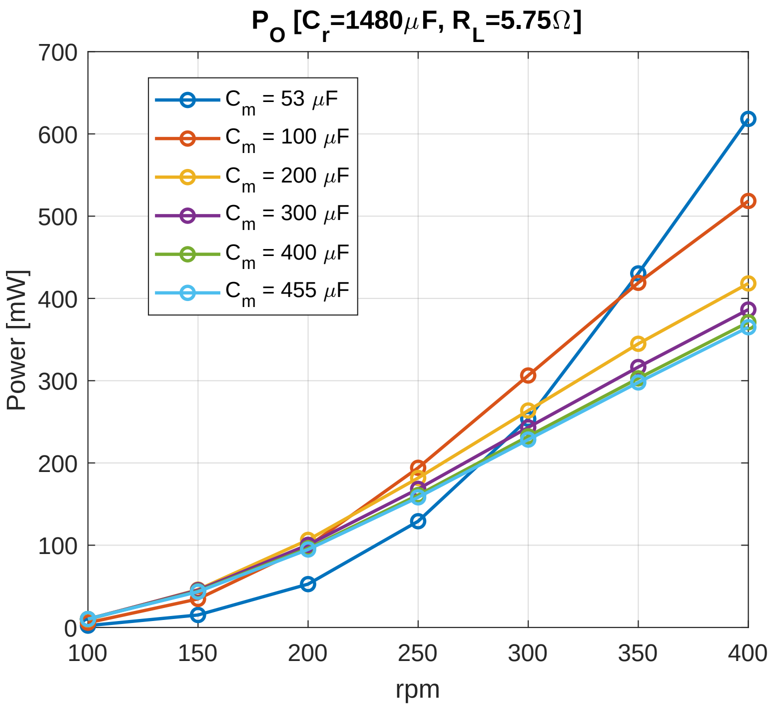


Figure 4.15: Combined output power of the first and second bridges across the target speed range, for six fixed values of the compensation capacitor C_m ($C_r = 1480$ μF, $R_L = 5.75$ Ω).

most PEE $\approx$ 0.32-0.52 depending on C_m, lower still than the star connection's peaks. This mirrors the crossover already seen in the frequency sweep of Figure

4.15. The reason is the same as for the star connection. η_{Rec} barely changes with C_m, so almost all of the swing in PEE comes from how closely C_m matches the coil at that speed, not from the rectifier's own efficiency.

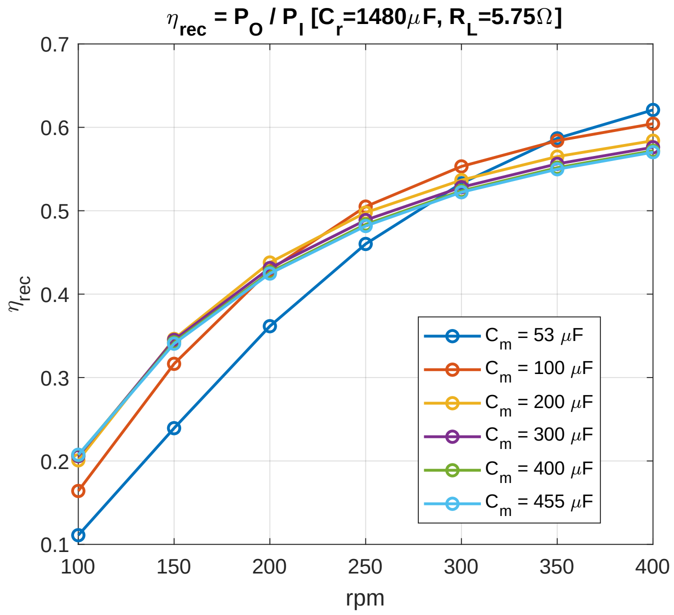


Figure 4.16: Rectifier efficiency η_{Rec} of the first and second bridges combined, across the target speed range, for six fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 5.75\ \Omega$).

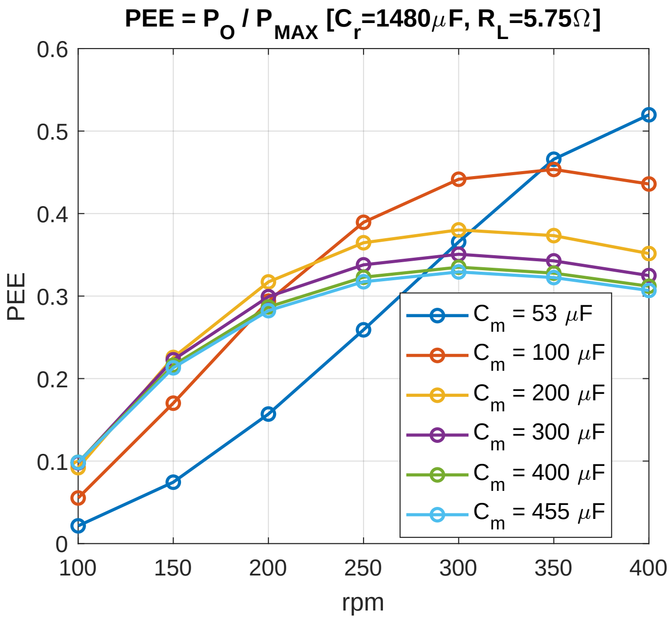


Figure 4.17: Power extraction efficiency (PEE) of the first and second bridges combined, across the target speed range, for six fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 5.75\ \Omega$).

4.3 Negative Voltage Converter (NVC)

Figure 4.18 shows the Simulink circuit used to simulate one of the three two-phase groups in the anti-series NVC connection of Section 3.3. The two antiphase pickup units are each modelled as an ideal sinusoidal source in series with the coil impedance and the current-sensing resistance introduced in Section 4.1, followed by the compensation capacitor C_m of (2.31), and connected in anti-series to the input terminals A and B of the NVC bridge. At the output, the bridge is connected to a filter capacitor C_r and the load resistor R_L in parallel, with a voltmeter measuring the resulting load voltage. The same circuit is used for all three NVC bridges, each fed with the corresponding pair of phases $k = 0, 3$; $k = 1, 4$; and $k = 2, 5$, from Table 4.1.

4.3.a Compensation Capacitor (C_m) Sweep

As with the FWR bridges, C_m was swept at 100 rpm for all three NVC bridges, with $C_r = 1480\ \mu$F and $R_L = 25.00\ \Omega$ fixed for this sweep. Figure 4.19 shows the output power of each bridge rising steeply for small C_m, reaching a maximum around $C_m = 260\ \mu$F, and then decreasing gently for larger values; all three bridges peak within this same narrow window of C_m. NVC1 delivers the most

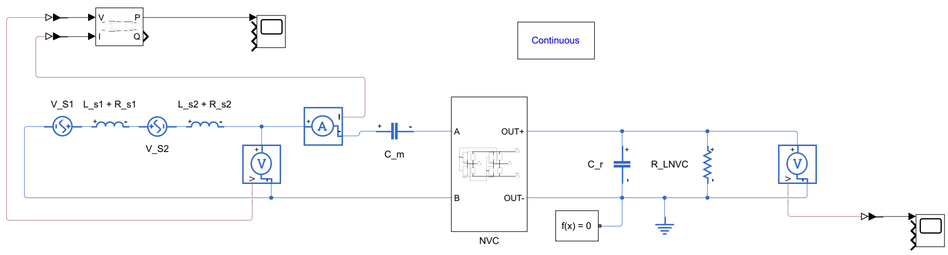


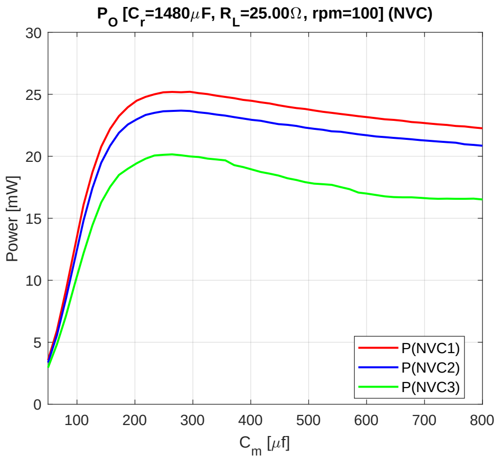
Figure 4.18: Simulink circuit used to simulate one two-phase group in the anti-series NVC connection, from the phase sources through the compensation capacitors C_m, the NVC bridge, to the output filter C_r and load R_L.

power, followed by NVC2, with NVC3 delivering noticeably less than either, about 20% less than NVC1 and 12% less than NVC2 at the peak, and falling off more steeply than the other two as C_m grows beyond the peak. The combined power of the three bridges, shown below, peaks at the same C_m chosen from the individual curves, confirming that $C_m = 260$ μF also maximises the total power.

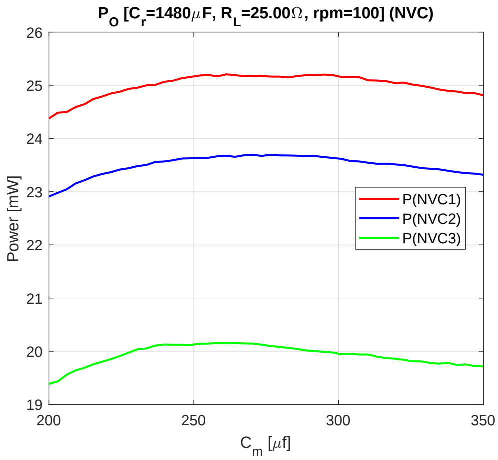
At 400 rpm, Figure 4.20 shows a much sharper peak than at 100 rpm, around $C_m = 38$ μF, after which the power falls off quickly before levelling out for larger C_m. The three bridges again peak within the same narrow window, but the gap between them is noticeably smaller than at 100 rpm: NVC1 leads, followed closely by NVC2 and then NVC3, only about a tenth below NVC1 at the peak. The combined power peaks at the same value, around 938 mW at $C_m = 38$ μF, confirming this as the best choice at this speed.

4.3.b Load Resistance (R_L) Sweep

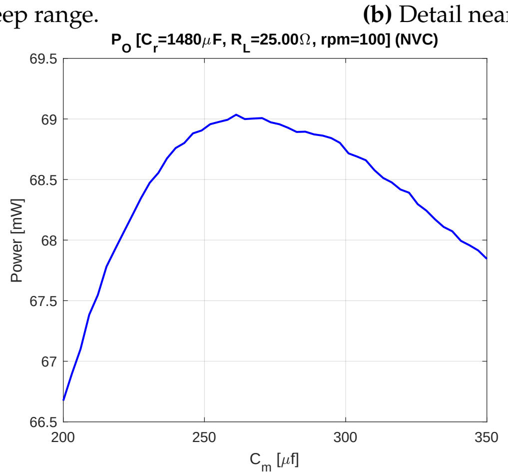
With $C_m = 260$ μF fixed at the value found optimal for 100 rpm in the previous sweep, and $C_r = 1480$ μF, the load resistance R_L was swept to find the value



(a) Full sweep range.



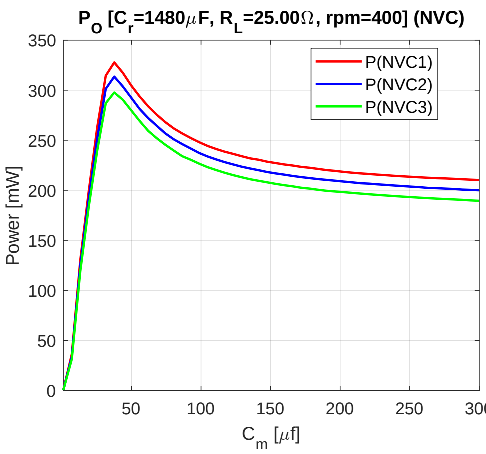
(b) Detail near the maximum.



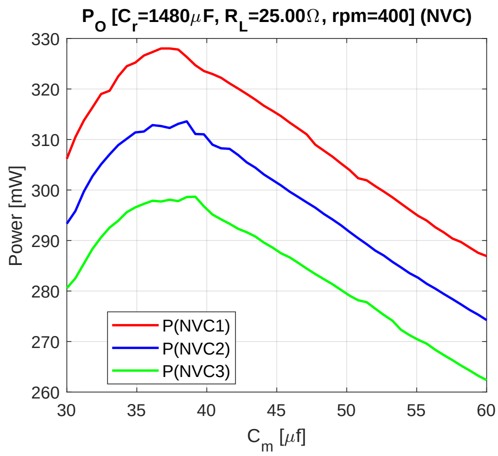
(c) Detail, combined power.

Figure 4.19: Output power of the three NVC bridges as a function of the compensation capacitor C_m at 100 rpm ($C_r = 1480$ μF, $R_L = 25.00$ Ω), shown individually (a, b) and combined (c).

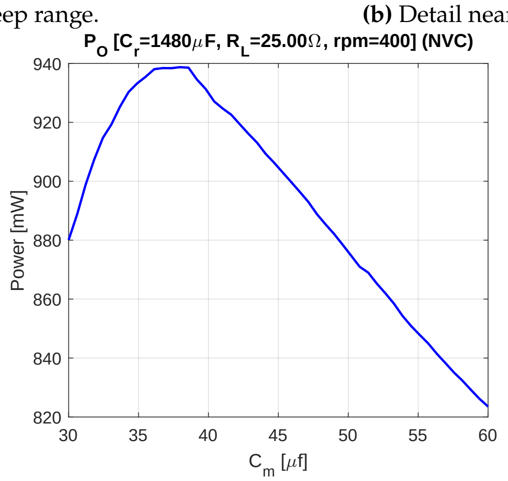
that maximises the output power at this speed. Figure 4.21 shows the resulting curves: NVC1 and NVC2 both peak around R_L = 25-26 Ω, while NVC3 keeps rising further, peaking only around R_L = 27-29 Ω. Despite this difference between the individual bridges, the combined power peaks around $R_L = 26$ Ω, staying close to its maximum up to about $R_L = 28$ Ω before decreasing. This value, $R_L = 26$ Ω, is therefore chosen as the best load resistance at this speed, refining the $R_L = 25.00$ Ω used above for the compensation capacitor sweep.



(a) Full sweep range.



(b) Detail near the maximum.

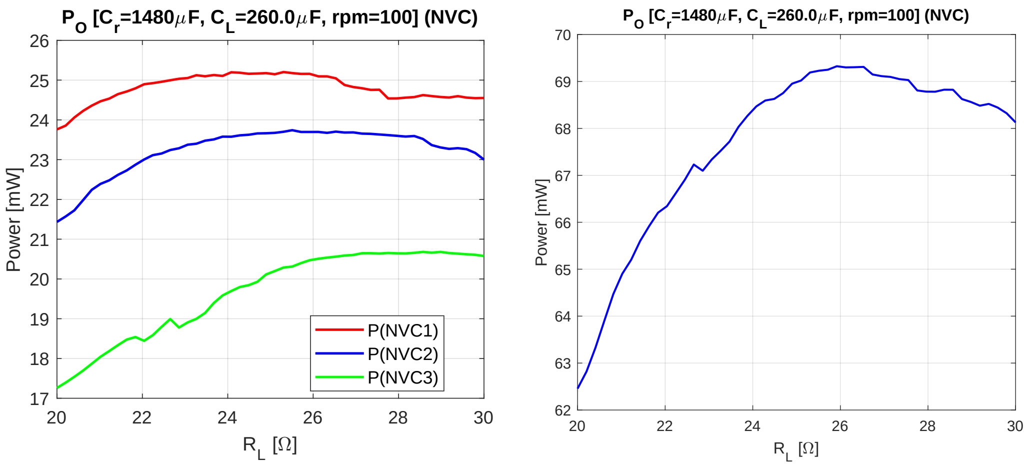


(c) Detail, combined power.

Figure 4.20: Output power of the three NVC bridges as a function of the compensation capacitor C_m at 400 rpm ($C_r = 1480$ μF, $R_L = 25.00$ Ω), shown individually (a, b) and combined (c).

4.3.c Frequency Sweep

Figure 4.22 sweeps six fixed values of C_m, from 38 μF (optimal at 400 rpm) to 260 μF (optimal at 100 rpm), across the whole target speed range, with $C_r = 1480$ μF and $R_L = 26.00$ Ω fixed, and plots the resulting combined output power of the three NVC bridges. At the low-speed end of the range, $C_m = 260$ μF delivers the most power, while $C_m = 38$ μF performs far worse than every other value. As with the FWR delta connection, the best performer over the middle of the range is not one of the two extreme values just found, but an intermediate one: $C_m = 100$ μF takes the lead by about 200 rpm and stays on top up to 300 rpm, exceeding even $C_m = 260$ μF. Above 300 rpm, a second intermediate value,



(a) Individual bridges.

(b) Combined power.

Figure 4.21: Output power of the three NVC bridges as a function of the load resistance R_L at 100 rpm ($C_r = 1480$ μF, $C_m = 260$ μF), shown individually (a) and combined (b).

$C_m = 50$ μF, rises sharply and becomes the best performer at 350 rpm, reaching about 780 mW against 715 mW for $C_m = 38$ μF and 685 mW for $C_m = 100$ μF. Only at 400 rpm does $C_m = 38$ μF finally take the lead, reaching about 920 mW, with $C_m = 50$ μF close behind at about 860 mW and every other value well below both. As with the FWR bridges, no single value of C_m is therefore best across the whole range, and since 100 rpm still sets the lower bound on the power available across the whole range, $C_m = 260$ μF, the value found optimal at 100 rpm in the compensation capacitor sweep, remains the preferred choice to keep this lower bound as high as possible.

4.3.d Efficiency

Figures 4.23 and 4.24 show the rectifier efficiency $\eta_{\text{Rec}} = P_{\text{out}}/P_{\text{in}}$ and the power extraction efficiency (PEE) for the same six values of C_m across the speed range. As with the FWR bridges, these NVC simulations do not include a DC-DC converter stage after the rectifier, so η_{Rec} is the quantity measured here; it coincides with the full PCE of (1.1) once such a stage, with its own efficiency $\eta_{\text{DC-DC}}$, is added. The η_{Rec} curves are far more sensitive to C_m than in the FWR case, especially at low speed: at 100 rpm, η_{Rec} ranges from about 0.24 for the smallest capacitor ($C_m = 38$ μF) to about 0.67 for the largest one ($C_m = 260$ μF), while at 400 rpm all six curves converge towards $\eta_{\text{Rec}} \approx 0.76$-$0.83$. Each PEE curve peaks at the speed where its C_m is closest to the matching value of (2.31), most notably $C_m = 100$ μF around 300 rpm and $C_m = 50$ μF around 350 rpm, the same two intermediate hand-offs seen in the power sweep. As with the FWR bridges, η_{Rec},

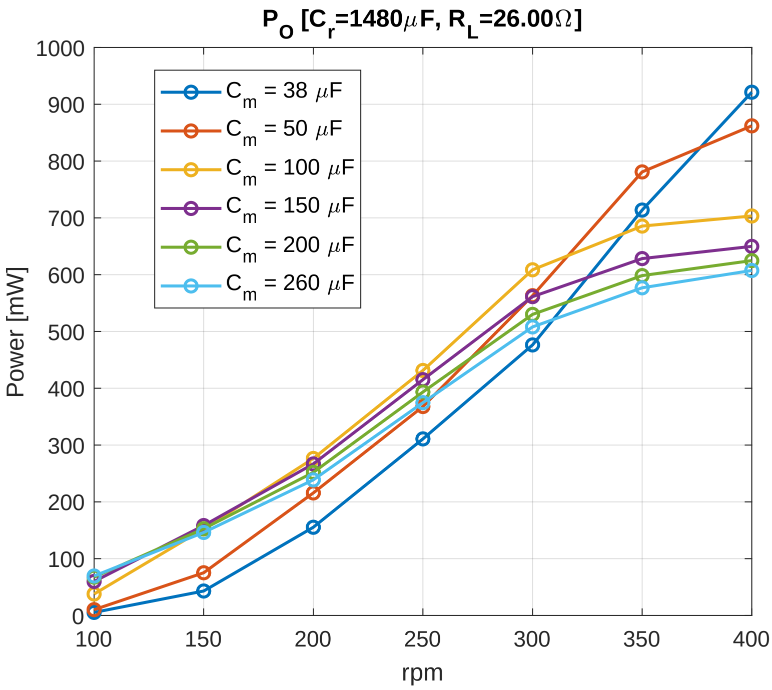


Figure 4.22: Combined output power of the three NVC bridges across the target speed range, for six fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 26.00\ \Omega$).

and therefore PCE, change far less with C_m than PEE does. Most of the swing seen in PEE therefore comes from how closely C_m matches the coil at that speed, not from a change in the rectifier's own efficiency.

4.4 Comparison Between Topologies

Sections 4.2 and 4.3 evaluate the FWR (star and delta connections) and the NVC independently, each with its own compensation capacitor and load resistance chosen to maximise the output power at 100 rpm, the speed that sets the lower bound on the power available across the whole target range. Since all three configurations rectify the same six pickup units, only regrouped differently (two three-phase groups for the FWR, three anti-series two-phase groups for the NVC), their output power and efficiency can be compared directly. Table 4.2 summarises the resulting operating points and performance of the three configurations at the two ends of the target speed range.

The NVC delivers the most output power of the three configurations at both ends of the target range, and by a wide margin at the low-speed end: at 100 rpm it delivers over six times the power of the FWR delta connection and nearly twice

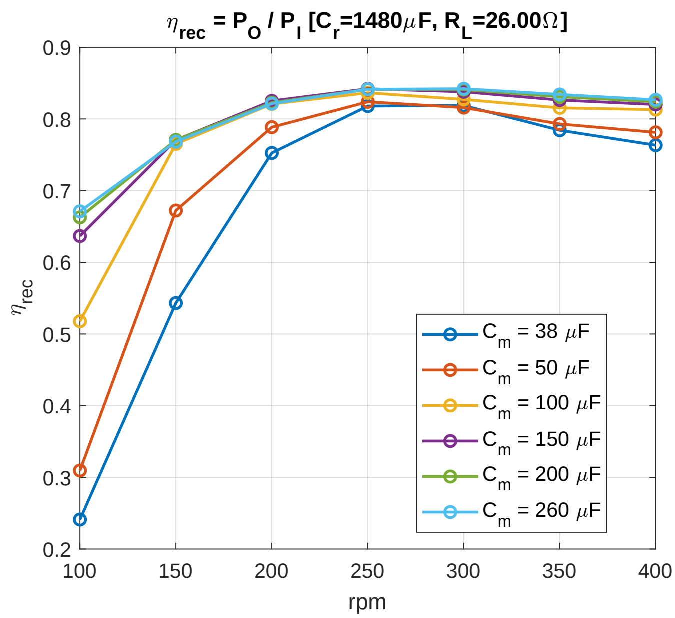


Figure 4.23: Rectifier efficiency η_{Rec} of the three NVC bridges combined, across the target speed range, for six fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 26.00\ \Omega$).

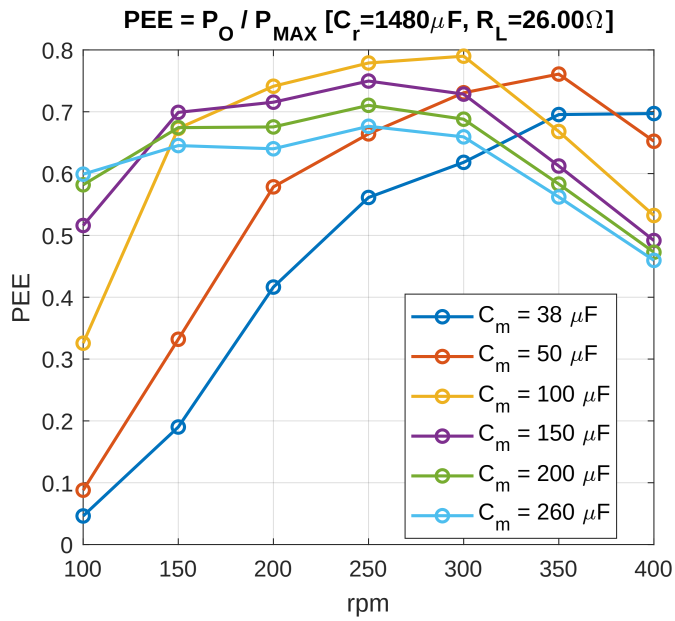


Figure 4.24: Power extraction efficiency (PEE) of the three NVC bridges combined, across the target speed range, for six fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 26.00\ \Omega$).

	FWR (Star)	FWR (Delta)	NVC
C_m at 100 rpm (μF)	490	455	260
C_m at 400 rpm (μF)	55	53	38
R_L (Ω)	14.25	5.75	26.00
P_O at 100 rpm (mW)	36.7	10.7	69
P_O at 400 rpm (mW)	430	365	610
η_{Rec} at 100 rpm	0.43	0.21	0.665
η_{Rec} at 400 rpm	0.71	0.57	0.82

Table 4.2: Comparison of the three rectifier configurations at their respective chosen operating points, using the C_m found optimal at 100 rpm for the output power and efficiency figures.

that of the FWR star connection, while at 400 rpm it still leads, delivering about 42% more power than the FWR star connection and 67% more than the delta connection. Between the two FWR connections, the star connection consistently outperforms the delta connection, consistent with Section 3.2: the star connection's line voltage is $\sqrt{3}$ times higher than the delta connection's, which more than makes up for its lower current. The delta connection's voltage is lower, so the FWR's fixed two-diode drop takes away a bigger part of it.

This same two-diode drop loss also explains the NVC's efficiency advantage. As shown in Section 3.3, the NVC bridge only replaces the diode drop with the much smaller MOSFET on-resistance once the voltage across its anti-series pair exceeds $V_{\text{off}} \approx 1.2$ V; below this threshold it falls back to the same diode-limited behaviour as the FWR. At 100 rpm, where the phase voltage is smallest, this shows up clearly in Table 4.2: $\eta_{\text{Rec}} \approx 0.665$ for the NVC against 0.43 and 0.21 for the FWR star and delta connections. At 400 rpm, where the phase voltage is large enough that the fixed diode drop matters proportionally less, the gap narrows, but the NVC still leads with $\eta_{\text{Rec}} \approx 0.82$ against 0.71 and 0.57. The power extraction efficiency (PEE) of all three configurations follows the same qualitative pattern already discussed individually in Sections 4.2 and 4.3: each curve peaks close to the single speed for which its chosen C_m best matches the coil, so this is not repeated here: it reflects the quality of the C_m match rather than a difference between topologies.

Overall, these simulations indicate the NVC as the preferred topology for the MP-VREH across the whole target speed range, delivering both the highest output power and the highest rectifier efficiency, with its advantage most pronounced at the low-speed end where the FWR's fixed diode drop is most significant.

5 Prototype Board Design

5.1 Overview

Chapter 3 introduced the FWR and NVC rectifier topologies used in this thesis, and Chapter 4 evaluated their Simulink models. This chapter describes the prototype board built to test these topologies on the real MP-VREH harvester, from the component choices behind it to its schematic and physical layout.

5.2 Components Used

Table 5.1 lists the key components used to build the FWR and NVC prototype board of Chapter 3. The same Schottky diode part is used both for the FWR's rectifying diodes and for the NVC's parallel diodes on its cross-coupled bridge. The NVC bridge itself also needs two N-MOSFETs and two P-MOSFETs.

The critical voltage $V_{\text{off}} \approx 1.2\,\text{V}$ above which the NVC bridge switches properly is set by the largest of its four MOSFET threshold voltages. Table 5.1 shows this is the N-MOSFET's $V_{\text{GS(th)}} = 1.2$ V, larger in magnitude than the P-MOSFET's $V_{\text{GS(th)}} = -1.0$ V.

5.3 Schematic

Figures 5.1-5.6 show the complete schematic of the prototype board, page by page. The board integrates both FWR three-phase groups (labelled FBR1, FBR2 in the schematic, i.e. Full bridge rectifier) and all three NVC bridges (NVC1, NVC2, NVC3), together with jumpers that reconfigure each coil between the FWR and NVC paths, the FWR groups between star and delta connection, and

Component	Specification	Characteristics
D_1-D_{10}	Schottky diode (CTS05S40-L3F)	$V_F = 0.31$ V @ $I_F = 0.1$ A
N_1, N_2	N-MOSFET (BSS138PW-115)	$R_{\text{DSon}} = 1\ \Omega$, $V_{\text{GS(th)}} = 1.2$ V, $V_{\text{SD}} = 0.75$ V
P_1, P_2	P-MOSFET (PMN27XPE-115)	$R_{\text{DSon}} = 27\ \text{m}\Omega$, $V_{\text{GS(th)}} = -1.0$ V, $V_{\text{SD}} = -0.7$ V

Table 5.1: Key components used on the FWR and NVC prototype board.

the compensation capacitor C_m of each coil, so that every configuration simulated in Chapter 4 can be tested on the same board.

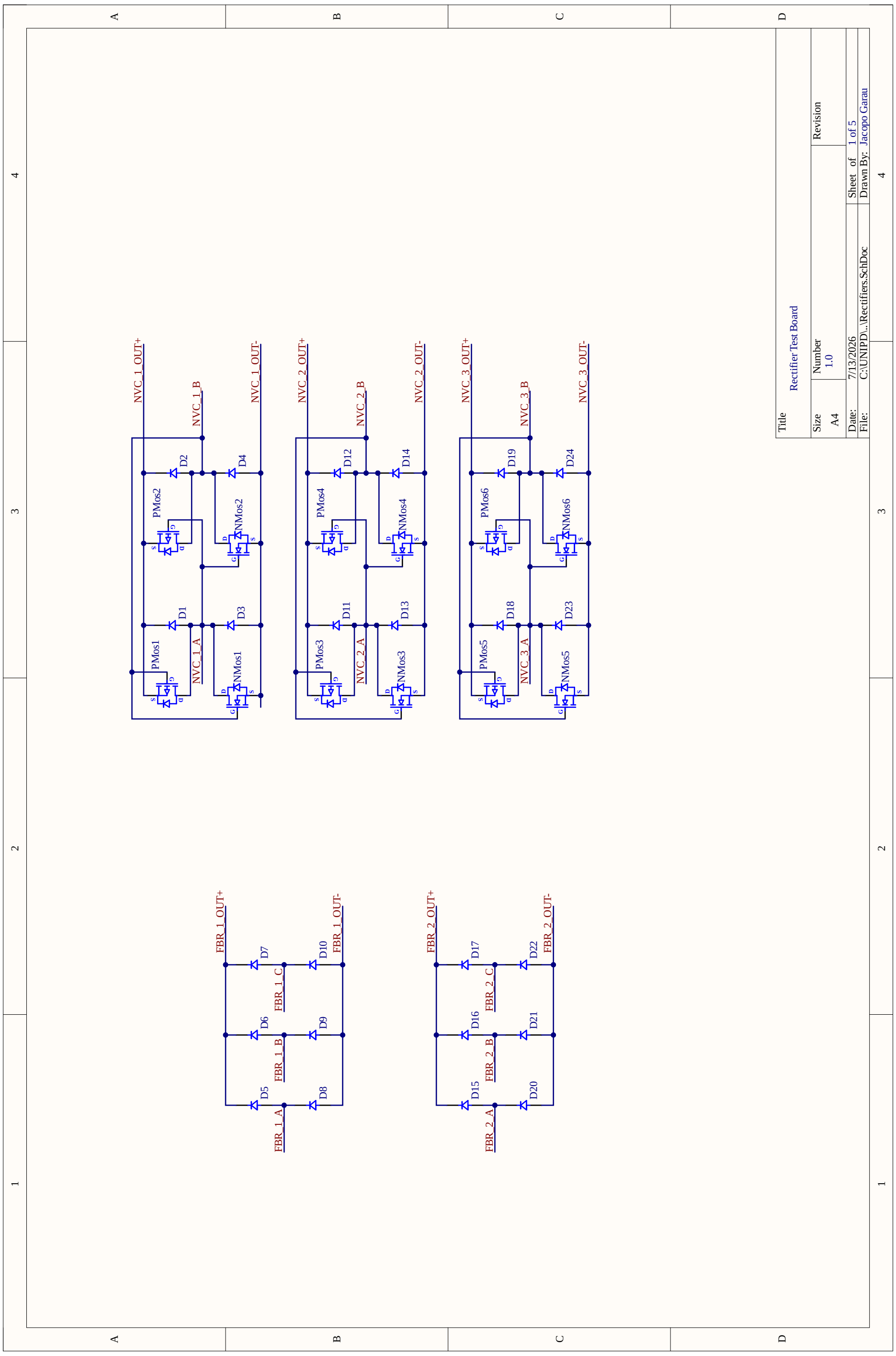


Figure 5.1: Rectifier bridges: the two FWR three-phase groups (labelled FBR1, FBR2 in the schematic) and the three anti-series NVC bridges (NVC1,NVC2,NVC3).

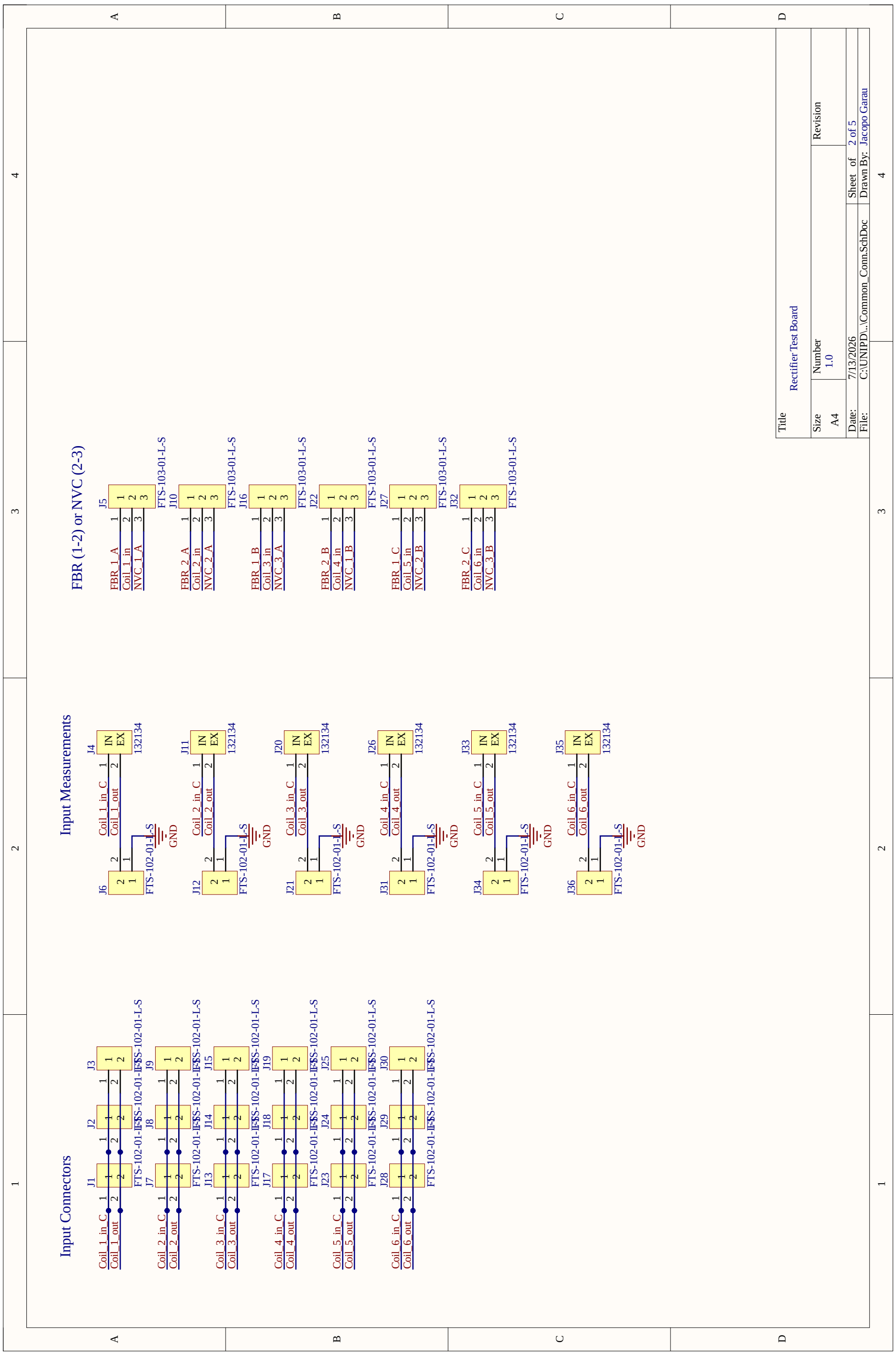


Figure 5.2: Common connectors: input connectors for the six coils, voltage-sensing measurement points, and the jumpers selecting whether each coil feeds the FWR or the NVC bridges.

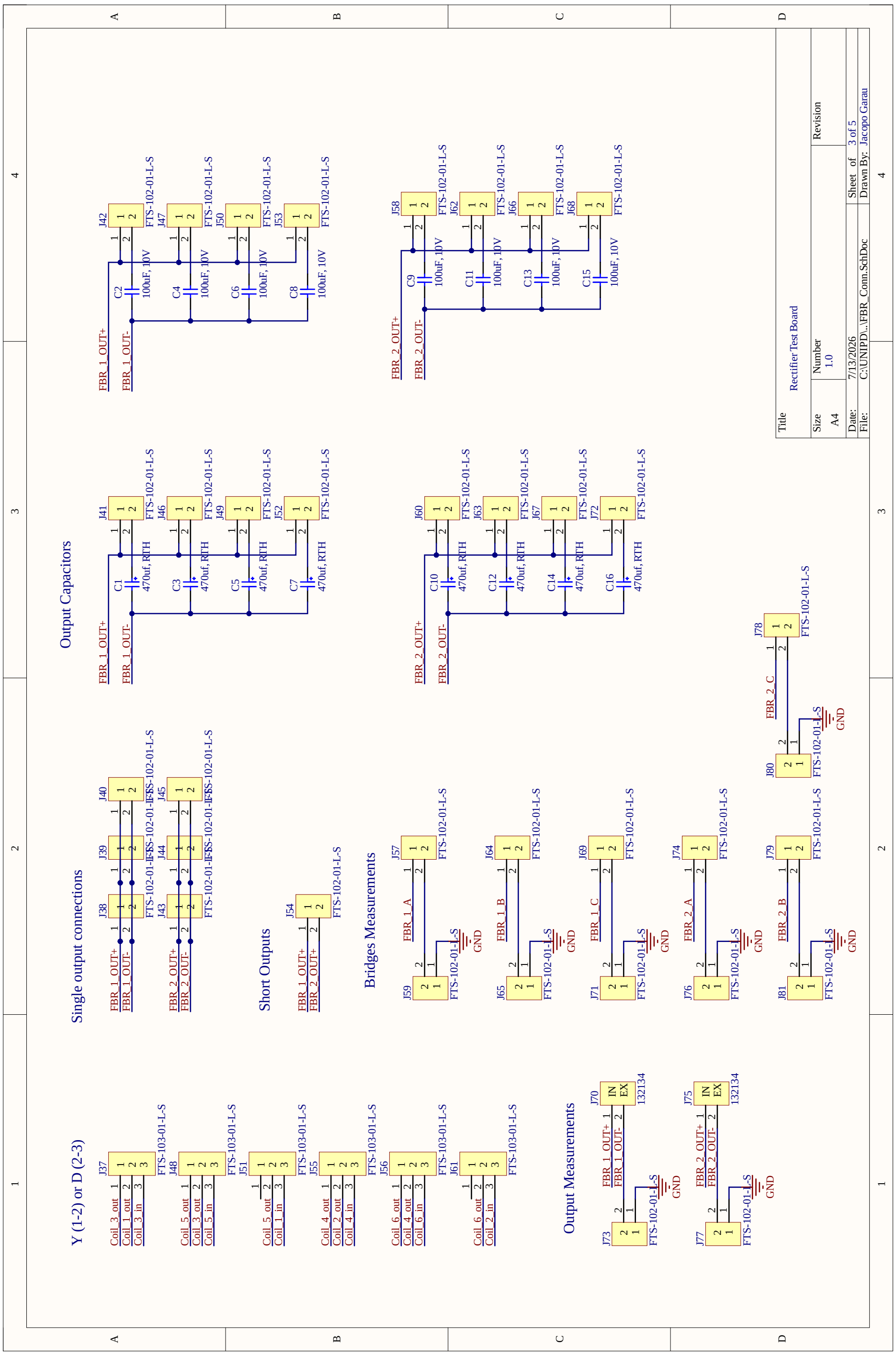


Figure 5.3: FWR connections: jumpers selecting the star or delta connection of each three-phase group, the output filter capacitors, and the bridge and output voltage measurement points.

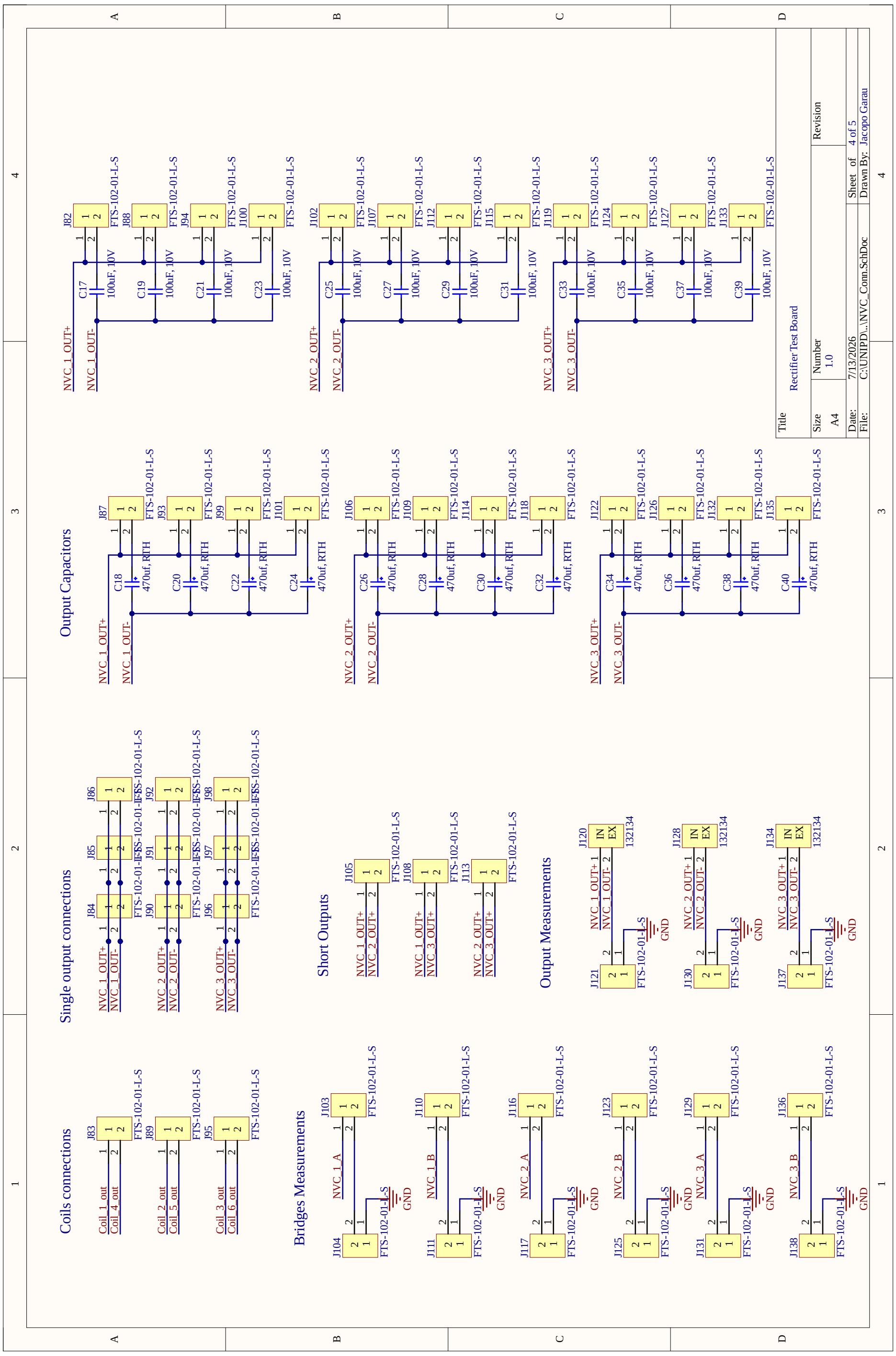


Figure 5.4: NVC connections: anti-series pairing of the six coils, the output filter capacitors, and the voltage measurement points for the three NVC bridges.

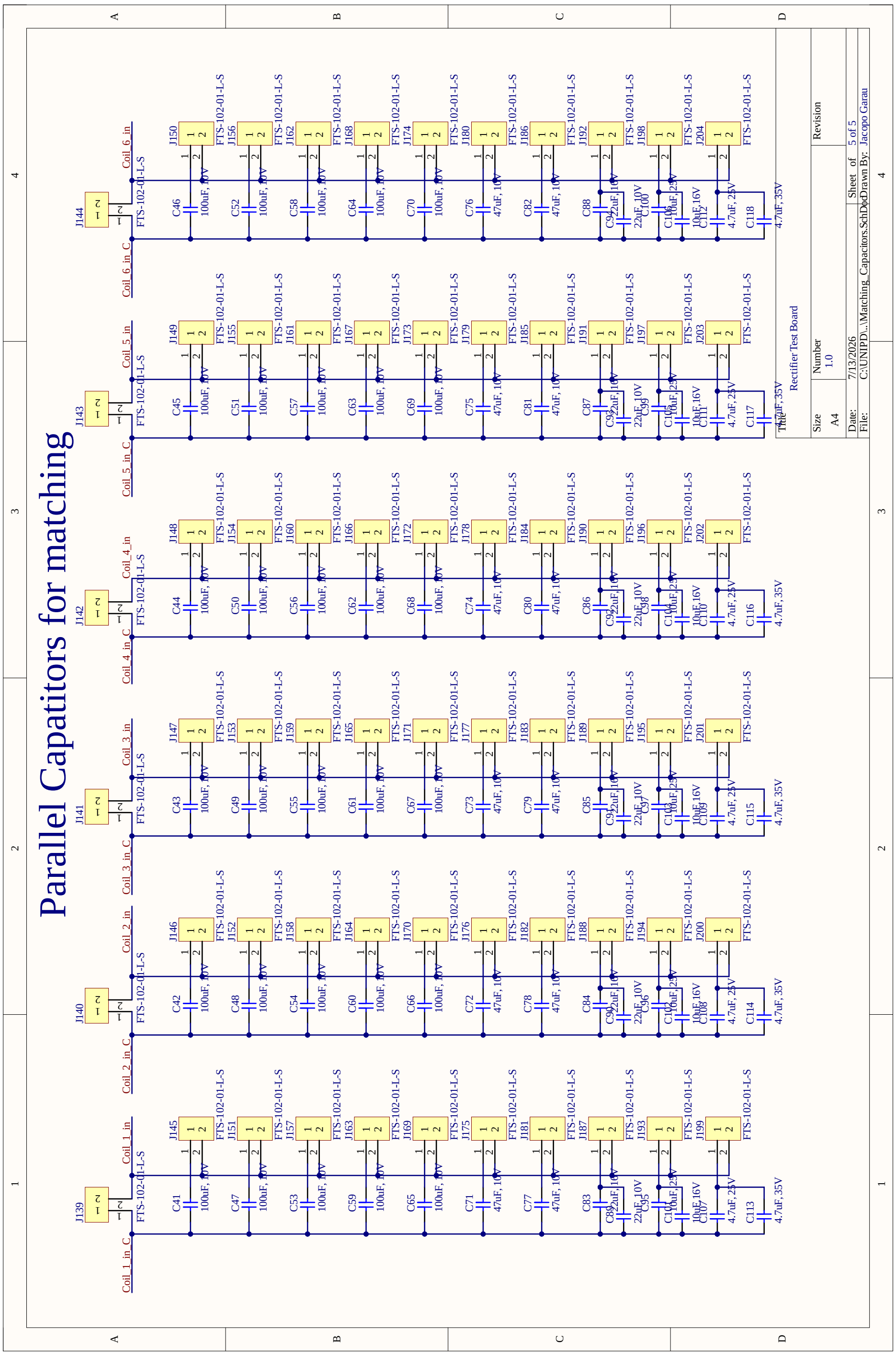
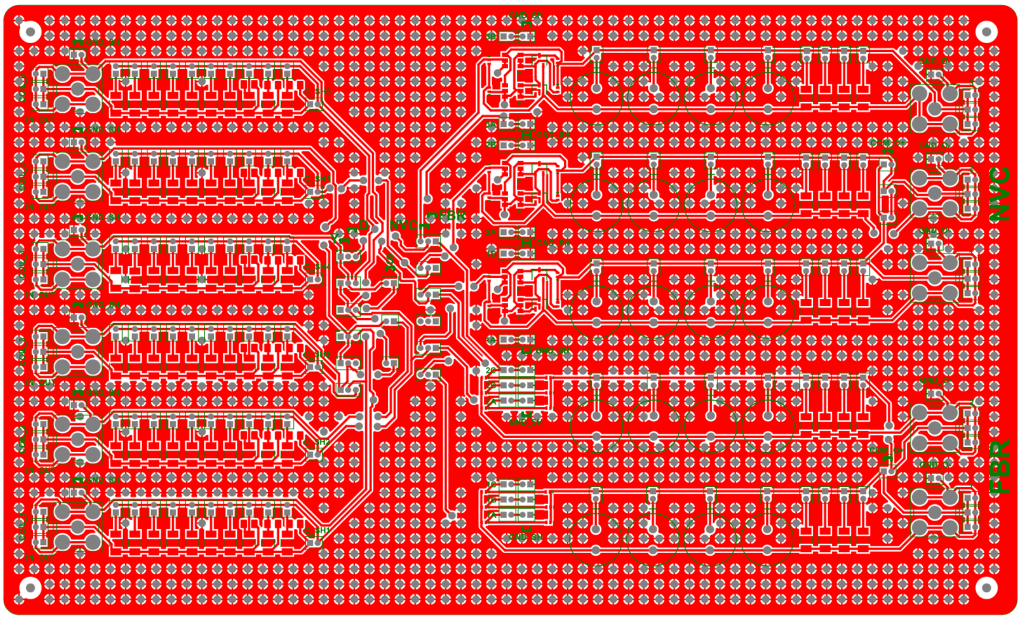
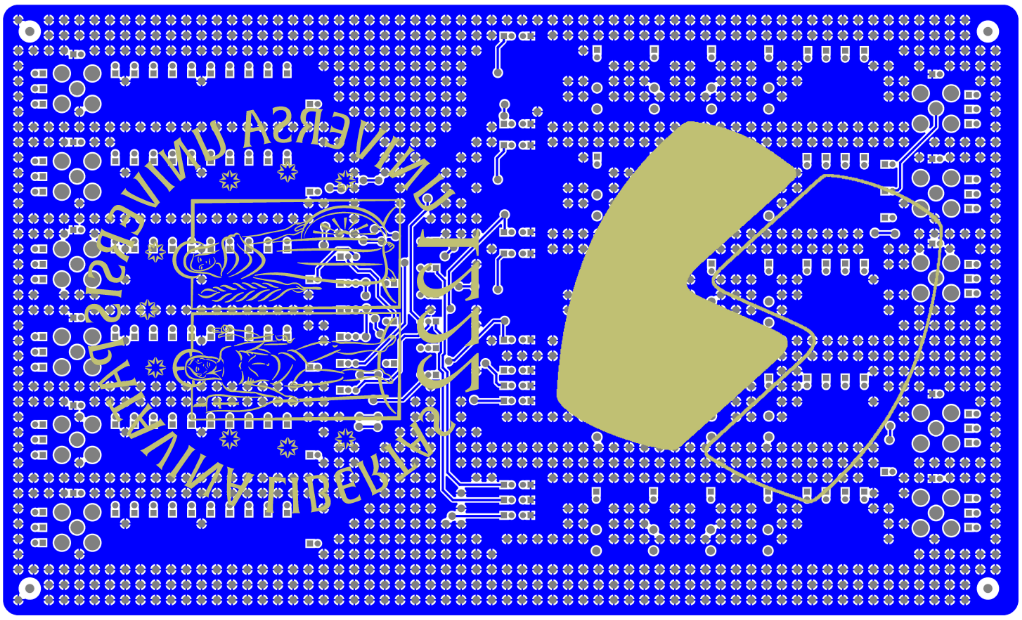


Figure 5.5: Matching capacitors: the jumper-selectable parallel capacitor bank used to set the compensation capacitor C_m of each coil.



(a) Top copper layer.



(b) Bottom copper layer.

Figure 5.6: PCB layout: top (a) and bottom (b) copper layers.

5.4 Board Assembly

Figure 5.8 shows a 3D rendering of the board generated directly from Altium Designer, before assembly. In the top view, the six coil connectors (COIL1-COIL6) sit on the left, each next to its own bank of matching capacitors; the Y/D and FBR/NVC selection jumpers sit in the middle; and the NVC and FBR rectifier sections, each with its own output connector and filter capacitors, sit on the right.

The PCB was ordered from a Chinese board manufacturer, while all components were sourced separately and the entire board was hand-soldered. Figure 5.7 shows one of the three NVC bridges during the soldering process: the four diodes, on the sides, are already soldered, while the MOSFET packages, on the right, still show unmelted solder sitting on their pins.

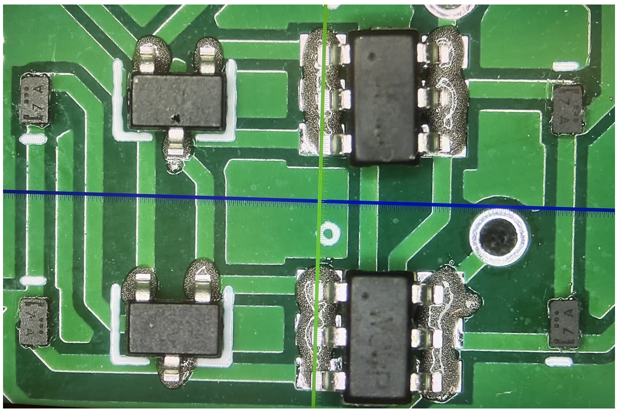
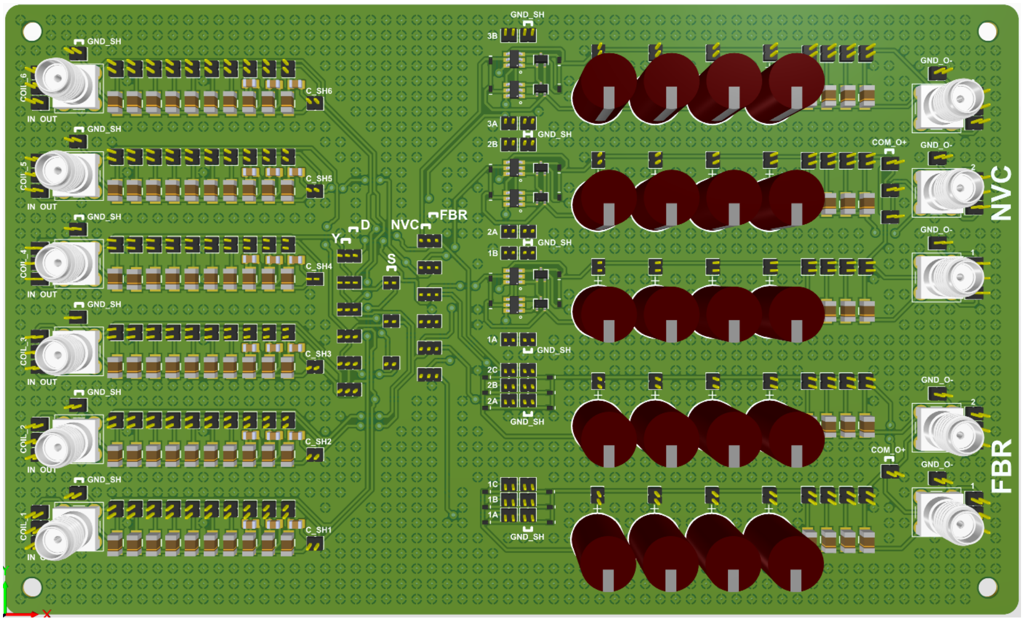
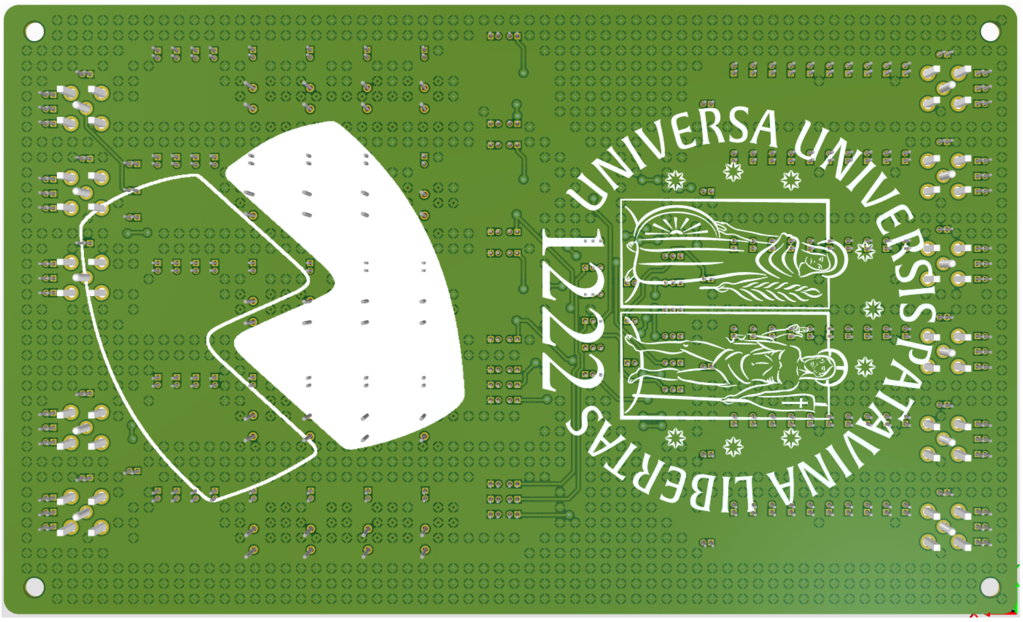


Figure 5.7: Microscope photo of one NVC bridge during hand assembly: the four diodes are already soldered, while the two packages on the right, together holding all four MOSFETs of the bridge, still have unmelted solder on their pins and still need to be soldered.



(a) Top view.



(b) Bottom view.

Figure 5.8: 3D rendering of the prototype board: top (a) and bottom (b) views.

6 Experimental Measurement Setup

6.1 Overview

The circuit simulations of Chapter 4 predict the behaviour of the FWR and NVC stages introduced in Chapter 3, but validating those predictions requires measuring the real prototype. Both the voltage and the current at each phase, and at the output of each rectifying stage, must be acquired at the same time in order to compute the electrical power. This chapter describes the measurement setup built for this purpose, from the acquisition instrument down to the current sensing elements and their calibration.

The sensor calibrations in this chapter rely on two precision instruments to provide known reference values, rather than the Saleae logic analyser of Section 6.3 used for the actual circuit measurements. A Keysight B2901A precision source-measure unit supplies the known reference currents used in Section 6.5. A Keysight 34470A precision digital multimeter, whose built-in four-wire resistance function is used in Section 6.6, provides voltage readings more accurate than the Saleae. Both instruments can be controlled directly from MATLAB, which makes it possible to automate long sequences of measurements, such as the many current-voltage points needed to build the lookup tables of Section 6.5, without manual control at each step.

6.2 Test Bench

The voltages and currents measured throughout this chapter all depend on the harvester's rotation speed, so every measurement must be repeatable at a known, controlled speed. To achieve this, the MP-VREH harvester was mechanically coupled to an induction motor driven by a variable-frequency inverter: the inverter sets the motor's frequency, and so its speed, directly. Figure 6.1 shows the resulting test bench, with the induction motor on the left coupled to the harvester on the right through a flexible shaft coupling. This coupling acts as an interface between the two shafts, whose purpose is to minimise the mechanical vibrations transmitted from the motor to the harvester. These vibrations strongly affect the repeatability of the measurements described in this chapter, keeping them to a minimum is essential. This makes it possible to repeat every measurement in this chapter at each rotation speed of interest (100-400 rpm).

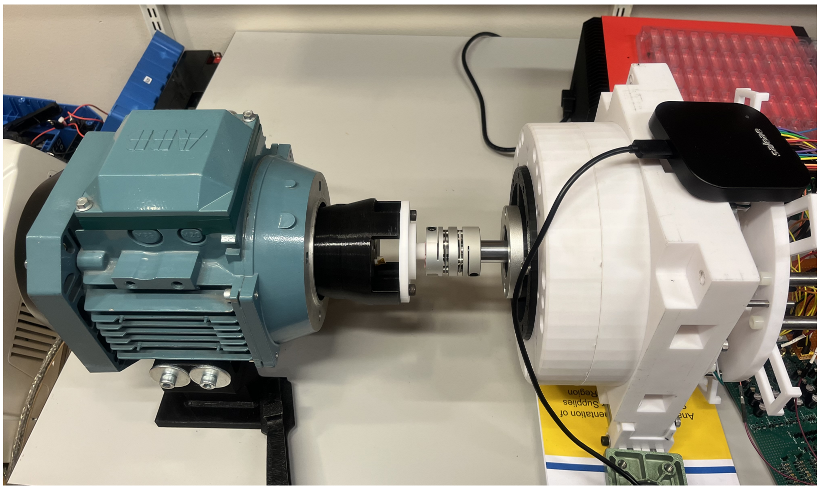


Figure 6.1: Test bench used for all measurements: the induction motor (left) coupled through a shaft coupling to the MP-VREH harvester (right, white housing); the motor speed, and so the harvester's rotation speed, is set by a variable-frequency inverter.

6.3 Data Acquisition System

All signals are acquired with a Saleae logic analyser, shown in Figure 6.2, configured for differential analog acquisition, on 16 channels.

For each generator, both the voltage, measured directly across its own terminals (e.g. across a coil or a rectifier output), and the current, measured by the corresponding current sensor of Section 6.4 inserted in series with that same branch, are acquired differentially, each through its own differential channel. Referencing every channel to a single common ground instead would not only short-circuit several of the measured points, most notably the two terminals of a series shunt resistor, which are not referenced to the same node as the rest of the circuit, but would also unbalance the six otherwise independent phases of the MP-VREH. Because all channels are sampled by the same instrument, the

Figure 6.2: The Saleae logic analyser used for data acquisition.

resulting waveforms are synchronous, which is what makes it possible to pair a phase voltage with its corresponding current and compute instantaneous power.

6.4 Current Sensing: Shunt Resistors and Instrumentation Amplifiers

The Saleae itself only measures voltage, so each current of interest must first be converted into a proportional voltage. This is done with a shunt resistor R_{shunt} inserted in series with the branch to be measured, so that the current $i(t)$ flowing through it produces a voltage drop

$$v_{\text{shunt}}(t) \;=\; R_{\text{shunt}}\, i(t). \tag{6.1}$$

Given the low phase currents expected from the simulations of Chapter 4, $v_{\text{shunt}}(t)$ is small and, since the shunt sits in series within the branch rather than at a fixed reference node, it is not measured with respect to a common ground on either side. A sensing circuit is therefore needed to amplify the small differential drop across the shunt to a level suitable for acquisition.

This role is carried by an INA114AU precision instrumentation amplifier connected across each shunt resistor. Figure 6.3 shows one of the six current-sensing boards actually used, built around the INA114AU, with its input terminals on the top, its gain-setting trimmer in the middle, and its output connector on the right; the sticker on the board records the actual resistance of its own shunt, measured as described in Section 6.6.

The INA senses the two shunt terminals differentially, with high input impedance

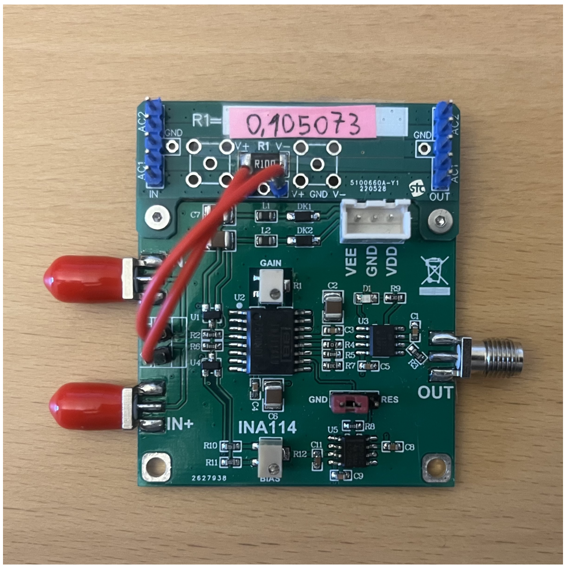


Figure 6.3: One of the six current-sensing boards used in this thesis, built around the INA114AU instrumentation amplifier.

and high common-mode rejection, and delivers an output voltage proportional to the shunt voltage. Its gain G is set by a single external resistor R_G according to

$$G \;=\; 1 + \frac{50\ \mathrm{k}\Omega}{R_G}, \tag{6.2}$$

so that the amplifier output is

$$v_{\text{out}}(t) \;=\; G\, v_{\text{shunt}}(t) \;=\; G\, R_{\text{shunt}}\, i(t). \tag{6.3}$$

The current sensor output voltage is then connected to the corresponding Saleae channel. In principle, the current could then be recovered by simply inverting (6.3), using the nominal values of R_{shunt} and R_G together with (6.2) to compute G; in practice, this is not accurate enough, and an individual calibration, described in Section 6.5, is required for every current sensor.

6.5 Current Sensors Calibration

Equation (6.3) gives the nominal, linear relationship between the sensor output and the current it measures, but a real current sensor does not follow it exactly. Component tolerances and offset voltage make every sensor, comprising its own shunt resistor and INA114AU, different from every other nominally identical sensor, and the response of the amplifier itself is not perfectly linear over the full input range. A single scale factor is therefore not accurate enough to describe a given sensor; instead, each sensor is calibrated individually by building a lookup table that maps its acquired output voltage directly to the true current.

Figure 6.4 plots, for each of the six current sensors of the MP-VREH (CS1 to CS6), the known current driven by the B2901A against the corresponding sensor output voltage. The six sensors use nominally identical shunt resistors and INA114AU amplifiers, but their slopes are clearly different, especially for CS2. A single scale factor could not capture this spread, so each sensor needs its own lookup table.

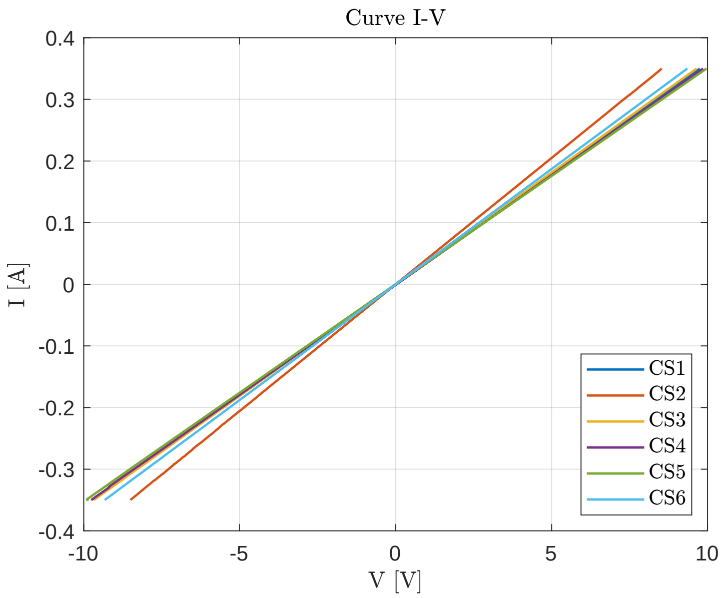


Figure 6.4: Calibration curves of the six current sensors of the MP-VREH, CS1 to CS6.

The calibration consists of driving the shunt resistor with a set of known reference currents, spanning the range expected in operation and simultaneously acquiring the corresponding sensor output voltage. Each (voltage, current) pair obtained this way becomes one entry of that sensor's lookup table. The current corresponding to any voltage acquired later during normal operation is then recovered by interpolating between the nearest entries of this table, rather than by inverting a single linear formula, which is what allows the non-linearity and sensor-to-sensor variation described above to be compensated for. Every current used later in this thesis is recovered from the acquired voltage through this per-sensor lookup table.

The B2901A, however, has a maximum output current lower than the peak phase current found in the simulations of Chapter 4, so it cannot itself reproduce, through the $R_{\text{shunt}} = 0.1\ \Omega$ shunt used on the board, the full range of shunt voltages that shunt produces in operation. The calibration shunt is therefore chosen ten times larger, $R_{\text{cal}} = 1\ \Omega$. What actually sets the operating point of the INA114AU is the shunt voltage of (6.1), not the current itself. With a shunt ten times larger, the same shunt voltage, and so the same amplifier operating point, is reached with a current ten times smaller, well within the B2901A's range.

The lookup table built this way, however, relates the acquired output voltage to the current actually driven through R_{cal} during calibration, not to the current that would flow through R_{shunt} in the circuit at the same shunt voltage. At equal shunt voltage, the two currents are related by the ratio of the two resistances,

$$i_{\text{shunt}} = i_{\text{cal}} \frac{R_{\text{cal}}}{R_{\text{shunt}}}, \tag{6.4}$$

so every entry of the lookup table is rescaled by this ratio before use. Because (6.4) is a direct multiplicative correction, its accuracy depends entirely on how precisely R_{cal} and R_{shunt} are known, and their nominal values are not sufficient for this purpose. Instead, the actual resistance of every calibration and circuit shunt is measured individually, as described in Section 6.6.

6.6 Shunt Resistor Characterisation

The scaling of (6.4) requires the actual resistance of the $R_{\text{cal}} = 1\ \Omega$ and $R_{\text{shunt}} = 0.1\ \Omega$ shunts, rather than their nominal values. At these low resistances, this cannot be obtained with an ordinary two-wire measurement, e.g. with a standard multimeter, which applies its test current and senses the resulting voltage through the same pair of cables. The resistance of these cables and the contact resistance where the probes touch the shunt terminals are then included in the measurement, and for a $0.1\ \Omega$ shunt they are typically comparable in magnitude to the resistance being measured, so a two-wire reading is not accurate enough.

Each shunt is instead characterised with the 34470A's built-in four-wire, or Kelvin, resistance function: the instrument sources a known current I through the shunt via one pair of cables, and senses the resulting voltage drop through a second, separate pair of cables, connected as close as possible to the shunt's own terminals. Because the sensing cables carry a negligible current into the multimeter's high-impedance input, the voltage drop across their own resistance and across the contact resistance at their tips is negligible, and the voltage V it measures is essentially the drop across the shunt element alone. The 34470A then computes the shunt's actual resistance directly from Ohm's law, getting rid of the cable and contact resistance errors that limit the two-wire measurement. This four-wire measurement is applied individually to every calibration shunt and every circuit shunt used in this thesis, and the resulting actual resistances, rather than the nominal $1\ \Omega$ and $0.1\ \Omega$ values, are the ones used in the scaling of (6.4).

A single reading of such a small resistor is still affected by the noise of the 34470A itself, so rather than relying on one measurement, a long sequence of repeated four-wire measurements is acquired automatically using a MATLAB script for every shunt in the system. Figure 6.5 illustrates this with one representative example, 2000 consecutive measurements of one shunt resistor: the top plot gives the measured resistance against acquisition number, together with the mean value and the $\pm 1\sigma$ band, and the bottom plot shows the same 2000 values as a histogram. This scatter is simply noise, which a single reading cannot cancel out. So, for every shunt, the resistance value used in the scaling of (6.4) is not any single acquisition, but the mean of a big sequence of measurements.

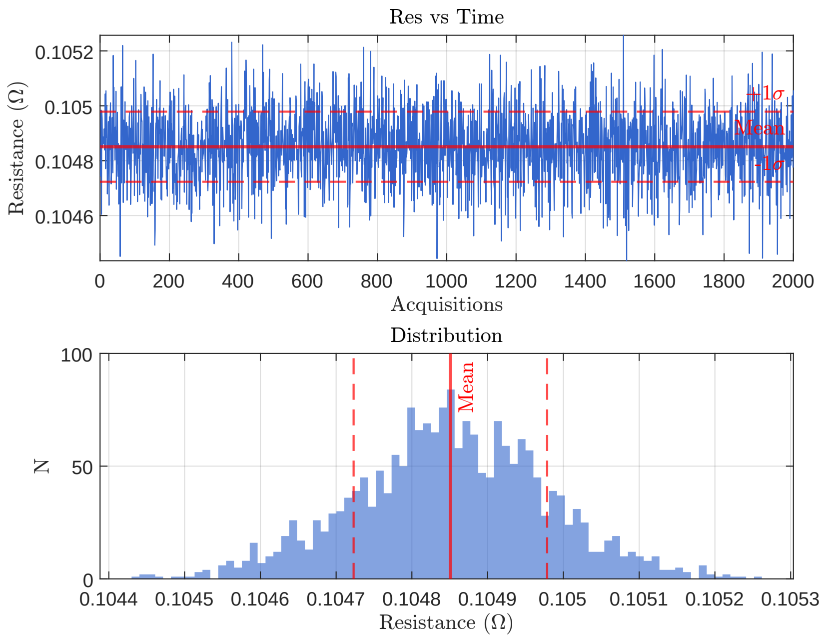


Figure 6.5: 2000 consecutive four-wire measurements of one of the shunt resistors, shown as a representative example of how the repeated measurements are distributed.

6.7 Load Resistor

The load resistance R_L connected to each rectifier output during the measurements of this chapter is set with a custom eight-decade programmable resistor board, shown in Figure 6.6. Each of its eight decades contributes one digit of the total resistance, selected independently with its own jumper; together they span the full range from $0\ \Omega$ to $9\,999\,999\ \Omega$ (about $10\ \mathrm{M}\Omega$) in steps of $0.1\ \Omega$. This lets R_L be set to the exact value required by each measurement, rather than being limited to a fixed set of discrete resistors. In practice, this precision is not really needed for this thesis: R_L is simply swept across a range of steps.

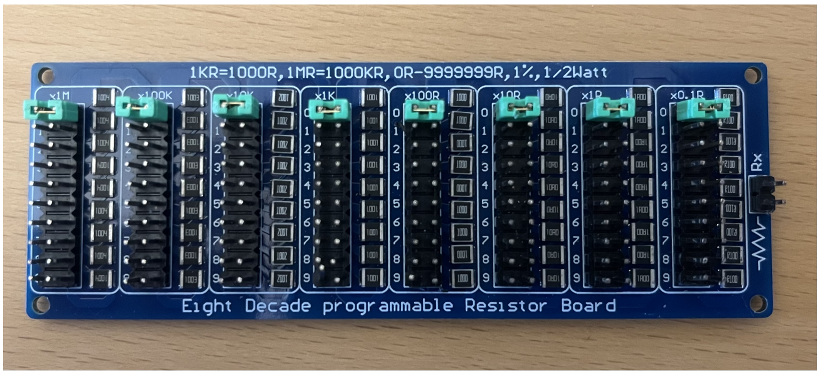


Figure 6.6: The eight-decade programmable resistor board used to set the load resistance R_L, with one jumper per decade selecting a digit from 0 to 9.

7 Experimental Measurement Results

7.1 Overview

This chapter presents the results of applying the measurement setup of Chapter 6 to the physical prototype, following the same experimental structure used for the Simulink simulations of Chapter 4: for each rectifier configuration, the compensation capacitor C_m and the load resistance R_L are first swept independently at 100 rpm, the speed that sets the lower bound on the power available across the target range, before sweeping the resulting configuration across the whole 100-400 rpm range and evaluating its rectifier efficiency. Unlike those simulations, which swept C_m at both ends of the target speed range to identify the two limiting values used in the frequency sweep, the compensation capacitor of the prototype was swept only at 100 rpm; the smaller value needed at the top of the range is instead identified directly within the frequency sweep itself, by comparing several fixed values of C_m across the whole range, as described below. As in Chapter 4, the filter capacitor is fixed at $C_r = 1480$ μF throughout this chapter, a large enough value to keep the output ripple as low as possible.

Throughout this chapter, R_L refers to the physical load resistor set on the bench, not the total resistance loading the bridge. As in the simulations of Section 4.1, the current-sensing chain and the connecting cables add a further series resistance of approximately 1.5 Ω, which is not included in the reported R_L values and must be added to them to obtain the effective load resistance seen by the rectifier.

Where the prototype allows it, the bridges of a configuration are measured not only individually, each loaded by its own resistor, but also already wired together in parallel onto a single shared load, giving a genuinely combined measurement rather than the sum of two or three independent ones. Both the individual and the combined figures are shown wherever available.

7.2 Full-Wave Rectifier (FWR)

7.2.a Star Connection

Compensation Capacitor (C_m) Sweep

With the load fixed at $R_L = 11\ \Omega$, C_m was swept at 100 rpm for both bridges of the star connection. Figure 7.1 shows the output power of FWR1 rising from 17.23 mW at $C_m = 330\ \mu$F to a broad maximum extending from about 530 to 730 μF, peaking at 20.03 mW at $C_m = 630\ \mu$F. FWR2 starts lower, at 15.40 mW, and rises more steeply, overtaking FWR1 around $C_m = 580\ \mu$F and peaking at the same $C_m = 630\ \mu$F (20.72 mW), before decreasing again alongside FWR1. The combined power of the two bridges therefore also peaks at $C_m = 630\ \mu$F (40.75 mW).

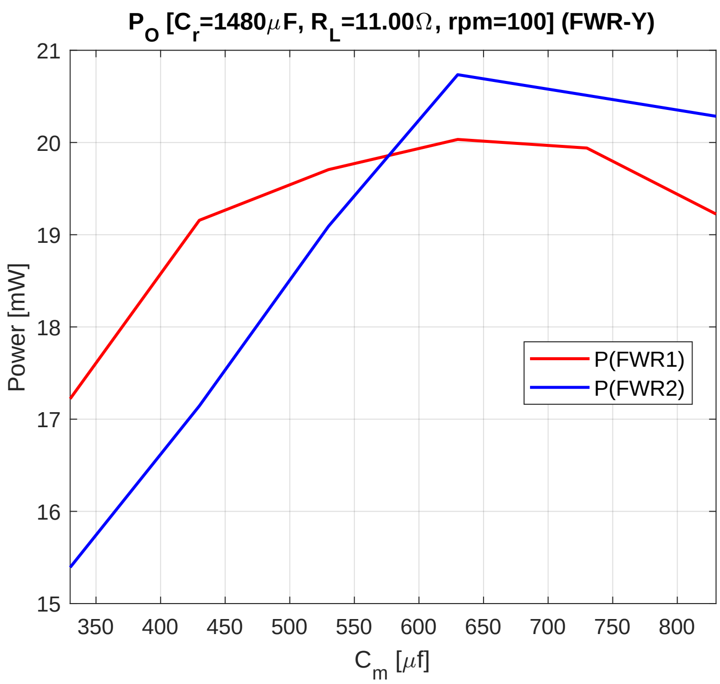


Figure 7.1: Output power of the star-connected first and second bridges as a function of the compensation capacitor C_m, measured at 100 rpm ($C_r = 1480\ \mu$F, $R_L = 11\ \Omega$).

Load Resistance (R_L) Sweep

With $C_m = 630\ \mu$F fixed, the load resistance of each bridge was swept individually at 100 rpm. Figure 7.2 shows both bridges peaking at $R_L = 16\ \Omega$ (21.24 mW for FWR1, 21.78 mW for FWR2), with a broad plateau extending from about 13 to 17-19 Ω depending on the bridge; $R_L = 16\ \Omega$ was chosen for each individually-loaded bridge, and is the value used in the frequency sweep and efficiency evaluation below.

The two bridges were then connected together onto a single shared load and swept again, over the narrower range expected for two similar loads in parallel. Figure 7.3 shows this combined measurement peaking at $R_L = 7\ \Omega$ (48.36 mW), with a plateau again extending from about 6 to 8 Ω. $R_L = 7\ \Omega$ is the value used whenever the two bridges are measured already connected in parallel, in the frequency sweep below.

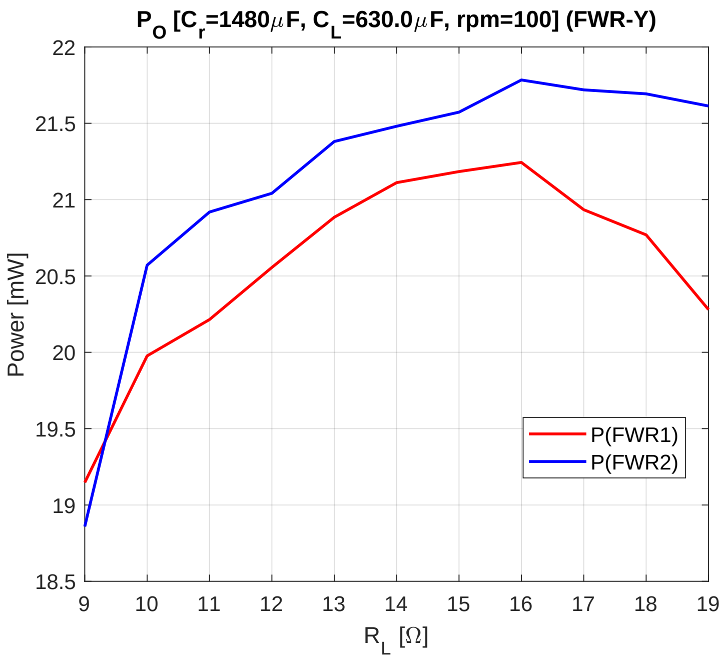


Figure 7.2: Output power of the star-connected first and second bridges, each individually loaded, as a function of its own load resistance R_L, measured at 100 rpm ($C_r = 1480\ \mu$F, $C_m = 630\ \mu$F).

Frequency Sweep

Figures 7.4 and 7.5 sweep seven fixed values of C_m, from 630 μF down to 47 μF, across the whole target speed range, with $R_L = 16\ \Omega$ fixed for each individually-loaded bridge. At 100 rpm, $C_m = 630\ \mu$F delivers the most power for both

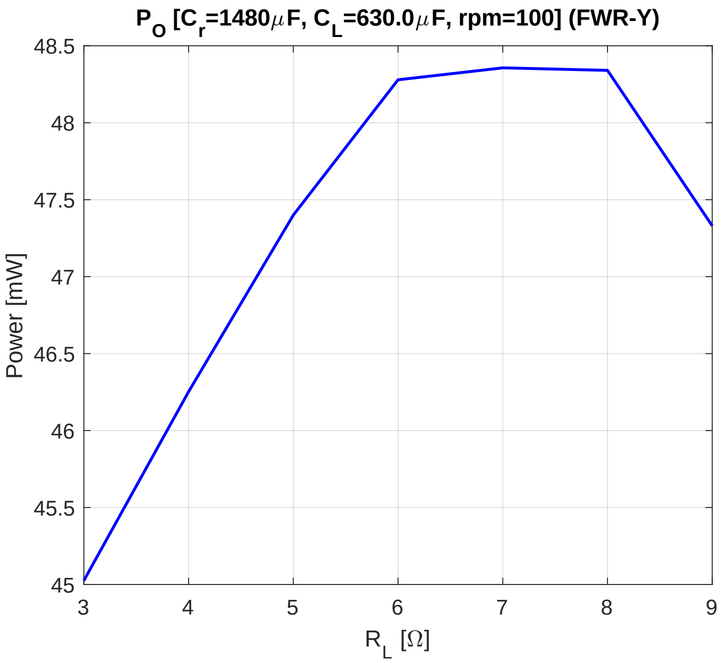


Figure 7.3: Output power of the star-connected first and second bridges, already connected in parallel onto a single shared load, as a function of that load resistance R_L, measured at 100 rpm ($C_r = 1480$ μF, $C_m = 630$ μF).

bridges, consistent with the compensation capacitor sweep above, while $C_m = 47$ μF performs the worst. As speed increases, the best value of C_m keeps getting smaller in a few discrete steps: for FWR1 it falls from 630 μF at 100 rpm down to 47 μF by 350-400 rpm, reaching 292.5 mW at the top of the range. FWR2 shows the same trend and also ends at $C_m = 47$ μF, reaching 327.6 mW at 400 rpm. As in the simulations of Section 4.2.a, no single value of C_m is therefore best across the whole range.

With the two bridges already connected in parallel onto the shared $R_L = 7$ Ω load found above, only the two extreme values of C_m were remeasured across the speed range. Figure 7.6 shows the same crossover seen for the individual bridges: $C_m = 630$ μF gives more power than $C_m = 47$ μF from 100 to 250 rpm (47.6 against 8.7 mW at 100 rpm, narrowing to 245.1 against 199.6 mW at 250 rpm). After that, $C_m = 47$ μF takes the lead and stays ahead for the rest of the range, reaching 677.2 mW against 433.3 mW at 400 rpm.

Efficiency

Figures 7.7 and 7.8 show the rectifier efficiency $\eta_{\text{Rec}} = P_{\text{out}}/P_{\text{in}}$ of each individually-loaded bridge, for the same seven values of C_m, across the speed range. As in the

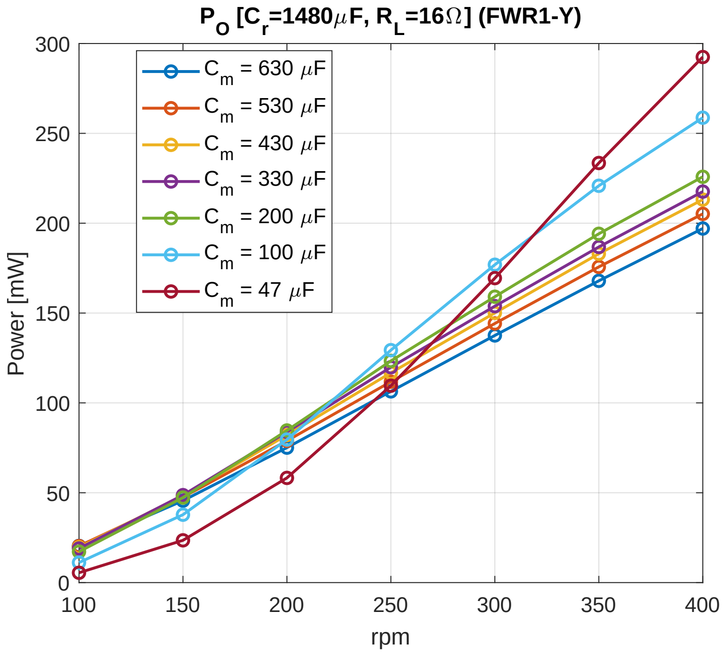


Figure 7.4: Output power of the first bridge (FWR1) of the star connection across the target speed range, for seven fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu\text{F}$, $R_L = 16\ \Omega$).

simulations, no DC-DC converter stage is included here, so η_{Rec} coincides with the full PCE of (1.1) once such a stage is added. Both bridges show the same qualitative behaviour: η_{Rec} is lowest for the smallest capacitor and highest for the largest one at 100 rpm, ranging from 0.275 to 0.408 for FWR1 and from 0.235 to 0.413 for FWR2; the seven curves then converge by 400 rpm to a narrow band, 0.548-0.567 for FWR1 and 0.52-0.57 for FWR2. The measured values at 100 rpm sit close to the $\eta_{\text{Rec}} \approx 0.20$-$0.43$ found in simulation (Section 4.2.a), but the values measured at 400 rpm are noticeably lower than the simulated $\eta_{\text{Rec}} \approx 0.70$-$0.76$.

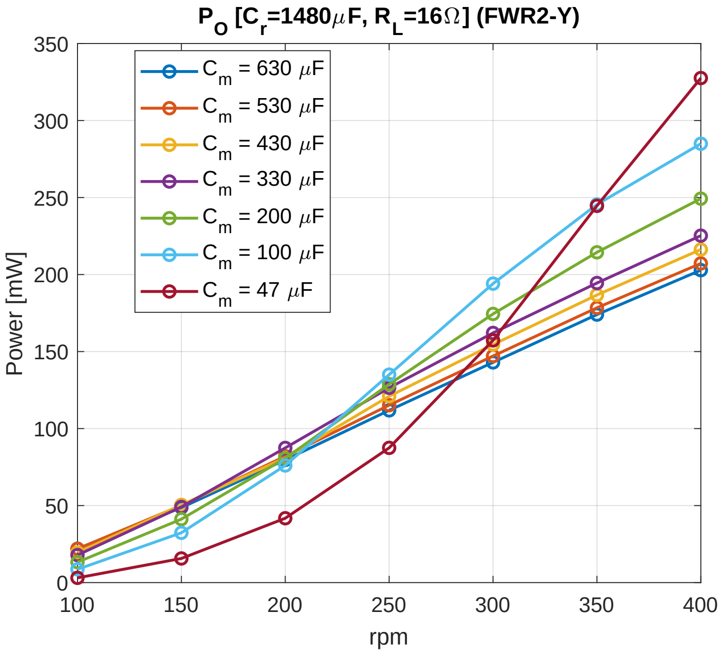


Figure 7.5: Output power of the second bridge (FWR2) of the star connection across the target speed range, for seven fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 16\ \Omega$).

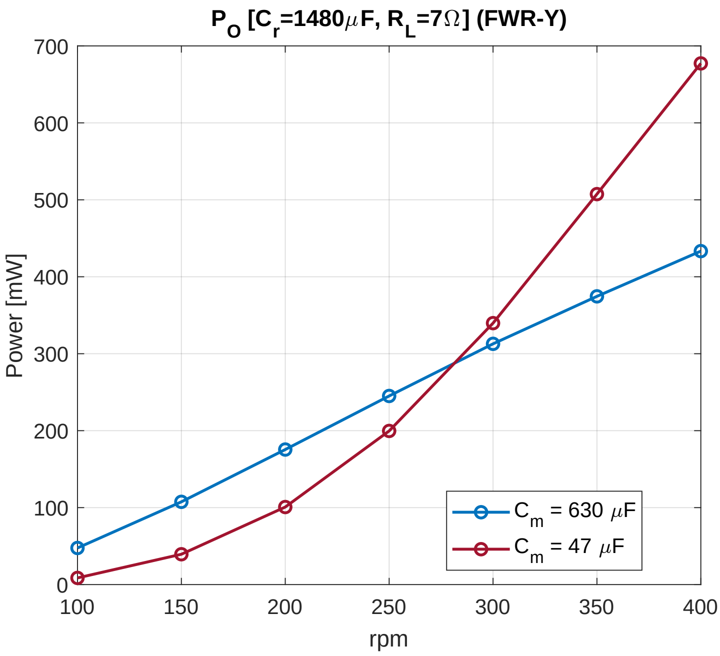


Figure 7.6: Combined output power of the star-connected bridges, already connected in parallel, across the target speed range, for the two extreme values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 7\ \Omega$).

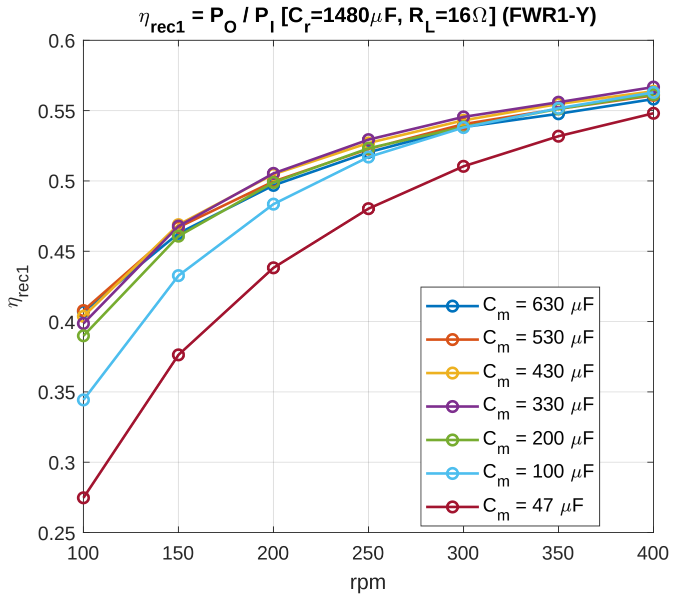


Figure 7.7: Rectifier efficiency η_{Rec} of the first bridge (FWR1) of the star connection across the target speed range, for seven fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 16\ \Omega$).

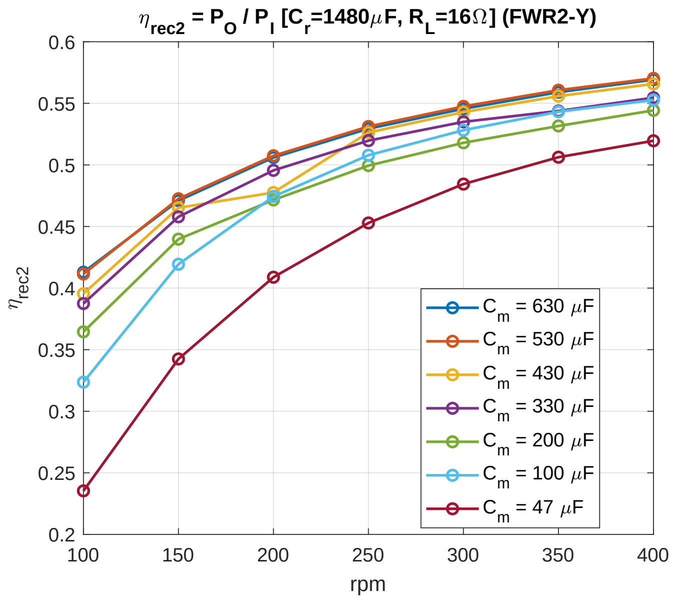


Figure 7.8: Rectifier efficiency η_{Rec} of the second bridge (FWR2) of the star connection across the target speed range, for seven fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 16\ \Omega$).

7.2.b Delta Connection

Compensation Capacitor (C_m) Sweep

With $R_L = 5\ \Omega$ fixed, C_m was swept at 100 rpm for both bridges of the delta connection. Figure 7.9 shows FWR1 peaking at $C_m = 630\ \mu$F (8.92 mW), with a plateau extending from about 530 to 630 μF, and FWR2 peaking at the same value (7.97 mW), with its own plateau extending slightly further, to about 730 μF. As with the star connection, both bridges share the same peak location here, and the combined power peaks at $C_m = 630\ \mu$F (16.89 mW), which was chosen and carried forward to the load resistance sweep below.

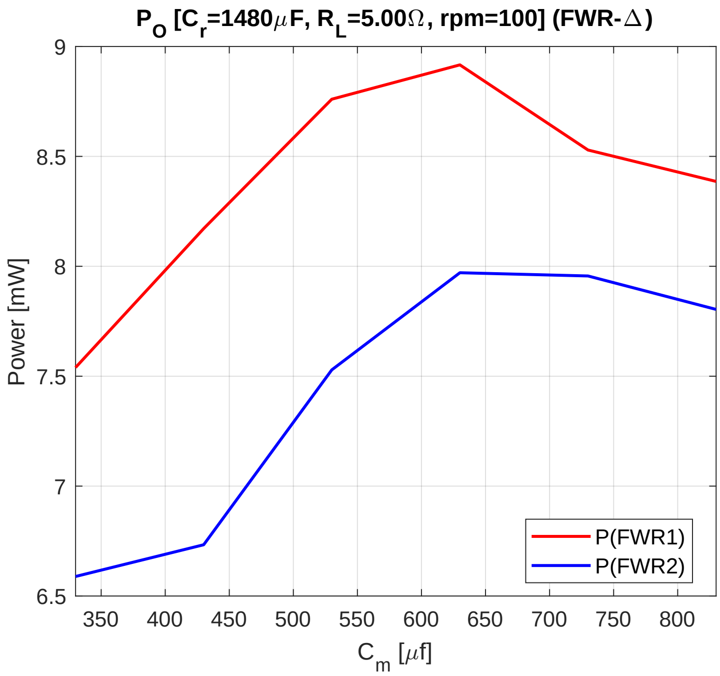


Figure 7.9: Output power of the delta-connected first and second bridges as a function of the compensation capacitor C_m, measured at 100 rpm ($C_r = 1480\ \mu$F, $R_L = 5\ \Omega$).

Load Resistance (R_L) Sweep

With $C_m = 630\ \mu$F fixed, the load resistance of each bridge was swept individually at 100 rpm. Figure 7.10 shows both bridges peaking at $R_L = 7\ \Omega$ (9.14 mW for FWR1, 7.97 mW for FWR2).

The two bridges were then connected together onto a single shared load. Figure 7.11 shows this combined measurement peaking at $R_L = 4\ \Omega$ (14.48 mW), close to half the 7 Ω found for each bridge individually, with a plateau extending from about 3 to 5 Ω; $R_L = 5\ \Omega$, within this plateau, is the value used whenever the two bridges are measured already connected in parallel, in the frequency sweep below.

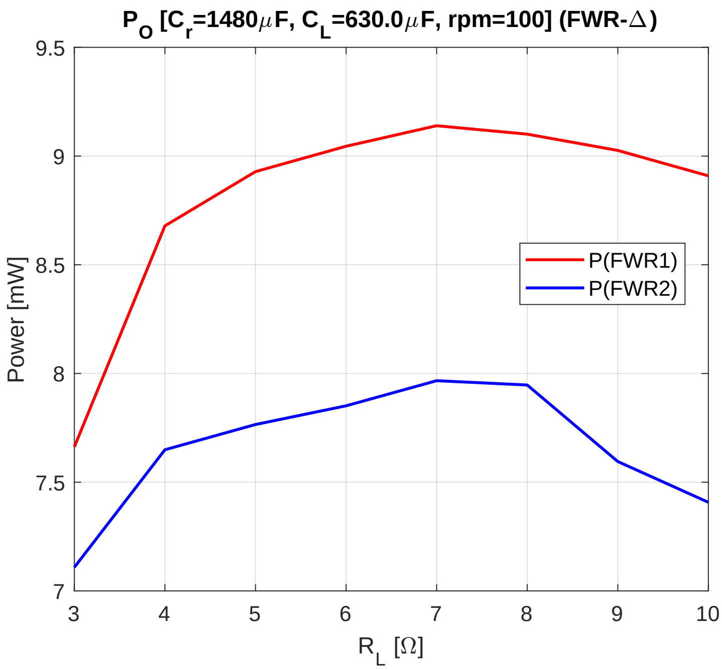


Figure 7.10: Output power of the delta-connected first and second bridges, each individually loaded, as a function of its own load resistance R_L, measured at 100 rpm ($C_r = 1480\ \mu$F, $C_m = 630\ \mu$F).

Frequency Sweep

Figures 7.12 and 7.13 sweep the same seven fixed values of C_m across the whole target speed range, with $R_L = 7\ \Omega$ fixed for each individually-loaded bridge. As speed increases, the best value of C_m again drops in steps. FWR1 goes from 630 μF down to 47 μF by 400 rpm, reaching 201.3 mW. FWR2 needs only a single step, from 630 μF to 330 μF at 150 rpm, reaching 196.1 mW at 400 rpm. FWR2's behaviour in Figure 7.13 is somewhat irregular compared to FWR1's, which may be explained by non-idealities such as an imperfect phase shift between the coils and a general imbalance across the six-phase system.

With the two bridges already connected in parallel onto the shared $R_L = 5\ \Omega$ load, Figure 7.14 again compares only the two extreme values of C_m. $C_m = 630\ \mu$F leads from 100 to 250 rpm, reaching 14.9 mW against 6.5 mW at 100 rpm, after which $C_m = 47\ \mu$F overtakes it and pulls ahead, reaching 368.2 mW against 310.7 mW at 400 rpm, the same crossover speed already seen for the star con-

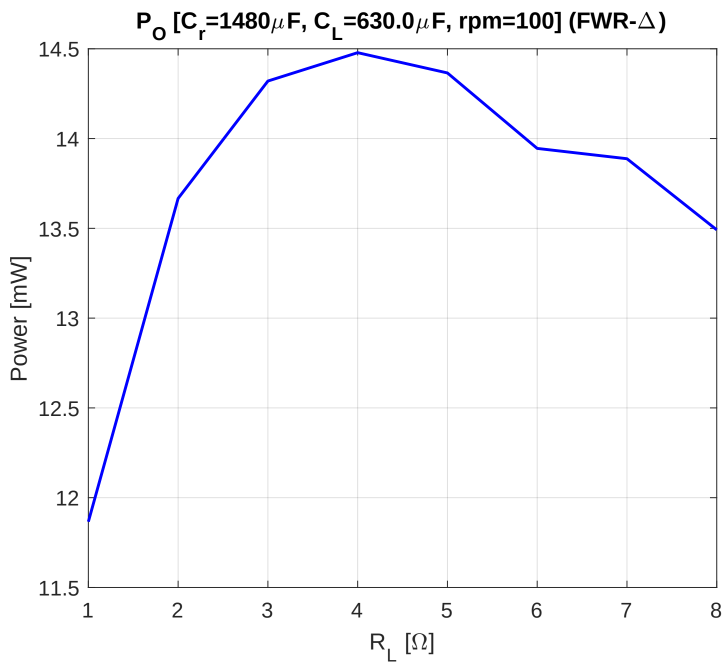


Figure 7.11: Output power of the delta-connected first and second bridges, already connected in parallel onto a single shared load, as a function of that load resistance R_L, measured at 100 rpm ($C_r = 1480$ μF, $C_m = 630$ μF).

nection.

Efficiency

Figures 7.15 and 7.16 show the rectifier efficiency of each individually-loaded bridge for the same seven values of C_m. At 100 rpm, η_{Rec} ranges from 0.182 to 0.301 for FWR1 and from 0.157 to 0.315 for FWR2; both ranges converge by 400 rpm to a narrow band, 0.484-0.509 for FWR1 and 0.482-0.512 for FWR2. These values are lower than the star connection's at both ends of the range (for example, 0.235-0.413 at 100 rpm, 0.52-0.57 at 400 rpm), consistent with Section 3.2: without the star connection's $\sqrt{3}$ voltage gain, the fixed two-diode drop loss of the FWR removes a larger fraction of the lower delta line voltage.

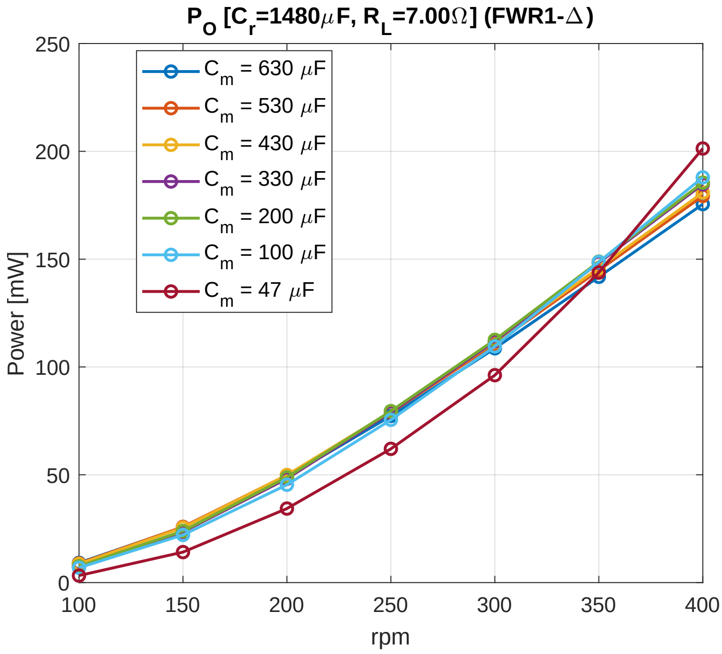


Figure 7.12: Output power of the first bridge (FWR1) of the delta connection across the target speed range, for seven fixed values of the compensation capacitor C_m ($C_r = 1480\,\mu$F, $R_L = 7\,\Omega$).

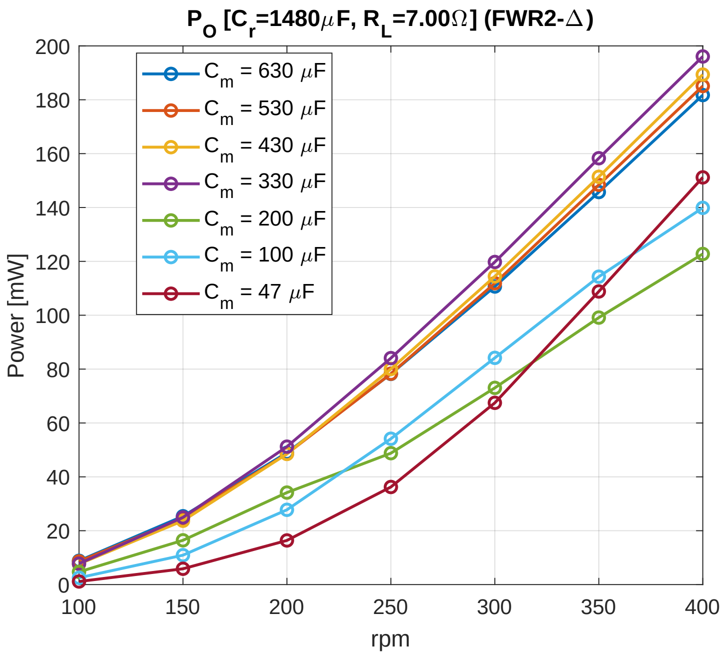


Figure 7.13: Output power of the second bridge (FWR2) of the delta connection across the target speed range, for seven fixed values of the compensation capacitor C_m ($C_r = 1480\,\mu$F, $R_L = 7\,\Omega$).

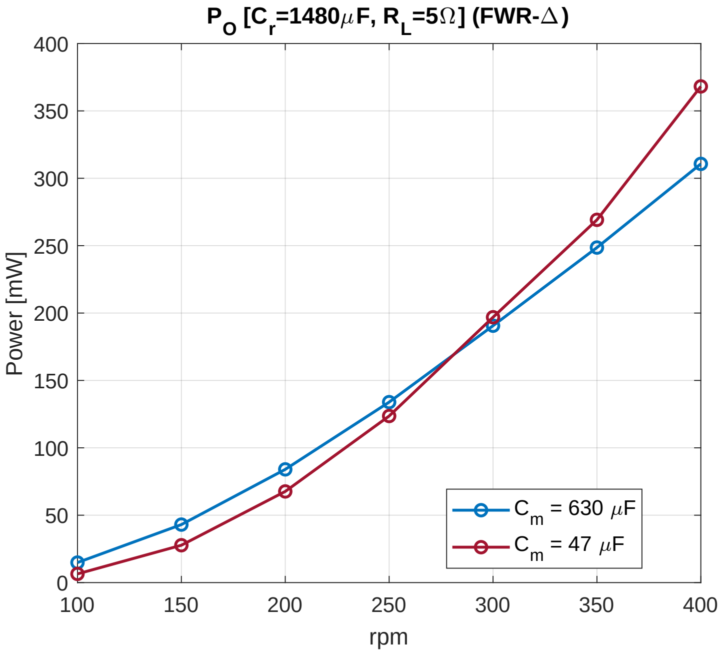


Figure 7.14: Combined output power of the delta-connected bridges, already connected in parallel, across the target speed range, for the two extreme values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 5\ \Omega$).

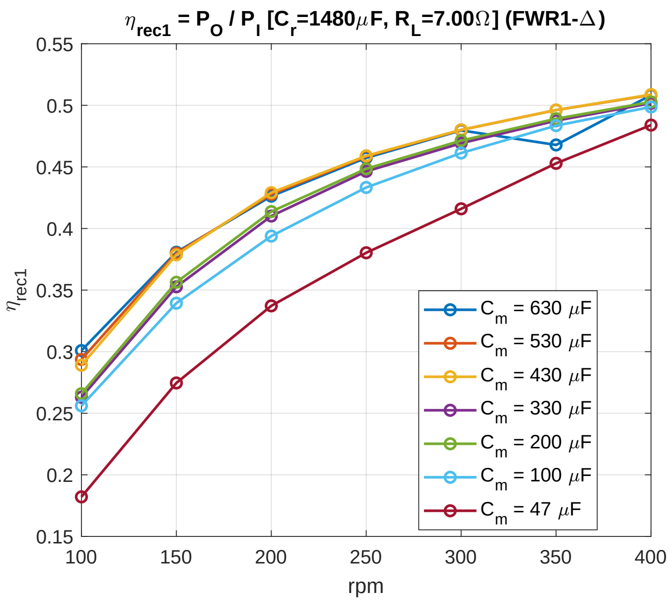


Figure 7.15: Rectifier efficiency η_{Rec} of the first bridge (FWR1) of the delta connection across the target speed range, for seven fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 7\ \Omega$).

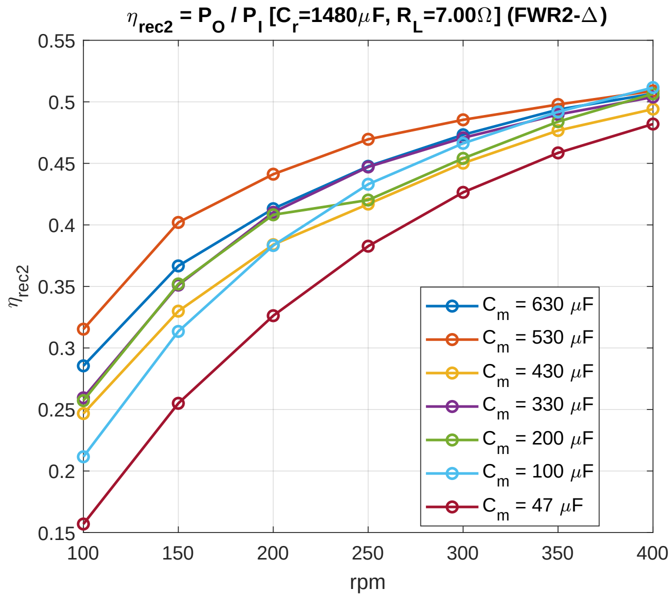


Figure 7.16: Rectifier efficiency η_{Rec} of the second bridge (FWR2) of the delta connection across the target speed range, for seven fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 7\ \Omega$).

7.3 Negative Voltage Converter (NVC)

7.3.a Compensation Capacitor (C_m) Sweep

With $R_L = 20\ \Omega$ fixed for each of the three bridges, C_m was swept at 100 rpm. Figure 7.17 shows all three bridges rising steeply for small C_m and peaking together around $C_m = 430\ \mu$F: NVC1 reaches 21.88 mW, NVC2 20.80 mW, and NVC3, delivering noticeably less than the other two, 17.96 mW, about 18% less than NVC1 and 14% less than NVC2 at the peak, closely matching the gap already seen in simulation (Section 4.3.a). All three bridges then decrease only very gently for larger C_m, NVC3 in particular staying within a few percent of its peak all the way to $C_m = 830\ \mu$F. The combined power of the three bridges peaks at the same value, $C_m = 430\ \mu$F (60.65 mW), which was chosen and carried forward to the load resistance sweep below.

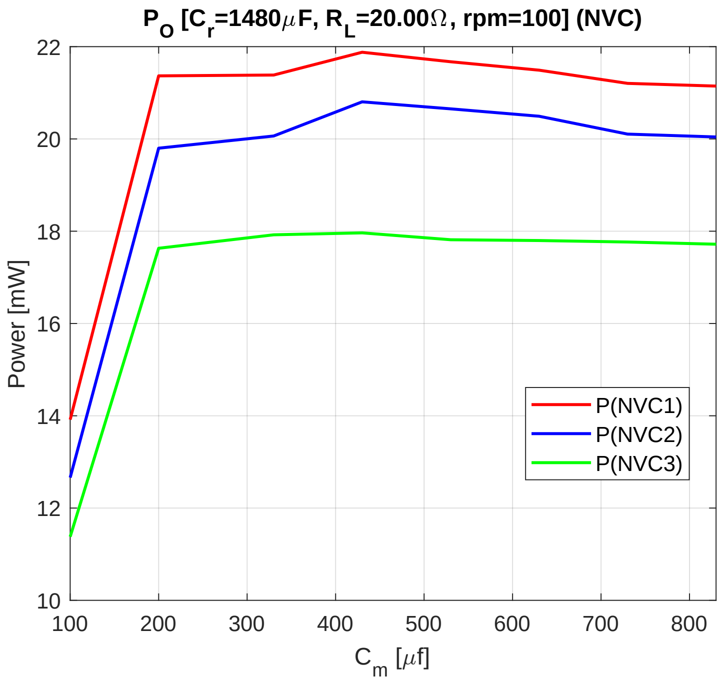


Figure 7.17: Output power of the three NVC bridges as a function of the compensation capacitor C_m, measured at 100 rpm ($C_r = 1480\ \mu$F, $R_L = 20\ \Omega$).

7.3.b Load Resistance (R_L) Sweep

With $C_m = 430\ \mu$F fixed, the load resistance of each bridge was swept individually at 100 rpm. Figure 7.18 shows all three bridges peaking within a few ohms of each other, NVC1 at $R_L = 24\ \Omega$ (21.86 mW), NVC2 and NVC3 both at $R_L = 21\ \Omega$ (20.71 mW and 17.30 mW respectively). The three bridges were then connected together onto a single shared load and swept again. Figure 7.19 shows this combined measurement peaking at $R_L = 6\ \Omega$ (589.6 mW). $R_L = 6\ \Omega$ is the value used whenever the three bridges are measured already connected in parallel, in the frequency sweep below. $R_L = 21\ \Omega$ was chosen for NVC2 and NVC3, and $R_L = 24\ \Omega$ for NVC1, for the individually-loaded measurements used in the same frequency sweep.

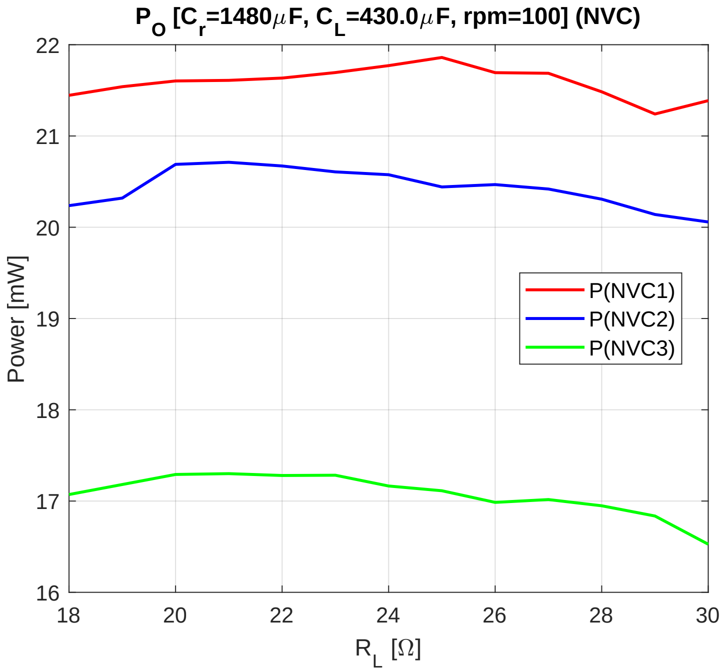


Figure 7.18: Output power of the three NVC bridges, each individually loaded, as a function of its own load resistance R_L, measured at 100 rpm ($C_r = 1480\ \mu$F, $C_m = 430\ \mu$F).

7.3.c Frequency Sweep

Figures 7.20-7.22 sweep five fixed values of C_m, from 430 μF down to 47 μF, across the whole target speed range, with $R_L = 24\ \Omega$, 21 Ω and 21 Ω fixed for NVC1, NVC2 and NVC3 respectively. As with the FWR bridges, the best-performing value of C_m shifts in stages as speed increases, rather than through a single crossover. NVC1 and NVC2 follow a similar progression, handing the lead from $C_m = 430\ \mu$F at 100 rpm, to 200 μF at 150 rpm, to 100 μF at 200 rpm (NVC2 already reaches $C_m = 47\ \mu$F by 250 rpm, one step ahead of NVC1, which does so only from 300 rpm), reaching 287.8 mW and 210.5 mW respectively at 400 rpm. NVC3 is different: it stays at $C_m = 100\ \mu$F from 150 rpm onward and

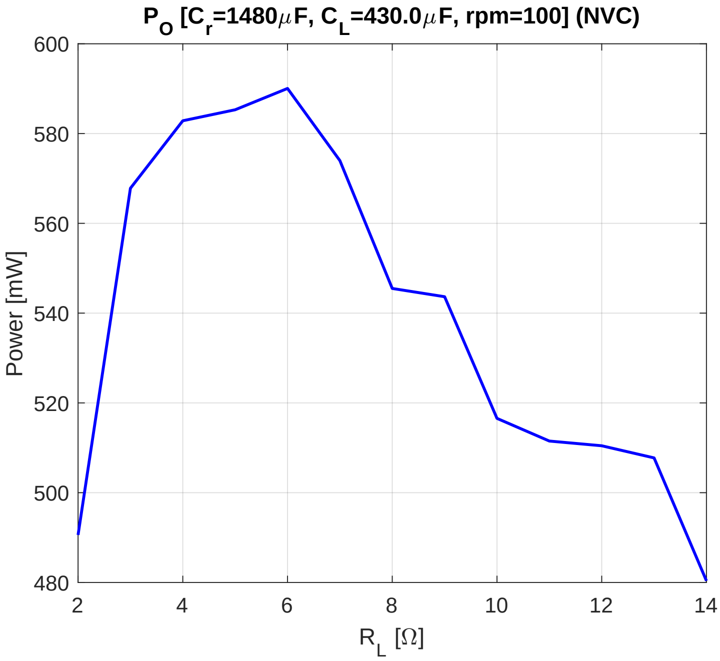


Figure 7.19: Output power of the three NVC bridges, already connected in parallel onto a single shared load, as a function of that load resistance R_L, measured at 100 rpm ($C_r = 1480$ μF, $C_m = 430$ μF).

never moves to $C_m = 47$ μF, reaching 174.0 mW at 400 rpm, compared to only 151.8 mW for $C_m = 47$ μF. As with FWR2's unusual behaviour in Section 7.2.b, this is probably another sign that the six-phase system is not perfectly balanced, likely because of small mechanical differences between the phases that then affect the electrical behaviour.

With the three bridges already connected in parallel onto a shared $R_L = 6$ Ω load, Figure 7.23 shows the same kind of behaviour in the combined power. $C_m = 430$ μF gives the most power at 100 rpm (57.7 mW). $C_m = 200$ μF then leads briefly at 150 rpm (121.6 mW), followed by $C_m = 100$ μF from 200 to 250 rpm (302.7 mW, barely ahead of $C_m = 47$ μF with 300.6 mW). From there on, $C_m = 47$ μF gives the most power, reaching 603.0 mW at 400 rpm against 511.1 mW for $C_m = 100$ μF. As with the FWR bridges, then, no single value of C_m works best across the whole range. $C_m = 430$ μF is still the value used, since it keeps the power as high as possible at the low end of the speed range.

7.3.d Efficiency

Figure 7.24 shows the rectifier efficiency of the three NVC bridges combined, already connected in parallel, for the same five values of C_m. At 100 rpm, η_{Rec}

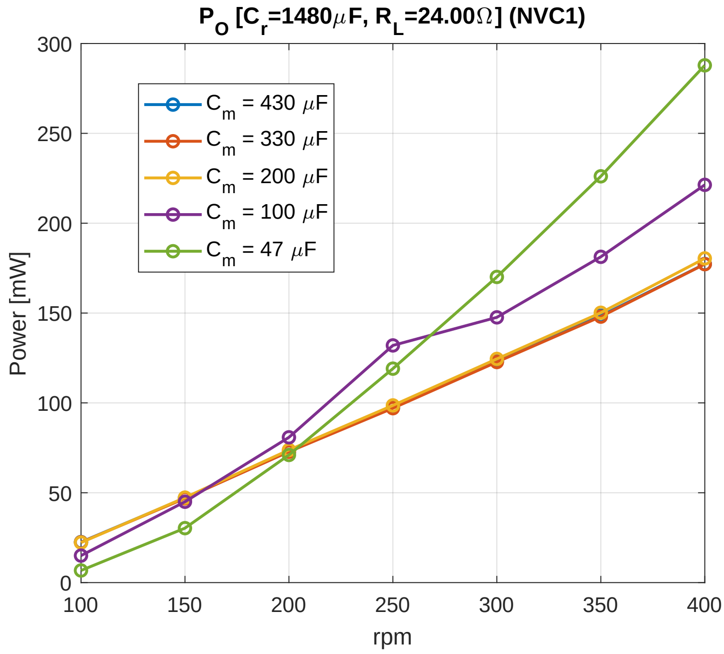


Figure 7.20: Output power of NVC1 across the target speed range, for five fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 24\ \Omega$).

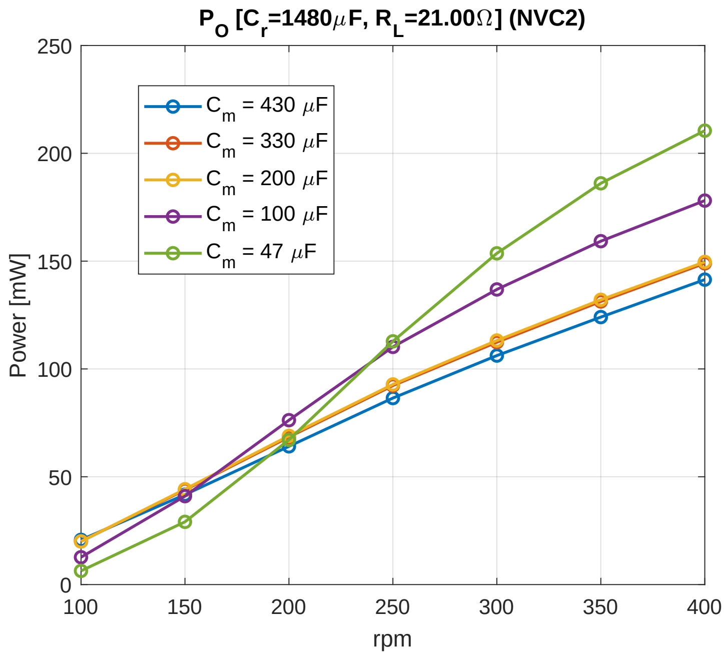


Figure 7.21: Output power of NVC2 across the target speed range, for five fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 21\ \Omega$).

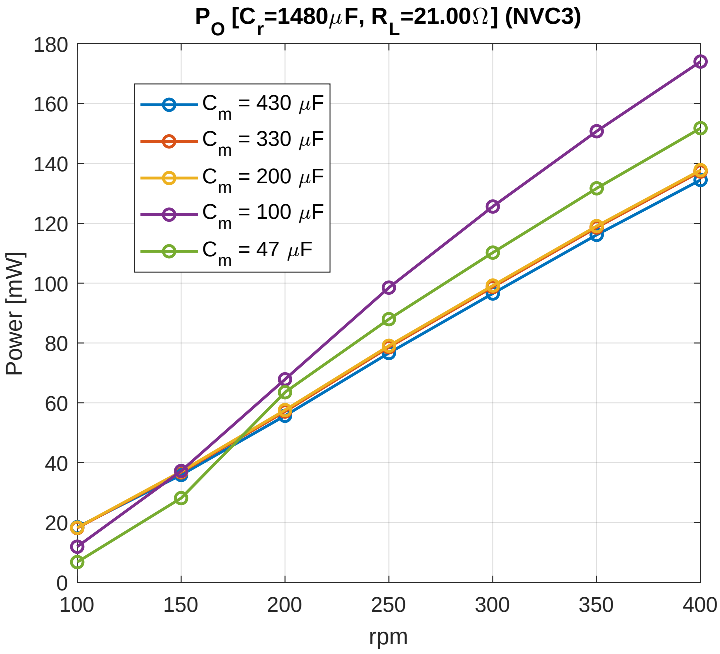


Figure 7.22: Output power of NVC3 across the target speed range, for five fixed values of the compensation capacitor C_m ($C_r = 1480$ μF, $R_L = 21$ Ω).

ranges from 0.461 for the smallest capacitor ($C_m = 47$ μF) to 0.568 for the largest one ($C_m = 430$ μF). At 400 rpm the five curves converge to a narrower band, $\eta_{\text{Rec}} \approx 0.663$-$0.701$. As with the FWR bridges, the curves are ordered consistently with C_m at 100 rpm and converge at high speed. The values measured at 100 rpm sit close to the $\eta_{\text{Rec}} \approx 0.24$-$0.67$ found in simulation (Section 4.3.d), while those measured at 400 rpm (0.663-0.701) fall short of the simulated $\eta_{\text{Rec}} \approx 0.76$-$0.83$.

7.4 Comparison Between Topologies

Sections 7.2 and 7.3 evaluate the FWR (star and delta connections) and the NVC independently, each with its own compensation capacitor and load resistance chosen to maximise the output power at 100 rpm and then kept fixed across the whole target speed range. All three configurations rectify the same six pickup units, only regrouped differently, so their measured output power and efficiency can be compared directly. Table 7.1 summarises the resulting operating points and performance of the three configurations at the two ends of the target speed range. Output power is taken from the bridges already connected in parallel onto a single shared load, the real combined measurement available for all three configurations; rectifier efficiency, for which this real combined measurement

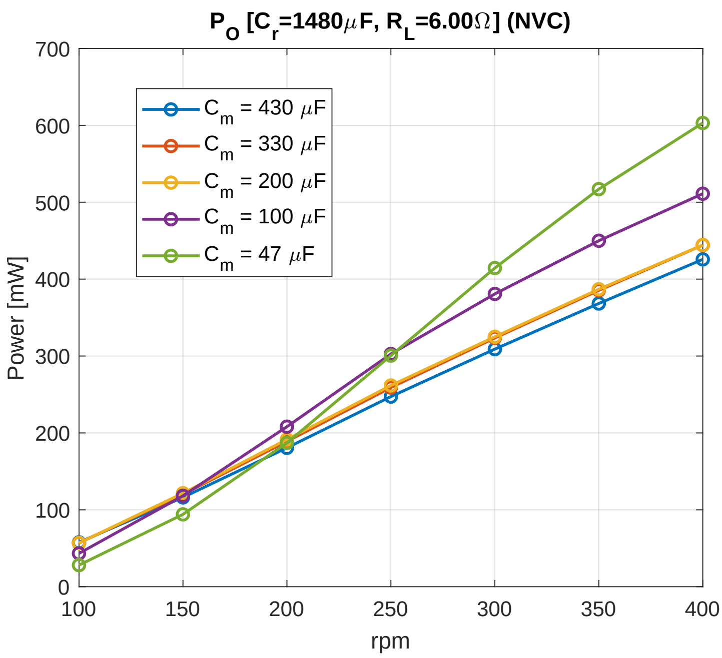


Figure 7.23: Combined output power of the three NVC bridges, already connected in parallel, across the target speed range, for five fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 6\ \Omega$).

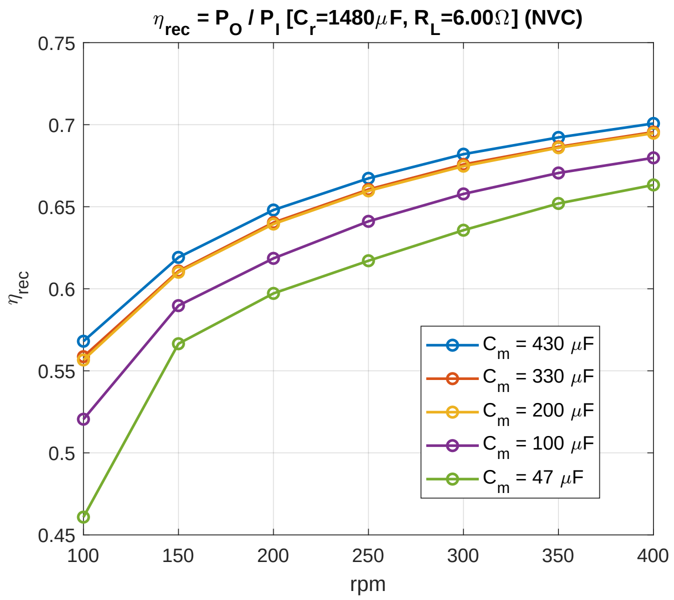


Figure 7.24: Rectifier efficiency η_{Rec} of the three NVC bridges combined, already connected in parallel, across the target speed range, for five fixed values of the compensation capacitor C_m ($C_r = 1480\ \mu$F, $R_L = 6\ \Omega$).

does not include the input power needed to compute it for the FWR connections, is instead taken as the mean value of the individual efficiency of each bridge, each loaded with its own optimal R_L (Sections 7.2.a-7.3).

	FWR (Star)	FWR (Delta)	NVC
C_m (μF)	630	630	430
R_L (Ω)	7	5	6
P_O at 100 rpm (mW)	47.6	14.9	57.7
P_O at 400 rpm (mW)	433.3	310.7	425.7
η_{Rec} at 100 rpm	0.410	0.308	0.568
η_{Rec} at 400 rpm	0.564	0.488	0.701

Table 7.1: Comparison of the three rectifier configurations, each at the compensation capacitor and load resistance found to maximise its output power at 100 rpm and kept fixed across the target speed range. R_L, and the output power it is measured at, refer to the bridges already connected in parallel onto a shared load; η_{Rec} is computed from the individually-loaded bridges instead (Sections 7.2.a-7.3).

At 100 rpm, the NVC delivers the most output power of the three configurations (57.7 mW), slightly ahead of the star connection (47.6 mW) and close to four times the delta connection (14.9 mW). At 400 rpm, the star connection takes the lead instead (433.3 against 425.7 mW for the NVC, under 2% apart), while the delta connection stays clearly behind both (310.7 mW). The star connection beats the delta connection at both speeds: its line voltage is $\sqrt{3}$ times higher, which more than makes up for its lower current, so the FWR's fixed two-diode drop removes a smaller share of it.

The rectifier efficiency follows a different ranking: the NVC leads by a wide margin at both speeds (0.568, 0.701), followed by the star connection (0.410, 0.534) and then the delta connection (0.308, 0.488). This follows the same two-diode drop mechanism discussed in Section 3.3: the NVC replaces the diode drop with the much smaller MOSFET on-resistance once its anti-series pair exceeds $V_{\text{off}} \approx 1.2$ V, giving it a real advantage at both ends of the range. This efficiency advantage does not carry over into output power at high speed, though: there, the star connection is the best performer instead, if only by a small margin over the NVC. Overall, the star connection and the NVC emerge as the two strongest candidates: the NVC for its higher efficiency, the star connection for its slightly higher power at high speed and its much simpler, diode-only circuit. The delta connection is the weakest of the three at every speed and in every metric. Chapter 8 relates these measured figures to the simulations of Chapter 4.

8 Simulation and Measurement Comparison

8.1 Overview

This chapter compares the Simulink simulations of Chapter 4 with the measurements taken on the physical prototype in Chapter 7, across the whole target speed range. For each of the three rectifier configurations, both the simulated and the measured frequency sweeps are taken at the compensation capacitor found to maximise the output power at 100 rpm and then kept fixed for the whole range (Sections 4.2-4.3, 7.2-7.3). The comparison in this chapter therefore reuses the frequency-sweep figures already presented there and summarises them in a table for each configuration.

The simulations always model each bridge of a configuration independently, loaded by its own resistor, and never a genuine electrical connection among bridges; because the simulated circuit is ideal, however, summing these independently simulated bridge outputs gives exactly the same result as connecting the same ideal bridges together onto a shared load. The simulated sum therefore already stands in for a simulated combined connection, and the fairest measured counterpart to it is the real combined measurement of Section 7.4: the bridges of each configuration already wired together in parallel onto a single shared load. This is the quantity used throughout this chapter, for both the output power and the rectifier efficiency. For the FWR star and delta connections, this real combined measurement is only available across the whole speed range for the two extreme values of C_m (Sections 7.2.a, 7.2.b); the rectifier efficiency of these two combined bridges was not recorded, so it is instead taken as the mean of the two individually-loaded bridges, each at its own optimal R_L, across the same speed range.

The compensation capacitor and the load resistance are each compared at their own chosen value too, rather than a common one shared between simulation and measurement, since each was found independently: simulation from the sweeps of Chapter 4, measurement from the sweeps of Chapter 7.

8.2 Full-Wave Rectifier (FWR)

8.2.a Star Connection

The measured compensation capacitor is $630\ \mu$F, well above the $490\ \mu$F found in simulation. Table 8.1 compares the resulting output power and rectifier efficiency across the target speed range, taken from Figures 4.8, 4.9, 7.6, 7.7 and 7.8.

rpm	P_O (mW)		η_{Rec}	
	Simulated	Measured	Simulated	Measured
100	36.7	47.6	0.430	0.423
150	94.3	107.5	0.541	0.475
200	165.8	175.6	0.608	0.507
250	240.0	245.1	0.650	0.529
300	308.8	312.7	0.674	0.544
350	372.3	374.6	0.691	0.556
400	430.0	433.3	0.710	0.565

Table 8.1: Star connection: simulated (Section 4.2.a) against measured (Section 7.2.a) output power and rectifier efficiency across the target speed range, each at its own chosen C_m and R_L ($490\ \mu$F simulated; $630\ \mu$F, measured).

The measured output power stays at or above the simulated value across the whole range, with the two converging closely from the middle of the range onward. The rectifier efficiency shows the opposite trend: close to the simulated value at low speed, it falls increasingly below it as speed increases. This widening gap is consistent with losses that grow with electrical frequency and so matter more at high speed, such as core losses in the pickup coils or frequency-dependent conductor losses, neither of which is represented in the coil model of Chapter 4, which uses a single frequency-independent series resistance R_{Coil} for every speed.

8.2.b Delta Connection

The measured compensation capacitor ($630\ \mu$F) is again well above the simulated one ($455\ \mu$F), the same direction seen for the star connection. Table 8.2 compares the resulting output power and rectifier efficiency, taken from Figures 4.15, 4.16, 7.14, 7.15 and 7.16.

rpm	P_O (mW)		η_{Rec}	
	Simulated	Measured	Simulated	Measured
100	10.7	14.9	0.210	0.279
150	43.7	43.2	0.340	0.379
200	95.0	84.2	0.425	0.441
250	158.3	134.1	0.482	0.480
300	228.5	190.8	0.522	0.508
350	298.0	248.7	0.550	0.515
400	365.0	310.7	0.570	0.511

Table 8.2: Delta connection: simulated (Section 4.2.b) against measured (Section 7.2.b) output power and rectifier efficiency across the target speed range, each at its own chosen C_m and R_L ($455\ \mu$F, $5.75\ \Omega$ simulated; $630\ \mu$F, $5\ \Omega$ measured).

Unlike the star connection, the delta connection's measured output power crosses from above the simulated value to below it as speed increases, then stays below it for the rest of the range. The rectifier efficiency shows the same crossover, only later, staying above simulation for longer before falling below it towards the high-speed end. As Table 8.2 shows, the delta connection is simply much less efficient than the star connection overall.

8.3 Negative Voltage Converter (NVC)

The measured compensation capacitor ($430\ \mu$F) is again well above the simulated one ($260\ \mu$F), the same direction seen for both FWR connections. Table 8.3 compares the resulting output power and rectifier efficiency, taken from Figures 4.22, 4.23, 7.23 and 7.24.

Unlike either FWR connection, the NVC's measured output power stays below the simulated value across the entire range, with the widest gap appearing around the middle of the range rather than at either end. The rectifier efficiency follows the same pattern, staying closest to the simulated value at the two ends of the range and falling furthest below it in the middle.

8.4 Discussion

Across all three configurations, the compensation capacitor found to maximise the output power at 100 rpm is higher in measurement than in simulation: 630

rpm	P_O (mW)		η_{Rec}	
	Simulated	Measured	Simulated	Measured
100	69.0	57.7	0.665	0.568
150	146.5	109.4	0.768	0.619
200	238.9	175.9	0.821	0.648
250	374.8	245.2	0.841	0.668
300	508.0	308.0	0.842	0.682
350	576.8	367.7	0.834	0.693
400	610.0	425.7	0.820	0.701

Table 8.3: NVC: simulated (Section 4.3) against measured (Section 7.3) output power and rectifier efficiency across the target speed range, each at its own chosen C_m and R_L (260 μF, 26.00 Ω simulated; 430 μF, 6 Ω measured).

against 490 μF for the star connection, 630 against 455 μF for the delta connection, and 430 against 260 μF for the NVC. Since C_m of (2.31) is inversely proportional to the coil inductance L_{Coil} at a given electrical frequency, this consistent shift is what would be expected if the real coil inductance of the prototype were somewhat lower than the nominal value used throughout the simulations of Chapter 4, rather than a coincidence specific to any one configuration. The ceramic capacitors used for C_m also lose part of their nominal capacitance as the voltage across them increases, so their real capacitance at the operating point is lower than their rated value; a higher rated C_m is then needed to reach the same effective capacitance assumed in simulation, pushing the measured optimum higher for this reason too.

The three configurations diverge in how their measured output power compares to simulation across the range. The star connection's measured power stays at or above the simulated value at every speed, with the two converging almost exactly from 250 rpm onward. The delta connection starts the same way at 100 rpm, but crosses below simulation by 200 rpm and stays there. The NVC, by contrast, is the only configuration whose measured power is below simulation across the whole range, with the widest relative gap of the three appearing around 300 rpm. The rectifier efficiency is more consistent among configurations: all three are measured below simulation for most of the range, and the star connection's efficiency gap in particular grows steadily with speed, consistent with a loss mechanism, such as core or frequency-dependent conductor losses, that is absent from the frequency-independent coil model of Chapter 4 and matters more at high speed. The NVC's efficiency gap instead peaks in the middle of the range and narrows at both ends, which points less to a single loss mechanism than to how closely a C_m fixed at its 100 rpm value continues to match the coil as speed changes.

Because the delta connection carries the least absolute power of the three configurations across the whole range, the same absolute measurement uncertainty, for instance from the current-sensing chain of Section 6.4 or from the load resistor of Section 6.7, affects its relative comparison more strongly than it does the other two configurations; this is consistent with the delta connection showing the largest relative deviations from simulation at the low-speed end of the range, where its absolute power is smallest.

Despite these gaps, the ranking among configurations found in simulation is largely confirmed by measurement: the NVC remains the most efficient of the three at every speed, and the delta connection the least, matching Section 4.4. The clearest exception is the NVC's output power at the high-speed end of the range, which simulation placed clearly above both FWR connections; in measurement, the NVC's early lead instead narrows to essentially nothing by around 250 rpm, after which the star connection edges slightly ahead for the rest of the range, with the delta connection well behind both, matching what Section 7.4 finds from the same combined measurement. Overall, the NVC and the star connection perform very similarly in measurement: the NVC delivers more power at the low-speed end of the range, while the star connection takes the lead at the high-speed end.

9 Power Management IC Compatibility

9.1 Overview

This chapter looks beyond the rectifier stage studied throughout the rest of the thesis, to the power management integrated circuit (PMIC) that would sit downstream of it in a complete energy-harvesting system, converting the rectified output into a regulated supply for the load. Since three different rectifier topologies were built and characterised (the star and delta connections of the FWR, and the NVC), each presents a downstream PMIC with a different open-circuit voltage as a function of rotational speed. Section 9.2 reports this voltage for all three topologies, Section 9.3 introduces five commercially available PMICs suited to sub-watt harvesting sources, and Section 9.4 compares the two to identify which topology-PMIC pairings are electrically compatible, and at which speeds, over the tested range.

9.2 Open-Circuit Voltage Across Topologies

Tables 9.1-9.3 report the open-circuit voltage V_{OC} measured at the output of each rectifier topology, i.e. with no load connected ($R_L = \infty$), together with the high-frequency ripple superimposed on it, over the same 100-400 rpm speed range used throughout Chapter 7. This is the voltage a downstream PMIC's input pin would see before any load is drawn, and is the quantity compared against each candidate PMIC's input voltage range in Section 9.4.

The star connection's open-circuit voltage grows the most steeply with speed, from 1.98 V at 100 rpm to 7.04 V at 400 rpm, consistent with the highest output power found for this connection in Chapters 7 and 8. The delta connection's

Speed (rpm)	V_{OC} (V)	Ripple (V)
100	1.98	0.012
150	2.83	0.012
200	3.67	0.011
250	4.51	0.012
300	5.32	0.012
350	6.17	0.011
400	7.04	0.012

Table 9.1: Open-circuit voltage of the FWR star connection.

Speed (rpm)	V_{OC} (V)	Ripple (V)
100	1.000	0.022
150	1.460	0.021
200	1.910	0.017
250	2.365	0.017
300	2.825	0.017
350	3.275	0.017
400	3.755	0.017

Table 9.2: Open-circuit voltage of the FWR delta connection.

Speed (rpm)	V_{OC} (V)	Ripple (V)
100	2.70	0.547
150	3.05	0.691
200	3.20	0.639
250	3.15	0.566
300	3.15	0.476
350	3.15	0.414
400	3.15	0.369

Table 9.3: Open-circuit voltage of the NVC.

open-circuit voltage grows over the same range but stays well below the star connection's at every speed, from 1.00 V to 3.755 V. The NVC's open-circuit voltage instead rises only up to 200 rpm, where it peaks at 3.20 V, and settles to 3.15 V for every higher speed tested; its ripple is also an order of magnitude larger than either FWR connection's, consistent with its active, switching mode of operation described in Section 3.3.

9.3 Candidate Power Management ICs

This section summarises five commercially available PMICs considered as candidates for the six-phase harvester, each specified in terms of the input voltage range it accepts from the rectifier stage and the outputs it regulates that range down to.

9.3.a DS-AEM13920_VR

Datasheet [5]

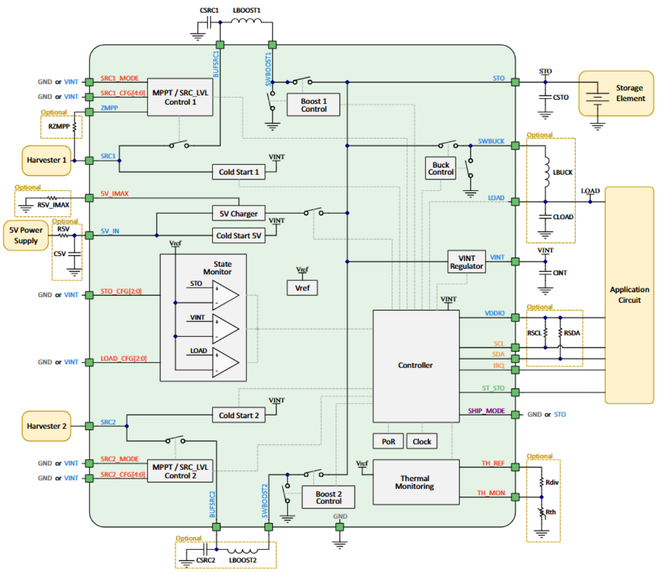


Figure 9.1: DS-AEM13920_VR block diagram.

The output voltage is regulated with a Buck converter connected to the storage element.

9.3.b DS-AEM10941_VR

Datasheet [4]

Symbol	Parameter	Range	Unit
V_{in}	Source input voltage	0.120-4.6	V
V_{cs}	Source cold-start voltage	0.275	V
V_{OC}	Open-circuit voltage tracked for MPP	0-5	V

Table 9.4: DS-AEM13920_VR input stage.

Symbol	Parameter	Range	Unit
V_{Buck}	Buck output voltage	0.6-2.5	V
I_{Buck}	Buck output current	0-100	mA

Table 9.5: DS-AEM13920_VR output stage.

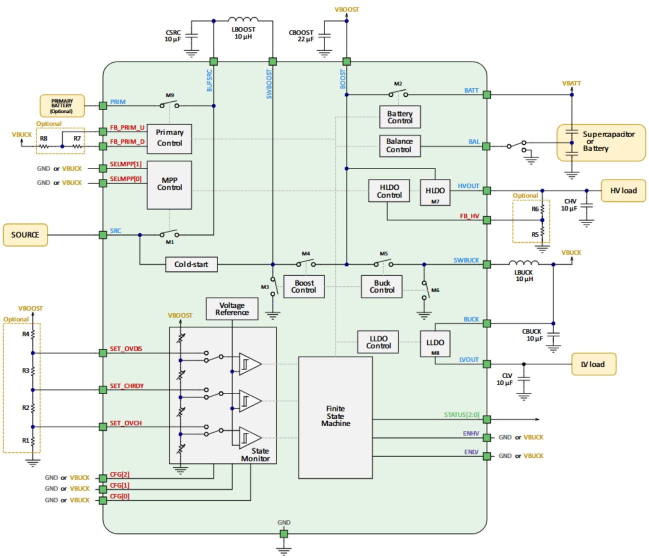


Figure 9.2: DS-AEM10941_VR block diagram.

Symbol	Parameter	Range	Unit
V_{in}	Source input voltage	0.05-5	V
V_{cs}	Source cold-start voltage	0.38	V
V_{OC}	Open-circuit voltage tracked for MPP	0.05-4.5	V

Table 9.6: DS-AEM10941_VR input stage.

The output voltage is regulated using a Buck converter with constant voltage (typ. 2.2 V) fed by the storage element. The IC additionally provides two LDO outputs for low-noise, high-stability supplies.

Symbol	Parameter	Range	Unit
V_{Buck}	Buck output voltage (typ. 2.2 V)	2-2.5	V
V_{LV}	LDO1 output voltage	1.2-1.8	V
I_{LV}	LDO1 output current	0-20	mA
V_{HV}	LDO2 output voltage	1.8-4.1*	V
I_{HV}	LDO2 output current	0-80	mA

* See the datasheet for additional details.

Table 9.7: DS-AEM10941_VR output stage.

9.3.c bq25570

Datasheet [15]

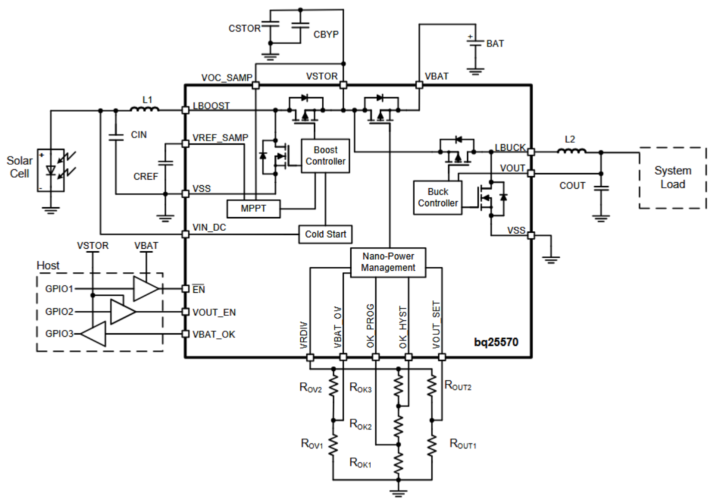


Figure 9.3: bq25570 block diagram.

The output voltage is regulated with a Buck converter connected to the storage element.

9.3.d SPV1050

Datasheet [14]

The SPV1050 can be configured either as a Boost or as a Buck-Boost converter at its input stage; Table 9.10 reports both.

Symbol	Parameter	Range	Unit
V_{in}	Source input voltage	0.1-5.1	V
V_{cs}	Source cold-start voltage	0.6	V
V_{OC}	Open-circuit voltage tracked for MPP	0.1-5.1	V

Table 9.8: bq25570 input stage.

Symbol	Parameter	Range	Unit
V_{Buck}	Buck output voltage (1.8 V $\pm$ 2%)	1.764-1.836	V
I_{Buck}	Buck output current	83-110	mA

Table 9.9: bq25570 output stage.

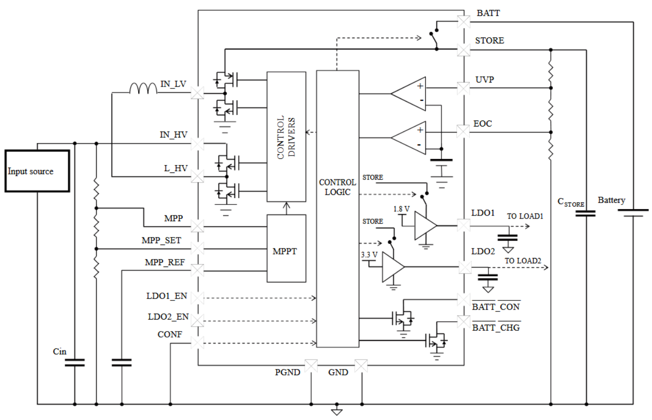


Figure 9.4: SPV1050 block diagram.

Symbol	Parameter	Range	Unit
Boost configuration			
V_{in}	Source input voltage	0.15-5.3	V
V_{cs}	Source cold-start voltage	0.55-0.58	V
V_{OC}	Open-circuit voltage tracked for MPP	0-5.3	V
Buck-Boost configuration			
V_{in}	Source input voltage	0.15-18	V
V_{cs}	Source cold-start voltage	2.6-2.8	V
V_{OC}	Open-circuit voltage tracked for MPP	0-18	V

Table 9.10: SPV1050 input stage.

Symbol	Parameter	Range	Unit
V_{LV}	LDO1 output voltage	1.8	V
I_{LV}	LDO1 output current	0-200	mA
V_{HV}	LDO2 output voltage	3.3	V
I_{HV}	LDO2 output current	0-200	mA

Table 9.11: SPV1050 output stage.

9.3.e SPV1040

Datasheet [13]

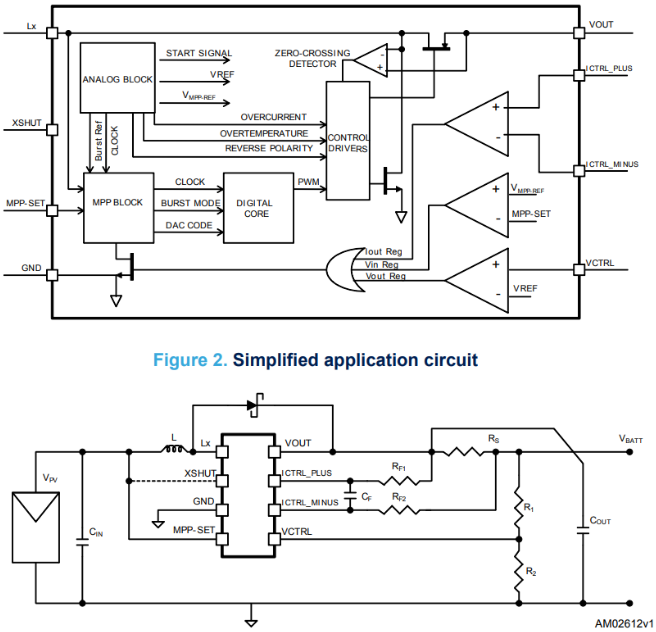


Figure 9.5: SPV1040 block diagram.

No output-stage specification is available for this device from the source material used for this chapter.

9.4 Compatibility Analysis

Table 9.13 compares each PMIC's input voltage range against the open-circuit voltage measured for each topology (Tables 9.1-9.3), to identify over which part of the 100-400 rpm tested range the two remain compatible. Since the delta connection's open-circuit voltage, including its ripple, never exceeds $3.755 + 0.017 = 3.772$ V and the NVC's never exceeds $3.20 + 0.639 = 3.839$ V, both

Symbol	Parameter	Range	Unit
V_{in}	Source input voltage	0.3-5.5	V
V_{cs}	Source cold-start voltage	0.8-2	V
V_{OC}	Open-circuit voltage tracked for MPP	0-5.5	V

Table 9.12: SPV1040 input stage.

stay within the input range of every one of the six PMIC configurations across the whole tested speed range. The star connection, reaching $7.04 + 0.012 = 7.052$ V including its ripple at 400 rpm, exceeds the input range of four of the six configurations above a certain speed, listed in the table; only the SPV1050's Buck-Boost configuration, with its 18 V upper limit, accommodates the star connection's open-circuit voltage across the entire tested range.

PMIC (configuration)	V_{in} range (V)	V_{cs} (V)	Star connection
DS-AEM13920_VR	0.120-4.6	0.275	up to 250 rpm
DS-AEM10941_VR	0.05-5	0.38	up to 250 rpm
bq25570	0.1-5.1	0.6	up to 250 rpm
SPV1050 (Boost)	0.15-5.3	0.55-0.58	up to 250 rpm
SPV1050 (Buck-Boost)	0.15-18	2.6-2.8	full range
SPV1040	0.3-5.5	0.8-2	up to 300 rpm

Table 9.13: Input-voltage compatibility of each PMIC configuration with the star connection over the 100-400 rpm tested range. The delta connection and the NVC are compatible with every configuration across the full range, and are omitted from the table.

Cold-start capability follows a different pattern, since it is set by the source cold-start voltage V_{cs} rather than the upper end of the input range, and must be met already at the lowest speed the harvester is expected to start from, 100 rpm. Against the V_{cs} values in Table 9.13, all three topologies exceed the cold-start threshold of the first four configurations (at most 0.6 V) already at 100 rpm, where the lowest of the three open-circuit voltages, the delta connection's, is still 1.00 V. The much higher threshold of the SPV1050's Buck-Boost configuration (2.6-2.8 V) is not met by any topology at 100 rpm: the star connection crosses it by 150 rpm (2.83 V), the NVC sits at the edge of the range already at 100 rpm (2.70 V) and clears it by 150 rpm (3.05 V), and the delta connection only clears it by 300 rpm (2.825 V). The SPV1040's threshold (0.8-2 V) sits closer to the three topologies' voltages at 100 rpm: the star connection is already at the top of the range (1.98 V), the NVC is above it (2.70 V), and the delta connection sits within the lower half of the range (1.00 V), only clearing its upper bound between 200 and 250 rpm.

Overall, the delta connection and the NVC can be paired with any of the five PMICs considered without additional input protection, cold-starting at the lowest speed tested for four of the six configurations. The star connection delivers the highest open-circuit voltage, and, as Chapters 7 and 8 show, the highest output power at higher speeds. To use it safely above 250-300 rpm, though, it would need either the wide-range SPV1050 Buck-Boost configuration, or a clamp added in front of a narrower-range device to protect its input.

10 Conclusions

10.1 Summary

This thesis addressed how to interface a six-phase MP-VREH harvester with downstream power-management electronics, comparing three candidate rectifying topologies, the FWR star connection, the FWR delta connection, and the NVC, across the target 100-400 rpm speed range. After introducing the harvester's operating principle and the candidate topologies (Chapters 2-3), each configuration was simulated (Chapter 4), built and measured on a physical prototype board (Chapters 5-7), and the simulated and measured results were compared (Chapter 8) before evaluating compatibility with commercial PMICs (Chapter 9).

10.2 Key Findings

The NVC is the most efficient of the three configurations at every measured speed, while the FWR delta connection is consistently the least efficient and delivers the least power. The FWR star connection, however, overtakes the NVC in raw output power from around 250 rpm onward, so the NVC and the star connection end up performing very similarly overall, each better suited to a different end of the speed range. The gaps between simulation and measurement are consistent with three causes: a real coil inductance somewhat lower than the nominal value used in simulation, the DC-bias derating of the ceramic compensation capacitors, and a loss mechanism that grows with electrical frequency and is not captured by the fixed-resistance coil model of Chapter 4, as discussed in Section 8.4.

10.3 Practical Implications

The delta connection and the NVC stay within the input voltage range of every PMIC configuration across the whole speed range, while the star connection, despite giving the highest power at high speed, exceeds the input range of four of the six configurations above a certain speed and needs either a wide-range buck-boost PMIC or an input clamp to be used safely (Section 9.4). These compatibility figures are based on open-circuit voltage alone; verifying the total power conversion efficiency of a full harvester-PMIC chain was outside the scope of this work and is left as future work.

10.4 Limitations

The coil model used throughout the simulations represents losses with a single frequency-independent series resistance. In addition, the prototype's mechanical housing is 3D-printed, making it more sensitive to vibration and to small shifts in the alignment between the pickup units and the rotor than a machined structure would be; this makes highly repeatable, precise measurements difficult and likely contributes to part of the residual scatter between simulation and measurement.

10.5 Future Work

Promising directions include an adaptive compensation network that tracks the optimal C_m across the whole speed range instead of a single fixed value, and testing the selected PMICs on hardware to measure the total power conversion efficiency of the complete chain from harvester to PMIC output, confirming whether the compatibility conclusions of Chapter 9 also hold in terms of delivered efficiency rather than voltage compatibility alone. Once the conversion system with the highest PCE has been identified, the next step is to determine which downstream technology to implement and how to put this harvesting system to practical, useful application.

Acronyms

AC Alternating Current.

CF Crest Factor.

DC Direct Current.

EMF Electromotive Force.

FBR Full-Bridge Rectifier.

FEA Finite Element Analysis.

FWR Full-Wave Rectifier.

IC Integrated Circuit.

IoT Internet of Things.

LDO Low-Dropout Regulator.

MMF Magnetomotive Force.

MOSFET Metal-Oxide-Semiconductor Field-Effect Transistor.

MP-VREH Multi-Phase Variable Reluctance Energy Harvesting.

MPP Maximum Power Point.

NVC Negative Voltage Converter.

PCB Printed Circuit Board.

PCE Power Conversion Efficiency.

PEE Power Extraction Efficiency.

PM Permanent Magnet.

PMIC Power Management Integrated Circuit.

RF Radio Frequency.

RMS Root Mean Square.

TENG Triboelectric Nanogenerator.

VD Voltage Doubler.

VREH Variable Reluctance Energy Harvesting.

WSN Wireless Sensor Network.

Bibliography

[1] Steven Anton and Henry Sodano. "A Review of Power Harvesting Using Piezoelectric Materials (2003-2006)". In: *Smart Materials and Structures* 16 (May 18, 2007), R1. DOI: 10.1088/0964-1726/16/3/R01.

[2] Stephen Beeby, M.J. Tudor, and Neil White. "Energy Harvesting Vibration Sources for Microsystems Applications". In: *Measurement Science and Technology* 17 (Dec. 1, 2006). DOI: 10.1088/0957-0233/17/12/R01.

[3] Rohan Dayal and Leila Parsa. "A New Single Stage AC-DC Converter for Low Voltage Electromagnetic Energy Harvesting". In: *2010 IEEE Energy Conversion Congress and Exposition*. 2010 IEEE Energy Conversion Congress and Exposition. Sept. 2010, pp. 4447–4452. DOI: 10.1109/ECCE.2010.5618446. URL: https://ieeexplore.ieee.org/document/5618446 (visited on 03/10/2026).

[4] e-peas. *DS-AEM10941 Datasheet*. e-peas. URL: https://e-peas.com/documents/AEM10941/DS-AEM10941.pdf.

[5] e-peas. *DS-AEM13920 Datasheet*. e-peas. URL: https://e-peas.com/documents/AEM13920/DS-AEM13920.pdf.

[6] Feng-Ru Fan, Zhong-Qun Tian, and Zhong Lin Wang. "Flexible Triboelectric Generator". In: *Nano Energy* 1.2 (Mar. 2012), pp. 328–334. ISSN: 22112855. DOI: 10.1016/j.nanoen.2012.01.004. URL: https://linkinghub.elsevier.com/retrieve/pii/S2211285512000481 (visited on 04/30/2026).

[7] Marco Grossi. "Energy Harvesting Strategies for Wireless Sensor Networks and Mobile Devices: A Review". In: *Electronics* 10.6 (Jan. 2021), p. 661. ISSN: 2079-9292. DOI: 10.3390/electronics10060661. URL: https://www.mdpi.com/2079-9292/10/6/661 (visited on 04/27/2026).

[8] Chen-Yu Hsieh, Mehrdad Moallem, and Farid Golnaraghi. “A Simple Bidirectional Bridgeless AC/DC Buck-Boost Converter for Automotive Energy Harvesting”. In: *IECON 2014 - 40th Annual Conference of the IEEE Industrial Electronics Society*. IECON 2014 - 40th Annual Conference of the IEEE Industrial Electronics Society. Oct. 2014, pp. 1937–1943. DOI: 10.1109/IECON.2014.7048767. URL: https://ieeexplore.ieee.org/document/7048767/ (visited on 03/10/2026).

[9] M Kroener, S K T Ravindran, and P Woias. “Variable Reluctance Harvester for Applications in Railroad Monitoring”. In: *Journal of Physics: Conference Series* 476.1 (Dec. 2013), p. 012091. ISSN: 1742-6596. DOI: 10.1088/1742-6596/476/1/012091. URL: https://doi.org/10.1088/1742-6596/476/1/012091 (visited on 05/04/2026).

[10] Bernardo León Ávila et al. “Energy Harvesting Techniques for Wireless Sensor Networks: A Systematic Literature Review”. In: *Energy Strategy Reviews* 57 (Jan. 1, 2025), p. 101617. DOI: 10.1016/j.esr.2024.101617.

[11] P.D. Mitcheson et al. “Energy Harvesting From Human and Machine Motion for Wireless Electronic Devices”. In: *Proceedings of the IEEE* 96.9 (Sept. 2008), pp. 1457–1486. ISSN: 0018-9219, 1558-2256. DOI: 10.1109/JPROC.2008.927494. URL: http://ieeexplore.ieee.org/document/4618735/ (visited on 04/30/2026).

[12] Shad Roundy, Paul K. Wright, and Jan Rabaey. “A Study of Low Level Vibrations as a Power Source for Wireless Sensor Nodes”. In: *Computer Communications* 26.11 (July 2003), pp. 1131–1144. ISSN: 01403664. DOI: 10.1016/S0140-3664(02)00248-7. URL: https://linkinghub.elsevier.com/retrieve/pii/S0140366402002487 (visited on 04/27/2026).

[13] STMicroelectronics. *SPV1040 Datasheet*. STMicroelectronics. URL: https://www.st.com/resource/en/datasheet/spv1040.pdf.

[14] STMicroelectronics. *SPV1050 Datasheet*. STMicroelectronics. URL: https://www.st.com/resource/en/datasheet/spv1050.pdf.

[15] Texas Instruments. *bq25570 Datasheet*. Texas Instruments. URL: https://www.ti.com/lit/ds/symlink/bq25570.pdf.

[16] Mengfei Wu, Ye Xu, and Sebastian Bader. “Multi-Phase Variable Reluctance Energy Harvester For Smart Bearing Hub Units”. In: *2024 IEEE 23rd International Conference on Micro and Miniature Power Systems, Self-Powered Sensors and Energy Autonomous Devices (PowerMEMS)*. 2024 IEEE 23rd International Conference on Micro and Miniature Power Systems, Self-Powered Sensors and Energy Autonomous Devices (PowerMEMS). Tons-

berg, Norway: IEEE, Nov. 18, 2024, pp. 46–49. ISBN: 979-8-3503-8020-0. DOI: 10.1109/PowerMEMS63147.2024.10814240. URL: https://ieeexplore.ieee.org/document/10814240/ (visited on 03/10/2026).

[17] Ye Xu, Sebastian Bader, and Bengt Oelmann. "A Survey on Variable Reluctance Energy Harvesters in Low-Speed Rotating Applications". In: *IEEE Sensors Journal* 18.8 (Apr. 2018), pp. 3426–3435. ISSN: 1558-1748. DOI: 10.1109/JSEN.2018.2808377. URL: https://ieeexplore.ieee.org/document/8300310 (visited on 05/04/2026).

[18] Ye Xu, Sebastian Bader, and Bengt Oelmann. "Design, Modeling and Optimization of an m-Shaped Variable Reluctance Energy Harvester for Rotating Applications". In: *Energy Conversion and Management* 195 (Sept. 1, 2019), pp. 1280–1294. ISSN: 0196-8904. DOI: 10.1016/j.enconman.2019.05.082. URL: https://www.sciencedirect.com/science/article/pii/S0196890419306375 (visited on 05/04/2026).

[19] Ye Xu et al. "System Implementation Trade-Offs for Low-Speed Rotational Variable Reluctance Energy Harvesters". In: *Sensors* 21.18 (Sept. 21, 2021), p. 6317. ISSN: 1424-8220. DOI: 10.3390/s21186317. URL: https://www.mdpi.com/1424-8220/21/18/6317 (visited on 03/10/2026).

[20] Liang Yu and Haoyu Wang. "A Single Stage AC/DC Converter for Low Voltage Energy Harvesting". In: *2016 IEEE Energy Conversion Congress and Exposition (ECCE)*. 2016 IEEE Energy Conversion Congress and Exposition (ECCE). Sept. 2016, pp. 1–6. DOI: 10.1109/ECCE.2016.7854782. URL: https://ieeexplore.ieee.org/document/7854782/ (visited on 03/10/2026).

[21] Liang Yu, Haoyu Wang, and Alireza Khaligh. "A Discontinuous Conduction Mode Single-Stage Step-Up Rectifier for Low-Voltage Energy Harvesting Applications". In: *IEEE Transactions on Power Electronics* 32.8 (Aug. 2017), pp. 6161–6169. ISSN: 1941-0107. DOI: 10.1109/TPEL.2016.2614995. URL: https://ieeexplore.ieee.org/document/7582535/ (visited on 03/12/2026).

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